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Fast response MPPT charger for the Técnico Solar Boat

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Introduction to the Research in

Electrical and Computer Engineering

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Abstract

Solar-powered vehicles, particularly boats, need effective energy conversion in rapidly changing conditions. The Maximum Power Point Tracker (MPPT) charger plays a critical role in energy conversion, ensuring that the solar panels operate at their Maximum Power Point (MPP). This work focuses on the design, simulation, and testing of a fast response MPPT charger for Técnico Solar Boat (TSB).

The developed prototype was designed to charge a lithium-ion battery of 43.2 V nominal voltage. The prototype is based on a synchronous boost converter operating at 100 kHz controlled by a Microcontroller Unit (MCU).

Different MPPT techniques were analyzed to identify a suitable solution for the dynamic operating conditions of the vessel. Balancing tracking speed, accuracy and complexity led to the implementation of a Perturb and Observe (P&O) algorithm. The converter and the Photovoltaic (PV) model were simulated in LTspice, and the tracker was tuned in MATLAB/Simulink. A custom Printed Circuit Board (PCB) was designed, manufactured and soldered by hand. Firmware on the MCU implements P&O with CC-CV charging, Controller Area Network (CAN) communication and a computer application for configuration.

Laboratory tests gave a peak efficiency of 96.84%, with tracking typically below 100 ms and 200 ms in the most demanding case, remaining below the 0.5 s and 1 s targets. Conducted noise and a 3 h thermal test stayed within safe limits. The charger followed the CC-CV profile in the laboratory and in an on-shore field test. At a cost of about 125 €, the board is a dedicated, configurable and low-cost MPPT charger for TSB.

Keywords

MPPT, MPPT topology, DC-DC Converter, Solar energy, P&O, Solar Boat

Resumo

Veículos alimentados a energia solar, em particular embarcações, necessitam de uma conversão de energia eficaz em condições climáticas de rápida variação. O carregador MPPT desempenha um papel importante nesta conversão, garantindo que os painéis operem no seu MPP. Este trabalho centra-se no projeto, simulação e ensaio de um carregador MPPT de resposta rápida para o TSB.

O protótipo foi projetado para carregar uma bateria de iões de lítio com tensão nominal de 43,2 V. O carregador proposto baseia-se num conversor boost síncrono que opera a 100 kHz, controlado por um MCU.

Foram analisadas diferentes técnicas de MPPT de modo a identificar uma solução adequada às condições de operação dinâmicas da embarcação. O compromisso entre a velocidade de seguimento, precisão e complexidade levou à implementação de um algoritmo P&O. O conversor e o modelo fotovoltaico foram simulados no LTspice, e o MPPT algoritmo no MATLAB/Simulink. Foi projetada, fabricada e soldada manualmente uma PCB dedicada. O firmware no MCU implementa P&O com carregamento CC-CV, comunicação CAN e uma aplicação para configuração.

Os ensaios de laboratório deram uma eficiência máxima de 96,84%, com um tempo de seguimento tipicamente inferior a 100 ms e de 200 ms no caso mais exigente, abaixo dos alvos de 0,5 s e 1 s. O ruído conduzido e um ensaio térmico de 3 h permaneceram dentro de limites seguros. O carregador seguiu o perfil CC-CV em laboratório e num ensaio de campo em terra. Com um custo de cerca de 125 €, a placa constitui um carregador MPPT dedicado, configurável e de baixo custo para o TSB.

Palavras Chave

MPPT, topologia MPPT, conversor DC-DC, energia solar, P&O, barco solar

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Acronyms

AC	Alternating current
BMS	Battery Management System
CAN	Controller Area Network
CC	Constant Current
CCM	Continuous Conduction Mode
CV	Constant Voltage
DC	Direct Current
DCM	Discontinuous Conduction Mode
ESL	Equivalent Series Inductance
ESR	Equivalent Series Resistance
GUI	Graphical User Interface
IC	Integrated Circuit
IncCond	Incremental Conductance
MCU	Microcontroller Unit
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracker
LUT	Look-Up Table
OC	Over Current
OTS	Off the shelf

P&O	Perturb and Observe
PCB	Printed Circuit Board
PV	Photovoltaic
R&D	Research and Development
SG01	São Gabriel 01
SR03	São Rafael 03
STC	Standard Testing Conditions
TSB	Técnico Solar Boat

1

Introduction

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This thesis presents the design, simulation, implementation and experimental validation of a custom MPPT charger for the photovoltaic system of TSB. The charger is intended to extract the maximum available power from the team's solar panels under the rapidly changing irradiance of a marine environment, while charging the low voltage battery of SG01 and providing telemetry through a CAN interface.

This chapter introduces the context of the work, the motivation for developing an in-house MPPT, the project objectives, and the organization of the remainder of this document.

1.1 Motivation

With environmental pollution intensifying worldwide, the transition to cleaner energy sources such as solar power has become increasingly urgent. In 1985, only 20.82% of the world's energy came from renewable sources. These numbers have been rising every year, and in 2024, they have reached 31.92% [11]. In 2024, 6.9% of global energy came from solar panels, and in Portugal this rose to 14.5%, demonstrating the importance and impact of solar energy [11]. Solar energy still plays a minuscule role, and it is listed behind the other sources of energy in terms of its contribution to meeting the world's energy demand. However, solar energy is becoming more relevant, with the cost of solar panels dropping significantly in the last decade.

Compared with other renewable sources, photovoltaic energy is advantageous owing to its availability, simplicity, low maintenance, environmental friendliness and reliability. More recently, it is becoming more relevant in the automotive industry, with solar powered cars, boats, and robots [12]. The automotive sector is one of the main contributors to global CO2 emissions. In 2018, 29% of total CO2 emissions in the EU came from the transport sector, with 4% coming from the maritime sector alone [13]. These challenges have directly motivated the development of the TSB project.

In 2015, TSB was created with the goal of designing and building solar powered boats to compete in international competitions while also contributing to innovation in the yacht industry. Since then, the project has grown, and several vessels have been built. It began with the construction of the first solar prototype, São Rafael 01, which had a lot of room for improvement, followed by São Rafael 02 and 03. The team then decided to approach hydrogen energy and autonomous driving, and therefore São Miguel 01, São Miguel02 and São Pedro 01 were built. More recently, a vessel was built to combine all the technologies used before and was named SG01 (Figure 1.1).

All these prototypes used solar energy to maximize their range and efficiency. In the first years, the energy produced was small, and the whole system was built using commercial Off the shelf (OTS) components. However, since the beginning of the TSB project, the team has maintained the spirit of innovation, and the desire to develop all the systems in-house has been present. In previous years, the team has built its own batteries, telemetry systems, flight controllers, fuel cell controllers, etc. The team

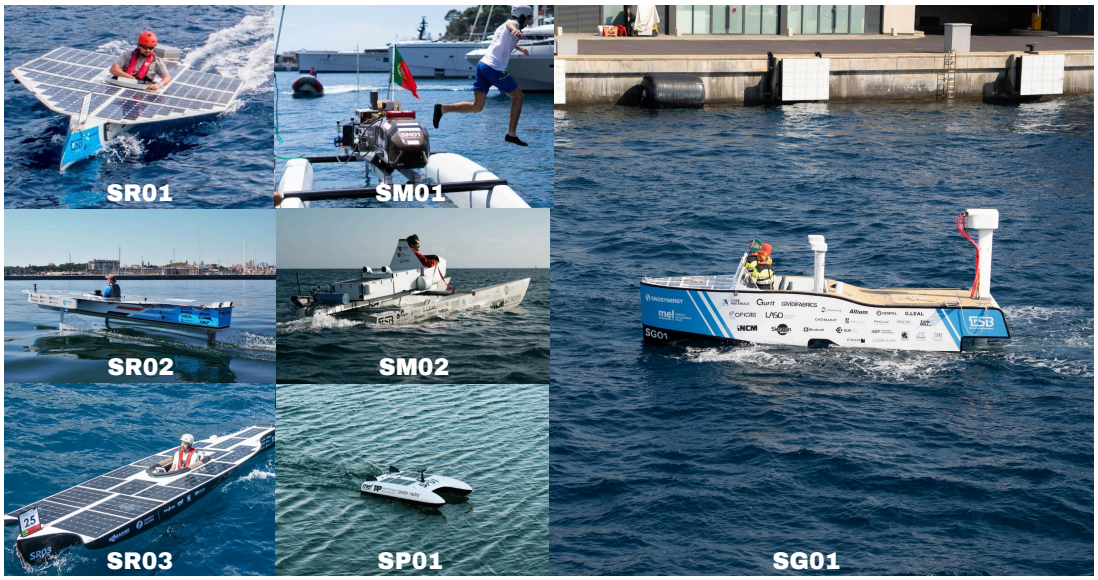

Figure 1.1: TSB vessels to date.

also designed more complex systems during the master's thesis. Until today a Battery Management System (BMS), dual motor system, Marine Propulsion System, a MPPT and more.

In 2020, the team started to build its own solar panels for São Rafael 03. To increase performance and the amount of information acquired by the solar panels' operation, in the same year, there was the first attempt to do a MPPT charger. This was the first step to achieve a fully functional prototype. The result was a working MPPT charger that was never implemented on any vessel because components constantly burned out, making it unreliable. Additionally, the project is outdated in its components and firmware and lacks configurability.

The MPPT charger is a fundamental subsystem in photovoltaic energy systems, responsible for maximizing the power extracted from solar panels by ensuring continuous operation at the MPP. A MPPT charger is typically a power converter that continuously monitors the voltage and current produced by the photovoltaic modules and adjusts their electrical operating point accordingly. By dynamically regulating these operating parameters, the MPPT algorithm compensates for variations in environmental conditions such as solar irradiance and temperature, which shift the MPP over time. As a result, the photovoltaic system can operate more efficiently and deliver maximum available power under changing conditions [14].

In the context of the TSB project, a solar system is important to increase the autonomy of the vessels. In the 2023/24 season, the team was able to reduce the consumption of SR03 from 1.5 kW to 300 W by using solar energy, decreasing 80 % of its consumption. With a custom-made MPPT, the team believes that it could reach even better performance while also increasing the data acquired about the system performance that would serve to improve the system.

Now, in the 2025/26 season, the relevance of the solar system decreased with our new vessel SG01, since it uses around 20 kW of constant power. So the main power source will be hydrogen. However, this vessel will still use solar energy to power auxiliary systems, as a reliable green source that keeps the boat powered in all situations.

1.2 Objectives

This project aims to design and implement a MPPT charger for the solar panels used in the TSB vessels. The MPPT charger converts the energy produced by the solar panel as efficiently as possible using a quality DC-DC converter and MPPT algorithms.

The main objectives of this project are:

- Study and understand the operation of solar panels and MPPT techniques;
- Choose the most suitable DC-DC topology for the system;
- Implement a fast response and accurate control algorithm;
- Design and simulate the hardware and software;
- Implement the system in hardware and develop a PCB;
- Test and validate the performance of the designed circuit;
- Ensure the safety and reliability of the MPPT charger for its integration in the TSB project;
- Provide performance data of the solar panel and the MPPT through a CAN communication interface;
- Develop a practical solution to change the prototype configuration.

By achieving these objectives, the project contributes to a more efficient solar system by maximizing the energy received from the solar panels and increasing the data available for performance tracking of the MPPT and solar panel. This data can be used to later improve the energy efficiency of the MPPT and draw conclusions about the manufacturing quality of the solar panels built by TSB members. A low-cost, efficient, and versatile MPPT solution can also provide long-term cost savings for the team, as the same design could be used across a variety of vessels.

In the context of TSB, such complex projects with high scientific rigor that involve simulations, PCB design, laboratory tests, and field tests are only possible in a master's thesis. So, for TSB, the main objective is to increase the team's knowledge of this subject by increasing the documentation about it. Also, due to a drastic event by another team, which burned all their MPPTs during the competition, the security of this system is the main concern.

1.3 Outline

This document is organized into seven chapters:

- **Chapter 1** introduces the context of the work and the motivation for developing a custom MPPT charger for TSB vessels. The objectives of the project are defined, and the structure of the document is outlined;
- **Chapter 2** describes the electrical architecture of SG01, including the high and low voltage systems, the photovoltaic installation, and the batteries used in this work. The contribution of the solar system to vessel autonomy is analyzed, and the time scales of environmental variations are estimated in order to define the required tracking speed of the MPPT algorithm;
- **Chapter 3** reviews the State-of-the-Art of photovoltaic systems and MPPT technology. It covers the operating principles of solar panels, the most common MPPT algorithms, non-isolated DC-DC converter topologies, lithium-ion battery charging techniques, and commercially available solutions;
- **Chapter 4** presents the implementation of the developed MPPT charger. The design of the synchronous boost converter, the selection of the processing unit and sensors, and the firmware architecture are described, including the charging state machine and communication interfaces;
- **Chapter 5** reports the simulations used to validate the converter design and the MPPT algorithm. A photovoltaic cell model and the boost converter are simulated in LTspice, while an analytical power-loss model is developed in MATLAB to estimate the efficiency and the contribution of each component. Additionally, using MATLAB/Simulink, the algorithm speed and accuracy were tested ;
- **Chapter 6** presents the experimental validation of the prototype. Three hardware iterations are assembled and tested in the laboratory, covering sensing accuracy, converter efficiency, thermal behavior, MPPT tracking performance, and field tests;
- **Chapter 7** summarizes the main conclusions of this work and identifies directions for future development.

2

System

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This chapter presents the electrical architecture of SG01, providing the context needed for developing and integrating the MPPT charger prototype. It presents the main subsystems relevant to solar energy conversion, such as the installed solar panels and battery. The impact of the PV system is analyzed, as well as the environmental changes.

2.1 Overview

SG01 is the most recent vessel built by TSB, designed to compete in the Sea Lab category of the Monaco Energy Boat Challenge. It uses hydrofoils to hydroplane and reduce drag. As it is TSB's largest and heaviest vessel to date, it required more power, addressed with a 130 kW motor combined with the first ever TSB High Voltage system (300-600 V), paired with a Low Voltage system (30-50 V) for the auxiliary electronics. Since solar panels alone could never meet the motor's power needs, the High Voltage system is designed to be powered by hydrogen instead, however, this is not yet implemented due to budget constraints, while the Low Voltage system relies on solar energy. Figure 2.1 shows a highly simplified diagram of the overall system architecture just described.

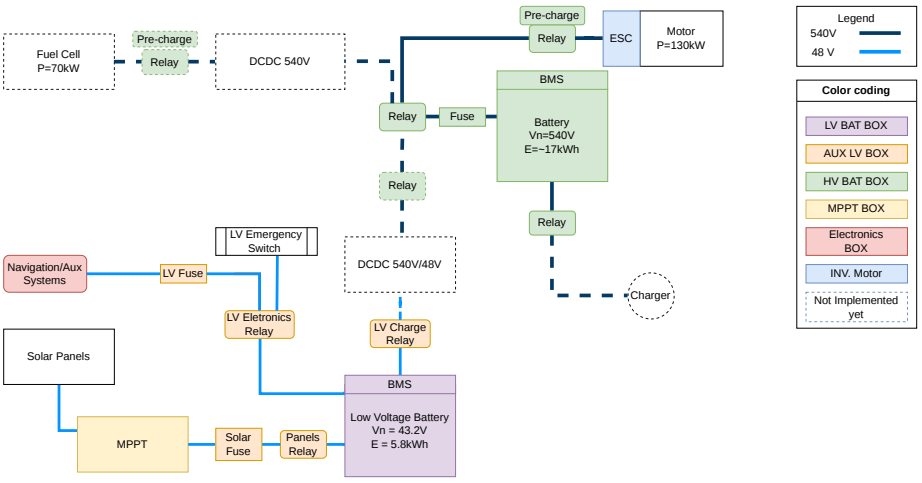


Figure 2.1: Simplified electrical schematic of SG01.

The low voltage system is designed to power all auxiliary system (dashboard, telemetry, flight control, electrical string, throttle, etc.). Some of these auxiliary systems are critical for the normal use of the vessel, for example, if the dashboard turns off, the pilot will not know the state of the high voltage system (temperature, voltage, state of charge, etc.). Or, even more critically, if the throttle does not work, the boat cannot electrically move.

Auxiliary systems are always changing and growing, so it is difficult to estimate their power consumption. For this work it is considered 1000 W of continuous consumption. It is not a perfect estimation

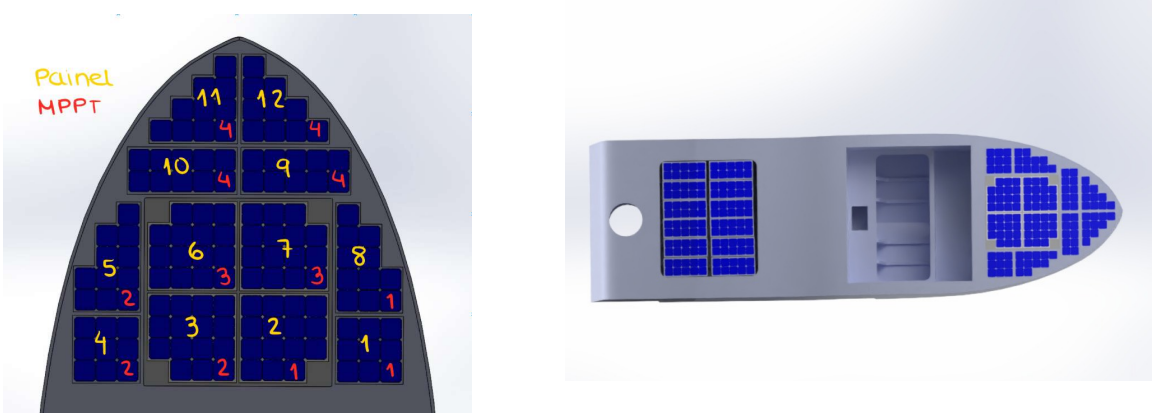
because the power consumption will depend on the driving mode (autonomous driving, hydroplaning, or normal drive), which can create different power profiles due to the actuators. Additionally, the cooling system (which is planned to be moved to the High Voltage system) has high power consumption that is not considered in this estimation.

To help keeping the low voltage system always powered, solar panels are used and connected to several MPPTs. In the future, there are plans to add a bidirectional DC-DC converter between the low voltage battery and the high voltage battery, enabling energy transfer in both directions.

2.2 PV system

The photovoltaic system of SG01 was designed to maximize the energy harvested from the available deck area while maintaining reliable operation under the varying environmental conditions encountered in marine applications.

The current installation consists of 140 Maxeon Gen 3 (Me3) photovoltaic cells distributed among ten custom manufactured solar panels. These panels connect to four independent MPPTs, as shown in Figure 2.2a. The distribution of panels among the MPPTs was carefully selected to reduce the effects of partial shading, temperature gradients, and non-uniform solar irradiance across the vessel surface. By electrically separating regions with different operating conditions, each MPPT can independently track the optimal operating point of its associated panels, increasing the overall energy yield.



(a) Distribution of solar panels and MPPT on SG01.

(b) Render of São Gabriel 01 with additional solar panels.

Figure 2.2: Solar panels installation on SG01.

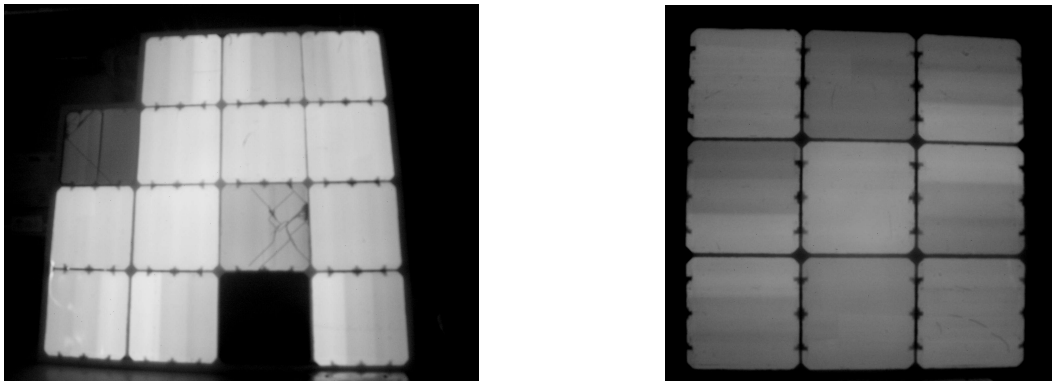
Each PV cell has an active area of approximately $155.06\ \mathrm{cm}^2$, resulting in a total photovoltaic area of $2.171\ \mathrm{m}^2$. Assuming Standard Testing Conditions (STC) (irradiance of $1000\ \mathrm{W/m}^2$ and $25\ ^\circ\mathrm{C}$), the photovoltaic array is capable of delivering approximately $526.4\ \mathrm{W}$ of peak power (theoretical value).

The PV energy is extracted by four commercial Genasun MPPTs, grouped in a dedicated MPPT box,

which convert the variable solar panel voltage into a stable charging voltage. This architecture allows each photovoltaic group to independently track its MPP, improving energy extraction under non-uniform illumination conditions.

To ensure high performance and shape configurability, the photovoltaic modules were manufactured in house using materials selected specifically for marine operation. This process begins by soldering all cells in a module together, then laminating them using vacuum and heat blankets.

Prior to installation, all panels underwent electroluminescence testing to detect manufacturing defects such as microcracks, broken cells, or interconnection failures, Figure 2.3. These tests were performed before integration into the vessel to ensure that only panels with acceptable electrical and structural integrity were installed. This quality control procedure reduces the likelihood of premature degradation and enables comparison before and after competitions.



(a) Worst results. **(b)** Best results.

Figure 2.3: Electroluminescence test results.

Although the current photovoltaic installation already contributes significantly to the low voltage energy budget, additional usable deck area remains available. Figure 2.2b presents a conceptual rendering with supplementary solar panels. Preliminary estimates indicate that the total photovoltaic power could be increased to approximately $977\ \mathrm{W}$, almost doubling the currently installed capacity. This upgrade would substantially reduce battery charging times and significantly increase the vessel's autonomy.

2.3 Battery

In recent years, TSB have been producing our batteries, achieving a lower cost and a highly customizable solution. For SG01 it was chosen lithium-ion cells due to their ease of use and quick assembly. The battery assembly is done by the team members, and it begins by placing the cells on a plastic holder to increase structural strength, then using a spot welder, the series and parallel connections are made with nickel tabs (Figure 2.4). In parallel, some thermistors are placed in different locations of the battery

to ensure a precise and robust temperature reading. In the end, the battery is placed in a case with a BMS and all necessary connections.

Figure 2.4: Low voltage battery assembly.

The current low voltage battery uses Samsung INR21700-50G cells in a 12s28p configuration. This configuration yields the specifications shown in Table 2.1.

Table 2.1: SG01 and SR03 battery specs.

Parameter	Unit	SG01 Battery	SR03 Battery	NAVICTUS Battery
Max Voltage	V	50.4	50.4	50.4
Nominal Voltage	V	43.2	43.2	43.56
Min Voltage	V	30	30	30
Configuration	–	12s28p	12s7p	12s1p
Number of cells	–	336	84	12
Energy Capacity	Wh	5866.56	1512	223.9
Capacity	Ah	135.8	35	5.140
Max charge current	A	135.8	42	5.3
Cutoff charge current	A	3.395	1.75	0.133
Max discharge current	A	271.6	175	15.9
Weight	kg	25.2	?	1
Internal resistance	$\mathrm{m}\Omega$	6	24	192

SG01 battery was designed for a different vessel which needed a substantially higher capacity. The battery is too heavy and overall overdimensioned.

For a more realistic impact analysis, the SR03 battery will be used. This is a much smaller battery with a reasonable capacity for this vessel. It uses Samsung INR21700-50s cells in a 12s7p configuration. This configuration yields the specifications shown in Table 2.1.

Additionally, to shorten the charging test time, another battery was used. This battery was provided by a startup called NAVICTUS and is a 12s1p pack with Samsung INR21700-53G cells.

2.4 PV system impact on autonomy

With SG01 battery (5844.56 Wh) and the maximum power that the solar panels can currently provide (526.4 W), the minimum time to fully charge the battery is 11 h and 7 min (5844.56 Wh / 526.4 W). This is a long time to charge a battery. There are two solutions to this problem: increase the power provided by the solar panels or decrease the battery capacity.

Considering the approach of reducing the battery capacity to $1512\ \mathrm{Wh}$, corresponding to the battery capacity of SR03, the battery can be fully charged in $2\ \mathrm{h}$ and $53\ \mathrm{min}$, which represents a much more reasonable charging time. Furthermore, increasing the power provided by the solar panels to $977\ \mathrm{W}$ reduces the charging time to $1\ \mathrm{h}$ and $33\ \mathrm{min}$, providing a highly suitable charging time for the battery.

The charging times are good metrics to evaluate the PV system, but the main goal is to maximize the time on water before full battery discharge. For this analysis, a consumption of $1000\ \mathrm{W}$ is assumed. Considering the original battery, it takes 5 h 51 min to fully discharge without solar panels, 12 h 21 min with solar panels, and approximately 10 days with the additional solar panels. As previously said, this battery is overdimensioned and will be changed in the future.

To better understand the impact of the PV system, the battery of SR03 will be considered. This battery takes 1 h and 30 min to fully discharge without solar panels, 3 h 12 min with solar panels, and approximately 2 days 18 h with the additional solar panels.

In theory, reducing the battery capacity while increasing the solar panel area could allow the battery to be maintained without the need for external charging. From another prescpective, the power cunsumption can be increased without the need to increase the battery capacity.

In practice, however, the solar panels will not continuously operate under STC conditions. Consequently, the power generated will generally be lower than the nominal value, and an appropriate margin must therefore be considered when sizing the battery and solar generation system.

2.5 Environmental changes

In engineering, the term “fast”, used in the title of this thesis, needs to be associated with a time scale. A process can only be considered fast when compared with another process with a longer characteristic time. Therefore, this section investigates the characteristic time scales associated with MPP variations caused by environmental changes, with particular emphasis on the vessel’s oscillatory motion and its implications for the required speed of the MPPT algorithm.

When analyzing MPP variations, the two main factors are temperature and irradiance. Temperature variations are generally slower than variations in irradiance. Although rapid temperature changes can occur, for example when water comes into contact with the photovoltaic panels, such events are considered unusual during normal operation and are therefore not considered in this analysis. Irradiation, by

contrast, can vary more rapidly as the angle of the solar panels changes with vessel roll and pitch.

The motion of a vessel subjected to waves can be described as a forced dynamic system. The vessel has characteristic natural frequencies associated with its different modes of motion, including roll, pitch and heave. The response to wave excitation depends on the relationship between the excitation frequency and the vessel's corresponding natural frequency. In particular, when the excitation frequency approaches a natural frequency, the response amplitude can increase significantly due to resonance. The vessel does not, however, necessarily oscillate at its natural frequency when subjected to excitation at another frequency, the steady state response occurs primarily at the excitation frequency, while the natural frequency determines the dynamic response and its amplitude.

Considering a simplified undamped free oscillation system, the natural frequency of a vessel can be calculated using the natural period Formula 2.1 where I is the mass moment of inertia and C is the restoring stiffness [15].

$$T = 2\pi\sqrt{\frac{I}{C}} \tag{2.1}$$

When applied to pitch and roll, the restoring stiffness is given by the mass (m), gravity (g) and metacentric height (GM_L for pitch and GM_t for roll). Since the inertia mass was not given, the equation was further simplified using the radius of gyration in pitch or roll (k_{yy} or k_{xx}), which is equivalent to Formula 2.2.

$$k = \sqrt{\frac{I}{m}} \tag{2.2}$$

$$T = 2\pi\sqrt{\frac{I}{mgGM}} \leftrightarrow T = 2\pi\sqrt{\frac{k_{yy}^2}{gGM_L}} \tag{2.3}$$

For pitch, $k = k_{yy}$ and $GM = GM_L$, whereas for roll, $k = k_{xx}$ and $GM = GM_T$.

This formula is used with results given by the hydrostatic simulations, which were provided by TSB, where GM_L is $21.515\ \mathrm{m}$ and GM_t is $1.867\ \mathrm{m}$. The radius of gyration is also calculated based on the longitude (L=8 m) and width/beam (B=2.5 m). Based on the International Towing Tank Conference (ITTC) recommendation [16], when the radius of gyration for pitch and roll is unknown, use 0.25L and 0.35B to 0.40B, respectively.

Based on the given values, the pitch and roll natural periods are 0.8654 s and 1.2853 s, respectively, which is equivalent to 1.1555 Hz and 0.7780 Hz. These values represent an estimate of the natural frequencies of the vessel.

As for the frequency of the waves, it has a wide range and depends on several factors. Studies on this subject show higher energy for waves with 1 s to 30 s periods, which is equivalent to 0.033 Hz to 1 Hz, Figure 2.5.

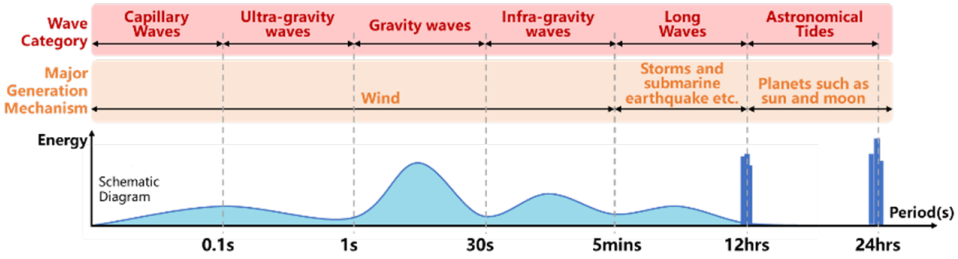


Figure 2.5: Classification of waves by period and respective energy [1].

The environmental excitation experienced by SG01 was further investigated using experimental data collected while the vessel was adrift at sea. Ten datasets containing the roll and pitch angles were collected and combined to obtain the frequency domain response shown in Figure 2.6. The results show significant spectral components at approximately 0.05 Hz and 0.5 Hz. The lower frequency likely reflects fast winds or transient oscillations and is consistent with the local forecast wave period. The higher frequency can be caused by the natural frequency of the vessel. The root causes were not investigated further, as this falls outside the scope of this project.

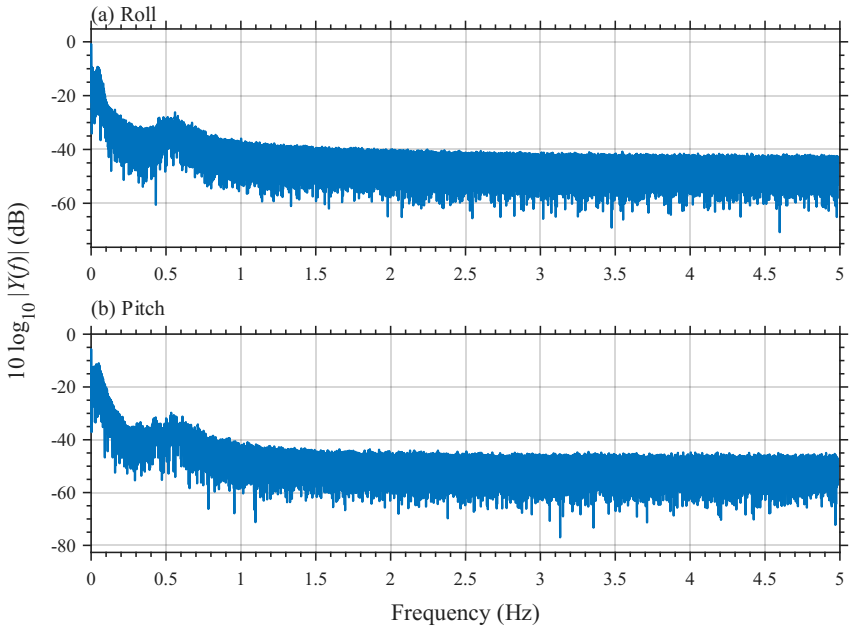


Figure 2.6: Measured oscillations from SG01.

Both theoretical and experimental results predict or measure frequency up to 1 Hz. In the context of the MPPT algorithm, the goal will be a minimum speed of 1 Hz or 1 s and a typical speed of more than 2 Hz or 0.5 s. These values will directly influence the sampling time and algorithm working frequency, as it will discuss later.

3

State-of-the-Art

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3.3 MPPT algorithms 18
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3.5 Battery charging techniques 26
3.6 Commercial Options 28

In this chapter, the main State-of-the-Art concepts regarding the engineering behind an MPPT charger and solar panels are explained. More specifically, it addresses key challenges such as: what is a solar panel, MPPT algorithms, DC-DC converter types, and battery charging techniques. Additionally, it contains a market analysis.

3.1 Solar panels

PV panels convert solar irradiance into electrical energy. A PV panel consists of multiple solar cells electrically connected in series and/or parallel to achieve the desired voltage and current levels. As in conventional electrical systems, the interconnection topology directly affects the panel's electrical characteristics: series connections increase output voltage, while parallel connections increase output current [17].

PV panels connected in series operate at higher voltage levels, reducing conduction losses in interconnecting cables and improving the efficiency of the DC-DC conversion stage. Moreover, power converters designed for high voltage and low current operation are generally less complex and more cost-effective than those operating at low voltage and high current. However, series configurations are more sensitive to partial shading or panel failure, as the output current of the entire string is limited by the weakest module [18]. In contrast, parallel configurations provide improved tolerance to shading and faults but result in higher current levels and increased conduction losses. In this project, the series/parallel configuration of the solar panels will not be changed, since TSB already defines it, but it is important to understand the implications of each configuration.

Each cell is made of a semiconductor material, usually silicon. This semiconductor is doped with phosphorus, a group V element, to create a negative type layer. On the other side, a layer is doped with boron, a group III element, to create a positive type layer. This creates a p-n junction, which is essential for the photovoltaic effect [3].

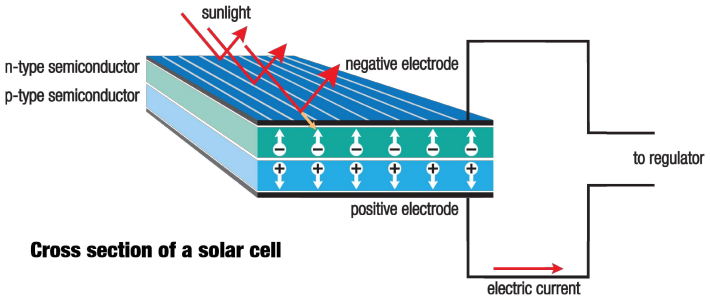


Figure 3.1: Principle of operation of a solar panel [2].

When sunlight reaches the solar cell, the semiconductor absorbs the photon's energy. This energy excites electrons, allowing them to escape their atomic bonds and create electron-hole pairs. The elec-

tric field at the p-n junction drives these free electrons toward the n-type layer and holes toward the p-type layer, generating an electric current when the cell is connected to an external circuit [3], Figure 3.1.

The energy generated results in voltage and current outputs that are not constant. Voltage and current are interdependent, therefore, a variation in one parameter leads to a corresponding variation in the other. This relationship is nonlinear and can be characterized by a current-voltage (I-V) curve. Furthermore, the power output also varies with operating conditions and can be represented by a power-voltage (P-V) curve. Figure 3.2 shows both curves. In both curves, there is a specific point where the power output is maximized, known as the MPP. This point corresponds to a unique combination of voltage and current, denoted as V_{mpp} and I_{mpp}, respectively. Operating the solar panel at this point ensures optimal energy extraction under a given condition.

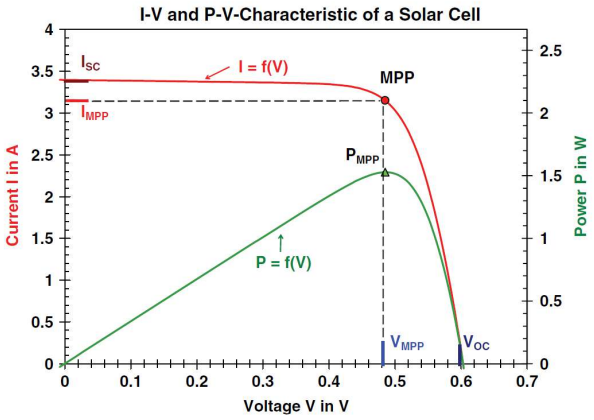
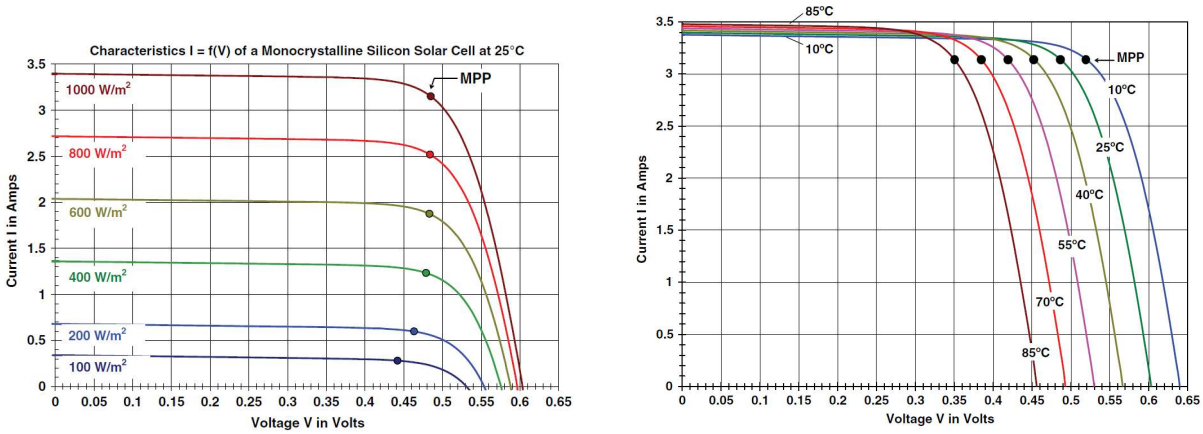


Figure 3.2: I-V and P-V curves of a solar panel [3].

The characteristic curves of a solar panel depend on several factors, including material quality, solar cell type, and environmental conditions such as temperature, solar irradiance, wind, and shade. It is noted that under partial shading conditions, it is possible to have multiple local maximums, but overall, there is still only one true MPP [19].

As shown in Figure 3.3a, changes in solar irradiance affect the solar panel's I-V and P-V curves. Higher irradiance produces more power mainly by increasing the panel's current output. In contrast, temperature has the opposite effect (Figure 3.3b). Higher temperatures decrease power output, mainly by reducing the panel's voltage output.

Both environmental conditions have a meaningful impact on the MPP of the solar panel and, therefore, on the system's power output. To maximize the power extracted by the solar panels, a classic power converter, such as a buck converter with a fixed duty cycle, is inefficient because neither the current nor the voltage remains constant under varying conditions. Instead, an MPPT charger is used, which will find the MPP and operate the solar panel at that point while sensing the changes caused by the environment.



(a) Cell temperature of $25\ ^{\circ}\mathrm{C}$ and variable irradiance.

(b) Irradiance of $1000\ \mathrm{W/m^2}$ and variable temperature.

Figure 3.3: I-V curves under different conditions [3].

3.2 Maximum Power Point Tracker (MPPT)

There are two main types of ”trackers”: mechanical trackers and electrical trackers. Both aim to maximize the energy extracted from a photovoltaic panel, but in different ways. Mechanical trackers are movable structures that follow the sun’s movement, maximizing the panel’s exposure to sunlight throughout the day. These devices are complex, expensive, and inefficient, so most research has focused on electrical “tracking” of the MPP.

An electrical Maximum Power Point Tracker (MPPT) charger is a device that is used to optimize the power output from solar panels by continuously tracking and adjusting the operation point of the panels to ensure they operate at their MPP.

To achieve this goal, an MPPT charger usually consists of a DC-DC converter, a microcontroller, and sensors (Figure 3.4). The sensors are used to measure the input/output variables of the control system (typically voltage and current). The microcontroller processes the information from these sensors and runs an algorithm to determine whether the MPP has been reached or how to act on the system to move toward the MPP. In the second case, the microcontroller generates a signal to control the DC-DC converter, which adjusts its operation accordingly.

The output can be connected to different types of loads, like batteries, DC loads or inverters. This project focuses on battery charging applications.

3.3 MPPT algorithms

There are plenty of MPPT algorithms, each one with its own advantages and disadvantages. The choice of algorithm depends on the specific application, the desired performance, and the available resources.

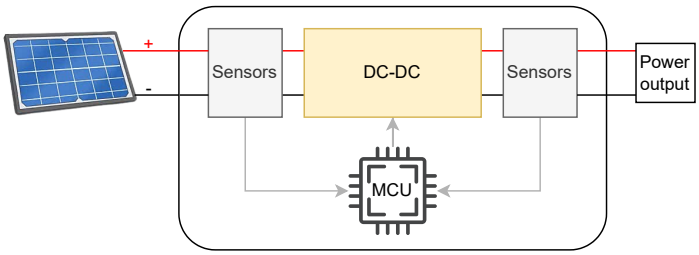


Figure 3.4: Simplified structure of an MPPT charger.

In this work, the aim is to extract the maximum possible energy from a solar panel installed on a moving boat, which produces I-V curve variations due to rapid changes in irradiance from inclination, orientation, and clouds, as well as temperature changes from waves and wind. For these reasons, tracking speed and accuracy are the most important factors to consider.

With this in mind, we briefly review the most relevant algorithms to identify the most suitable option for this project.

3.3.1 Constant Voltage (CV)

The Constant Voltage (CV) algorithm is the simplest and most inefficient method. It works by operating the solar panel at a fixed voltage. This reference voltage is used to calculate the duty cycle of the DC-DC converter to maintain the solar panel's operating point.

Well known variants improve performance by setting the reference voltage to a fraction of the open-circuit voltage (typically 71-78%) [19]. This voltage is obtained by briefly disconnecting the panel from the load, increasing reference accuracy and efficiency. Another variant uses a fraction of the short-circuit current as the reference [20].

Although both methods are simple and cost-effective, they are inherently inefficient because they must interrupt energy production during measurement phases. However, alternative implementations address this limitation through proxy measurement techniques employing dummy cells or diodes, whose physical properties closely approximate those of standard solar cells, thereby enabling continuous power generation during measurement acquisition [19] [21]. Still, the MPP is not always located at the same percentage of the open circuit voltage or short circuit current, leading to a lack of accuracy and therefore lower efficiency [14].

3.3.2 Perturb and Observe (P&O)

The P&O method is widely used in commercial products and is the basis of many advanced algorithms. Its popularity lies in its simplicity, low cost, and ease of implementation.

As its name indicates, the algorithm perturbs the voltage of a PV array and observes the resulting effect on the output power [22]. If an increase in voltage leads to an increase in output power, the operating point is moving toward the MPP, and the algorithm continues to increase the voltage. Conversely, if output power decreases, the operating point moves away from the MPP, and the algorithm reverses the perturbation direction [4, 23]. Figure 3.5 shows the algorithm flowchart.

The main drawback of this algorithm is that it oscillates around the MPP, causing power losses. This oscillation can be reduced by decreasing the perturbation step size, but this also reduces the algorithm's tracking speed. This results in an inherent trade-off between tracking speed and accuracy [19, 20].

Another well-known issue of P&O is that it can get confused in rapidly changing environmental conditions, like fast irradiance changes caused by clouds. In this case, the algorithm can misinterpret the power change caused by environmental variation as its own perturbation, leading it to move away from the MPP instead of toward it [19]. For example, if the perturbation is in the wrong direction but irradiance increases, the power output increases, and the algorithm continues perturbing in the wrong direction (Figure 3.6).

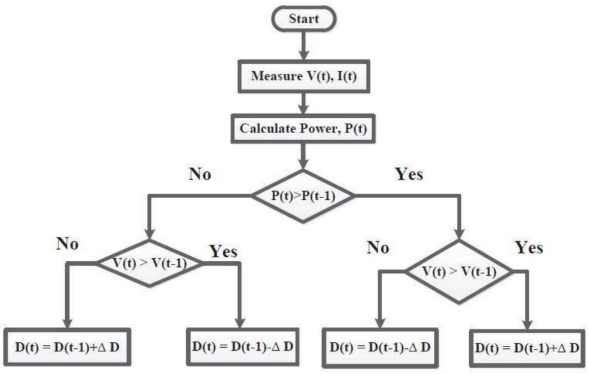


Figure 3.5: Flowchart of the classic version of Perturb and Observe algorithm [4].

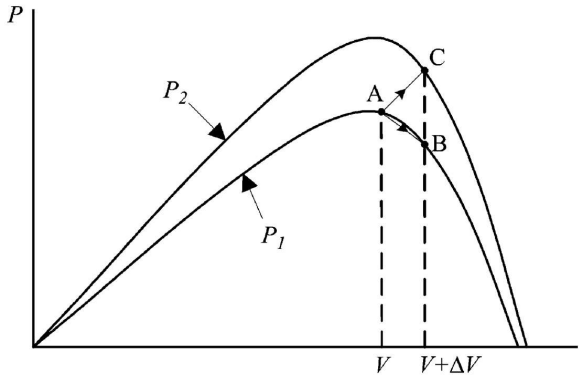


Figure 3.6: Divergence of P&O from MPP as shown in [5].

To address this limitation, an additional condition can be incorporated into the algorithm that evaluates two consecutive measurements of ΔP and ΔV. Specifically, when the signs of consecutive ΔP measurements do not follow the signs of ΔV, this indicates that the observed power variation results from environmental disturbances rather than the algorithm's perturbation. Consequently, the voltage adjustment is suppressed. This refined variant is designated the two-point algorithm or improved P&O method [17].

There are other variations of P&O algorithm that improve its performance, like Variable Step Size P&O (VSS-P&O), the three-point P&O, a-factor P&O, and more. Unfortunately, due to the scope of this work, it is not possible to explain all of them.

3.3.3 Incremental Conductance (IncCond)

The IncCond algorithm is another widely used MPPT method due to its accuracy and ability to track the MPP under rapidly changing environmental conditions.

This algorithm is based on the fact that at the MPP, the derivative of power with respect to voltage is zero. Given the solar panel output voltage and current, the algorithm can calculate the conductance (I/V) and the incremental conductance (dI/dV) [24].

The following formulas in Equation 3.1 show the equation on which this algorithm is based on.

$$\frac{dP}{dV} = \frac{d(V \cdot I)}{dV} = I + V\frac{dI}{dV} \rightarrow \frac{1}{V} \times \frac{dP}{dV} = \frac{I}{V} + \frac{dI}{dV} \tag{3.1}$$

At the MPP point, the slope of the P-V curve is zero, which means that the negative of the conductance, $-I/V$, is equal to the incremental conductance, dI/dV, (Figure 3.8). The algorithm compares these two values to determine if the operating point is at, to the left, or to the right of the MPP [20] [19]:

$$\begin{aligned} \frac{dP}{dV} = 0 \quad &\Longleftrightarrow \quad \frac{dI}{dV} = -\frac{I}{V} && \text{at MPP} \\ \frac{dP}{dV} > 0 \quad &\Longleftrightarrow \quad \frac{dI}{dV} > -\frac{I}{V} && \text{left of MPP} \\ \frac{dP}{dV} < 0 \quad &\Longleftrightarrow \quad \frac{dI}{dV} < -\frac{I}{V} && \text{right of MPP} \end{aligned} \tag{3.2}$$

Based on these conditions, the algorithm adjusts its operating point (increasing or decreasing the voltage) using a variable step size based on the instantaneous conductance relative to the incremental conductance, Figure 3.7. For example, as suggested in [19], the step size can be calculated using a proportional integral (PI) controller with zero as input reference and error defined as:

$$e = \frac{I}{V} + \frac{dI}{dV} \tag{3.3}$$

Comparative analysis shows that the IncCond algorithm performs better than the P&O method. This outcome is expected, as the Incremental Conductance technique was developed specifically to address the limitations inherent in the P&O approach [24]. For example, the IncCond method eliminates the oscillation problem around the MPP that is characteristic of the P&O method. Additionally, IncCond performs better under rapidly changing environmental conditions, such as sudden irradiance variations, because it uses derivative and variable step size [17]. The downside is increased complexity and, therefore, more parameters, which increases simulation and tuning time. Also, it uses fractions, which can be very demanding for MCUs.

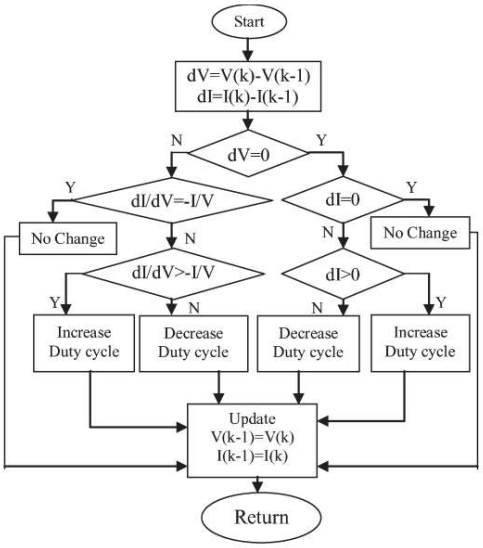


Figure 3.7: Flowchart of the IncCond method with direct control (for a buck converter) [6].

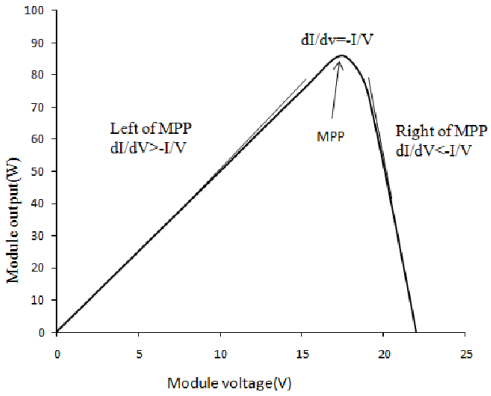


Figure 3.8: IncCond method principle.

3.3.4 Look Up Table (LUT)

The Look-Up Table (LUT) method is a simple and fast MPPT algorithm that relies on pre-calculated/measured data to determine the optimal operating point of a solar panel [20].

The LUT table contains several entries organized by voltage and current. Each of these entries contains the optimal duty cycle of the DC-DC converter, which is pre-calculated or measured for a specific voltage and current. This approach allows the algorithm to reach the MPP in one clock cycle by measuring the solar panel voltage and current, finding the closest entry in the LUT table, and applying the corresponding duty cycle to the converter. The entries of the table can be calculated through a mathematical model of the solar panel with different conditions (temperature and irradiance) or measured in a laboratory.

Table 3.1 visually represents a 2D LUT based only on current and voltage. Each entry represents the duty cycle used in each case.

The main advantage of this method is its speed since it can reach the MPP in one clock cycle. This makes it suitable for applications that require fast tracking or use weak MCUs. However, it is not very accurate [4].

To improve the accuracy, the LUT table needs to be large, which increases the required memory resources. Also, by adding the temperature and/or irradiance to the table would increase the accuracy, making it almost 100% accurate, but this would increase the size of the table exponentially. In extreme cases, the memory requirements became expensive and slow. There is a trade-off between size, speed, and accuracy that needs to be considered when using this method. Also, the LUT method does not adapt

Table 3.1: LUT matrix representation with $m \times n$ elements.

	V_1	$\cdots$	V_k	$\cdots$	V_m
I_1	D_{11}	$\cdots$	D_{1k}	$\cdots$	D_{1m}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_j	D_{j1}	$\cdots$	D_{jk}	$\cdots$	D_{jm}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_n	D_{n1}	$\cdots$	D_{nk}	$\cdots$	D_{nm}

to the aging of the solar panel or changes in its characteristics over time, unless the table is updated periodically through new measurements, which adds maintenance time.

3.4 Types of DC-DC converters

The DC-DC converter is a crucial component of an MPPT charger because it adjusts the solar panel operating point to match the MPP. This choice is important to the project because it directly affects the overall system's efficiency, cost, and complexity.

DC-DC converters fall into two main categories: isolated and non-isolated converters. The main difference between these two types of converters is the presence or absence of galvanic isolation between the input and output circuits [25].

3.4.1 Non-isolated versus isolated DC-DC converters

Galvanic isolation is usually achieved by using a transformer, which provides electrical isolation between the input and output sides of the converter. This isolation is important in applications where safety is a concern, such as in medical devices or industrial equipment, as it helps to prevent electrical shock and damage to sensitive components. In solar energy systems and E-mobility, isolated converters are often used when the solar panel is connected to the grid [26], as they help protect against voltage spikes and other electrical disturbances. Also, non-isolated converters can provide much higher/lower gain, which is useful for applications where solar panels MPP voltage differs greatly from the battery or load voltage (typically medium/high voltages, which is not the case).

In contrast, non-isolated converters do not provide galvanic isolation between the input and output circuits. However, they generally offer higher efficiency, lower cost, reduced size, and lower circuit complexity (Table 3.2). They also pose fewer thermal management challenges and are simpler to design. These advantages, along with the fact that galvanic isolation is not required for this application, led to excluding isolated DC-DC converters from this project.

Table 3.2: Comparison between non-isolated and isolated converters.

Type	Non-isolated Converters	Galvanic Isolated Converters
Circuit Complexity	Low	High
Efficiency	High	Medium
Size	Compact	Large
Electromagnetic Interference (EMI)	Medium/High	High
Switching Frequency	Medium	Medium
Thermal Management	Easier	Moderate
Control Complexity	Moderate	Moderate
Gain	Near 1	Very High/low
Cost	Low	High

Legend: undesired | medium | good

3.4.2 Non-isolated DC-DC converters

Modern non-isolated conversion circuits generally use one of three basic topologies: buck, boost, or buck-boost converters. They are ”basic” in the sense that only one switching element is needed. Each topology provides a specific result, such as voltage step-down, voltage step-up, or hybrid mode [8].

The **boost converter** is a step-up DC-DC converter widely employed in MPPT controllers when the required output voltage is higher than the solar panel open circuit voltage. It uses an inductor to store energy when the transistor is in the ON state and a diode in reverse bias (Figure 3.9b). When the transistor is on the OFF state, the inductor releases the stored energy to the output through the diode, increasing the output voltage [27].

This topology has been extensively documented in the literature and is distinguished by its inherent simplicity, cost-effectiveness, and superior efficiency characteristics [12] [4] [24]. Depending on the design variation used and constraints, it can achieve conversion efficiencies up to 98% [28].

A limitation of this topology is that it cannot emulate an impedance lower than the load impedance, which prevents it from reaching values close to the short-circuit current of the PV module. Consequently, it is not compatible with all solar panels or MPPT algorithms [28, 29].

The **buck converter** is a step-down DC-DC converter widely used when the solar panel voltage is higher than the battery or load voltage. Similar to the boost converter, it uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. However, in this case, the input connects to the output when the transistor is in the ON state. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, decreasing the output voltage (Figure 3.9a) [28].

This converter also has a similar downside to the boost converter. The buck converter can not emulate a smaller impedance than the load impedance, and therefore, it can not reach values near the open circuit voltage of the PV module [28] [29].

Buck-Boost converters are step-up and step-down DC-DC converters that can operate in both modes. This makes them more versatile than the previous two topologies, but also more complex and

less efficient [28]. This converter uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, increasing or decreasing the output voltage (Figure 3.9c) [8].

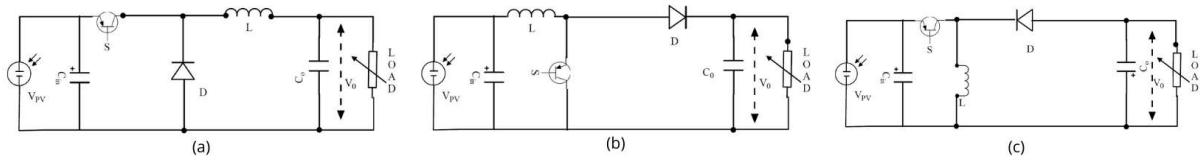


Figure 3.9: Buck (a), Boost (b) and Buck-Boost (c) DC-DC converters.

The conventional buck-boost converter produces an output voltage inverted relative to the input, which creates challenges for sensor measurements, driving circuits, and isolation, consequently, this option was excluded for the current project [27].

An enhanced variant of the buck-boost converter is the CUK converter, illustrated in Figure 3.10a. The main characteristic of the CUK converter is that the currents through inductors L_1 and L_2 are equal, allowing the use of a common magnetic core, which helps reduce current ripple. However, a major drawback of this topology is that the entire load current must pass through the coupling capacitor C. In addition, the converter produces an inverted output voltage [27].

The issue of output voltage polarity inversion can be addressed by using a single-ended primary inductor converter (SEPIC), as shown in Figure 3.10b. Nevertheless, its main drawbacks are the high output current and voltage ripple which increases losses and electromagnetic interference (EMI) [30]. The ZETA converter, shown in Figure 3.10c, mitigates these limitations by improving output current characteristics while maintaining the same output voltage polarity as the SEPIC converter [23] [27].

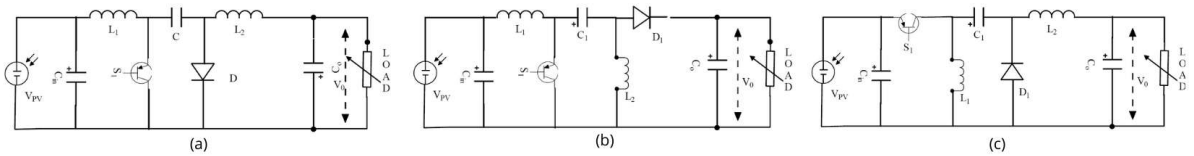


Figure 3.10: CUK (a), SEPIC (b) and ZETA (c) DC-DC converters.

3.4.3 Comparison between non-isolated DC-DC converters

All the non-isolated DC-DC topologies explained before have their own advantages and disadvantages. The choice of topology depends on the specific application, desired performance, and available resources. To help with this choice, Table 3.3 shows a comparison between the most relevant characteris-

tics of each topology [30]. The most significant advantages of each topology are highlighted according to the project's aim.

Table 3.3: Comparison of non-isolated DC-DC topologies.

Topology	Boost	Buck	Buck-Boost	CUK	SEPIC	ZETA
Electrical performance						
Efficiency	High	High	High	Medium	Medium	Medium/High
EMI	Low	Low	High	Medium	Medium	Medium
Input current ripple	Low	High	High	Very low	Medium	Low
Output current ripple	High	Low	High	Low	Medium	Very low
Output voltage ripple	Medium	Low	Medium	Medium	High	Low
Structural characteristics						
Circuit complexity	Low	Low	Low	Medium	Medium	Medium
Size	Low	Low	Low	Medium	Medium	Medium
Storage elements	1 L	1 L	1 L	2 L, 1 C	2 L, 1 C	2 L, 1 C
Switch stress	No	Yes	No	Yes	Yes	Yes
Functional aspects						
Type	Step-Up	Step-Down	Step-Down/Up	Step-Down/Up	Step-Down/Up	Step-Down/Up
Versatility	Low	Low	Medium	Medium	High	High
Inverted output	No	No	Yes	Yes	No	No
Reliability	High	High	High	Medium	Medium	Medium
Economic						
Cost	Low	Low	Low	Medium	Medium	Medium

Legend: good important

For MPPT controllers, the choice is mainly governed by the ability to operate at the MPP, which depends on the converter type (Step Up, Down, or Down/Up). Efficiency, cost (related to the number of storage elements), and output voltage ripple are also relevant to this choice.

The buck converter can be excluded from this comparison since, in this project, the solar panel voltage is lower than the battery voltage, therefore, a step-down converter is not suitable. However, when choosing between the boost and the step-up/down converters, there is a trade-off. The boost converter is simpler, cheaper, and more efficient. On the other hand, step-up/down converters are more versatile because they can operate over a wide range of input voltages, which is useful since solar panel size and battery voltage can change over the next few years.

Since the solar panel voltage is always lower than the battery voltage in this project, the boost converter is the most suitable topology because of its simplicity, cost-effectiveness, and high efficiency. However, if versatility were to become a priority, the ZETA converter would be the preferred choice among the step-up/down topologies, owing to its non-inverted output voltage and low output current ripple, which are advantageous for battery charging applications.

3.5 Battery charging techniques

As the demand for electronic devices and E-vehicles increases, the need for efficient, compact, and lightweight batteries has emerged. Among many existing technologies, lithium-ion batteries have one of

the best energy-to-weight/volume ratios and, at this moment, it is the technology used in TSB batteries. But like every other battery technology, they must be charged properly to ensure safety and longevity [7].

To charge these batteries properly, the MPPT algorithm can operate alone if an additional converter is used in series. But in most of MPPT chargers, the MPPT and the battery charging unit are integrated in the same converter. This way, the MPPT algorithm can adjust the solar panel operating point to extract maximum power, while the battery charging technique ensures the battery charges properly. A proper combination of both is essential to ensure both maximum energy extraction and safe battery charge.

Several charging techniques are available for lithium-ion batteries. For the application considered in this work, the most relevant approach is the constant-current constant-voltage (CC-CV) charging method [31].It is also widely used in commercial chargers because of its simplicity and effectiveness [32]. This method consists of two main phases (Figure 3.11). In the first phase, the battery is charged at a constant current until it reaches its maximum voltage. In the second phase, the voltage is held constant at this limit while the current gradually decreases as the battery approaches full charge. Once the current drops below a certain threshold, the charging process is terminated to prevent overcharging [32].

When employing the CC-CV charging technique with a MPPT controller, the initial Constant Current (CC) phase is regulated by the MPPT algorithm, which extracts the maximum available power from the solar panel, limited by the maximum charging current. If the maximum charge current is reached, it operates at CC. Once the CV phase is reached, the CC/MPPT function is disabled, and the power converter increases the solar panel voltage to progressively reduce the charging current while keeping the output voltage constant.

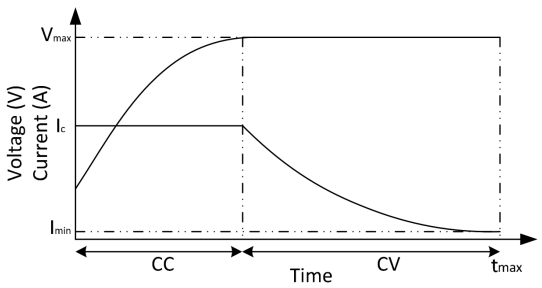


Figure 3.11: Charging profile of CC/CV [7].

The literature has proposed more advanced high-performance battery charging techniques to reduce charging time without increasing charging current, thereby preserving battery lifetime. However, these techniques generally introduce higher switching losses and increased system complexity. In the present application, the current generated by the photovoltaic panels is less than half of the maximum allowable battery charging current, consequently, the charging current is inherently limited by the solar panels

rather than by the charging strategy. Furthermore, battery lifetime is not a critical concern in this project, as the battery is expected to undergo a limited number of charge cycles, not exceeding 20 cycles per year.

Therefore, the CC-CV charging method is considered the most suitable technique for this project, owing to its simplicity, effectiveness, and widespread adoption in commercial battery chargers.

Other charging techniques fall outside the scope of this work, however, some methods are briefly discussed in the references provided [31] [7] [32].

3.6 Commercial Options

To assess the feasibility of adopting a OTS solution, this section reviews existing commercial options for solar battery charging with MPPT capability. Both specialized ICs and complete commercial MPPT chargers are considered, and their electrical, functional, and economic characteristics are evaluated against the requirements of this project. The objective is to identify whether a commercially available solution can satisfy the constraints of the application or, alternatively, to justify the need for a custom designed converter and control system.

3.6.1 Specialized ICs

Several commercially available Integrated Circuit (IC) are specifically designed for solar battery charging and MPPT applications. These devices are relevant to this project since, if suitable, they could significantly reduce design complexity by integrating functions such as DC-DC conversion, protection mechanisms, and an MPPT algorithm.

Table 3.4 summarizes four commonly used commercial ICs. Characteristics marked in red indicate incompatibilities with the project requirements, leading to the exclusion of the corresponding devices.

The BQ24650RVAT is a stand-alone synchronous buck battery charge controller that implements a high-efficiency battery charger and provides safety features. By adding a MCU to run a MPPT algorithm and current and voltage sensors, this IC can track the MPP of a solar panel, making it appropriate for applications with 12/24 V batteries and high-power panels, such as electric vehicles. However, it does not support the required 48 V battery voltage and is therefore unsuitable for this work.

The MAX20801TPBA+ and SPV1040TTR target low-power IoT applications with small batteries and solar panels. Both devices cannot meet the 48 V output requirement, and limited publicly available information on tracking speed further restricts performance evaluation.

The LT8491IUKJ#PBF offers a wide input and output voltage range, buck-boost operation, high efficiency, communication capabilities, and CC-CV charging. Despite these advantages, its reported MPPT tracking speed of up to 2 s is insufficient for the dynamic conditions of this application. Additionally,

its I^2C interface would require an additional microcontroller to provide CAN communication, increasing system cost.

Table 3.4: Comparison of commercial specialized ICs.

Model	BQ24650RVAT	MAX20801TPBA+	SPV1040TTR	LT8491IUKJ#PBF
Electrical specifications				
Input voltage	5 to 28 V	1.5 to 18 V	0.3 to 5.5 V	6 to 80 V
Output voltage	12 to 24 V	12.4 V	-0.3 to 5.5 V	1.3 to 80 V
Max output current	Externally limited	12 A	1.8 A	10 A
Switching frequency	600 kHz	Unknown	100 kHz	100 to 400 kHz
Electrical efficiency	95%	99.1% (max)	80 to 95%	95 to 99%
MPPT and control				
Algorithm	User input	Unknown	Unknown	P&O
Tracking accuracy	User dependent	99.9% (max)	Unknown	Unknown
Tracking speed	User dependent	Unknown	Unknown	1 to 2 s
Functional aspects				
Topology	Synchronous Buck	Synchronous Buck	Synchronous Boost	Buck-Boost
Communication	No	No	No	I^2C
Configurability	No	No	No	Yes
Charge profile	CC-CV	Unknown	Unknown	CC-CV
Extra hardware needed				
Transistors	Yes	No	No	Yes
Sensors	Yes	No	No	No
Storage elements	Yes	Yes	Yes	Yes
Economic				
Price	5.25 €	4.50 €	3.02 €	15-20 €

Legend: unsuitable

3.6.2 Commercial MPPT chargers

Several commercial MPPT chargers can address the problem under analysis without requiring additional hardware. However, each solution presents limitations when evaluated against the project requirements. Table 3.5 summarizes selected commercial MPPT chargers, with red-marked characteristics indicating incompatibilities.

The GVB-8-Li-CV(50.4) is a high-performance and fast MPPT charger, particularly suited for dynamic environments such as marine applications. It has been successfully used by TSB in previous projects, demonstrating high efficiency and reliability. Nevertheless, its lack of configurability, fixed output voltage, no communication and high cost restrict its suitability for this application. Despite these drawbacks, it serves as a valuable reference design.

The Smart Solar MPPT 100/20, developed by Victron Energy, offers high efficiency, fast tracking, and integrated communication interfaces. However, it uses a buck topology that requires higher input voltages, making it incompatible with the small and distributed solar panel arrays used in the TSB application.

Other commercial solutions, such as the Rover Lite and MS4840N, provide acceptable performance but rely on communication protocols incompatible with the project requirements and offer limited technical documentation.

The Reboost V0.2.1 is an open source MPPT charger originally developed for automotive applications. While it provides flexibility and near adequate specifications, its reliance on expensive and discontinued components limits its use.

Overall, none of the evaluated commercial MPPT chargers fully satisfy the technical, economic, and configurability requirements of this project, motivating the development of a custom solution.

Table 3.5: Comparison of commercial MPPT controllers.

Model	**GVB-8-Li-CV (50.4 V)**	**Smart Solar MPPT 100/20**	**Reboost V0.2.1**	**Rover Lite**	**MS4840N**
General information					
Brand	Gensun	Victron Energy	TPEE	Renogy	BougeRV
Type	Boost	Buck	Synchronous Boost	Boost	Boost
Control and communication					
Communication	No	VE.can and Bluetooth	Yes	Bluetooth and RS485	Bluetooth
Configurable	No	Yes	Yes	Yes	Yes
GUI	No	Yes	Yes	Yes	Yes
MPPT and charging					
Prog. battery voltage	No	Yes	Yes	non-lithium batteries	Yes
Tracking speed	15 Hz / 66.6(6) ms	Fast (unknown)	Programmable	Unknown	Unknown
Tracking efficiency	99%+ typical	Unknown	Programmable	99%	Unknown
Electrical efficiency	96-99% typical	98% peak	Unknown	97%	Unknown
Charger profile	CC-CV	Unknown	Programmable	Unknown	CC-CV
Hardware implementation					
MCU	ATtiny461A-U	Unknown	STM32G474	Unknown	Unknown
MCU Details	8-bit, RISC, 64 MHz	–	32-bit, Arm, 170 MHz	–	–
Transistor	FZT951	Unknown	GS61008T	Unknown	Unknown
Transistor type	PNP BJT	Unknown	GaN FETs	Unknown	Unknown
Economic					
Price	240 €	100-200 €	Components: 250-280 € PCB: 20-100 € Total: 270-380 €	300-350 €	120-160 €

Legend: unsuitable

4

Implementation

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This chapter presents the implementation of the MPPT charger prototype. The charger is based on a synchronous boost converter controlled by an STM32 MCU, with dedicated sensing, protection, communication and configuration features.

The chapter first describes the DC–DC converter, including its operating principles and the dimensioning of the main power components. The processing unit and its configuration are then presented, followed by the voltage, current, temperature and irradiance sensing circuits. Finally, the firmware architecture is described, covering the charging state machine, communication interfaces and the graphical user interface used to configure the device.

4.1 DC-DC converter

Since the solar panels voltage of SG01 are always lower than the battery voltage, a step-up converter is required. This section presents its operating principles and the dimensioning of the main components.

4.1.1 Operation principles

The DC-DC converter chosen for this project was a boost converter. This topology uses one switching device, one inductor, one output capacitor, and a diode, Figure 4.1. The switching device is used to control the output voltage by switching on and off at a specified frequency (fsw) and duty cycle (D). The inductor is used to store energy when the switch is on (stage 1) and release it when the switch is off (stage 2), while the output capacitor is used to smooth the output voltage.

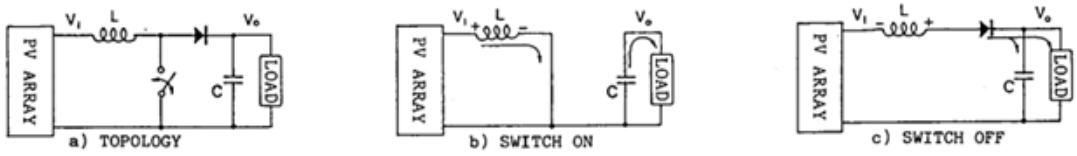

Figure 4.1: Classic boost converter topology and stages [8].

Analyzing the boost converter separately in its two stages, it is possible to define the voltage drop across the inductor (VL), as a function of the input voltage (Vin) and output voltage (Vout) as follows:

$$V_L = \begin{cases} V_{in} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} & \text{if } t \in [D \cdot T, T] \end{cases} \tag{4.1}$$

With equation 4.1 and the fact that at the stationary state the average voltage across the inductor is zero, it is possible to define the output voltage as a function of the input voltage and duty cycle as follows:

$$V_{Lavg} = \frac{1}{T}(\int_0^{DT} V_{in}dt + \int_{DT}^{T}(V_{in} - V_{out})dt) = 0 \Rightarrow V_{out} = \frac{V_{in}}{1-D} \tag{4.2}$$

Equation 4.2 is one of the most important equations in boost converter design, as it defines the relationship between the input voltage, output voltage and duty cycle. As the duty cycle increases, the output voltage also increases, which is the main principle behind boost converters. For our use case, it is also important to analyze the relationship between the input and output current. Considering no power losses, it is possible to define the input current as follows:

$$P_{in} = P_{out} \Leftrightarrow V_{in} \cdot I_{in} = V_{out} \cdot I_{out} \Leftrightarrow I_{in} = \frac{V_{out}}{V_{in}} \cdot I_{out} \Leftrightarrow I_{in} = \frac{I_{out}}{1-D} \tag{4.3}$$

However, these equations are only valid for ideal components since they do not take into account the parasitic elements of the components, such as the inductor resistance, the drain-source resistance, and the voltage drop across the diode (there are other parasitic components that also affect performance). In a real boost converter, these parasitic elements will cause a voltage drop and therefore reduce the output voltage.

The influence of the transistor internal resistance ($R_{DS_{(on)}}$), the inductor resistance (R_L) and the diode voltage drop (V_d) can be briefly evaluated by modifying Equation 4.1 as follows:

$$V_L = \begin{cases} V_{in} - R_{DS_{(on)}} \cdot I_{DS} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} - V_d & \text{if } t \in [D \cdot T, T] \end{cases} \tag{4.4}$$

In this case, at the stationary state, the average voltage across the inductor is $R_L \cdot I_L$ and therefore the output voltage can be defined as:

$$V_{Lavg} = \frac{1}{T}(\int_0^{DT} (V_{in} - R_{DS_{(on)}} \cdot I_{DS})dt + \int_{DT}^{T}(V_{in} - V_{out} - V_d)dt) \Leftrightarrow \tag{4.5}$$

$$\Leftrightarrow R_L \cdot I_L = \frac{1}{T}[T \cdot D \cdot (V_{in} - R_{DS_{(on)}} \cdot I_{DS}) + T \cdot (1-D) \cdot (V_{in} - V_{out} - V_d)] \Leftrightarrow \tag{4.6}$$

$$\Leftrightarrow V_{out} = \frac{V_{in} - D \cdot R_{DS_{(on)}} \cdot I_{DS} - R_L \cdot I_L}{1-D} - V_d \tag{4.7}$$

Equation 4.7 shows the 3 causes of losses in a boost converter. The most significant is usually the voltage drop across the diode (typically 0.3 V to 0.7 V), which reduces the efficiency through a conduction loss of $V_d \cdot I_{out}$. This can be largely avoided by using a synchronous boost converter (Figure 4.2), which replaces the diode with a transistor. In this case, the voltage drop is only $R_{DS,on} \cdot I$, which is usually below 0.1 V.

As for the transistor and inductor internal resistance, it cannot be removed entirely, so it must be

taken into account when designing the boost converter. Therefore, it is important to select components with low internal resistance to minimize losses and increase converter efficiency.

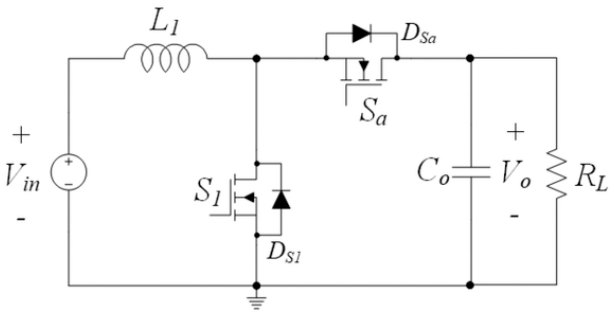

Figure 4.2: Synchronous boost converter topology [9].

4.1.2 Dimensioning

Let us first define the design goals of the synchronous boost converter. The converter must step up the voltage from the solar panels, which may consist of up to 40 cells (Maxeon Gen 3 from SunPower) connected in series, a minimum of 20 cells in series, and is optimized for 35 cells (Table 4.1). The converter must charge a lithium-ion battery with a nominal voltage of 43.2 V. The switching frequency was defined as 100 kHz. This relatively high frequency allows the use of smaller inductors and capacitors, while still avoiding excessive switching losses.

Table 4.2 and 4.1 summarizes the system requirements.

Table 4.1: PV panel parameters at MPPT.

Configuration	V_{mppt} (V)	I_{mppt} (A)	P_{mppt} (W)	V_{oc} (V)	I_{sc} (A)
40 cells	24.68	6.05	149.31	29	6.374
35 cells	21.875	6.05	132.34	25.375	6.374
20 cells	12.34	6.05	74.66	14.5	6.374

Table 4.2: Converter design constraints.

Parameter	Value
Battery voltage (nominal)	43.2 V
Battery voltage (max)	50.4 V
Battery voltage (min)	30 V
ΔV_o	0.1 V
ΔV_i	0.5 V
Efficiency	$> 90\%$

Since the load is a battery, the output ripple voltage must be limited to avoid aging and heating. Also, the output voltage ripple must be low to avoid unexpected behavior in the electronic components

connected to the battery. As for the input (the solar panel), the voltage ripple must be low for easier measurement and precise control. Therefore, the values presented in Table 4.2 were selected.

4.1.2.A Inductor

The inductor is one of the most important components in the boost converter since it is responsible for storing energy and releasing it to the load.

The choice of inductor strongly influences converter performance. High inductance will generate low current ripple, however it generates losses due to its internal resistance. On the other hand, a smaller inductance will generate high current ripples, high stress in components, and high magnetic interference. These current ripples can cause the system to operate in Discontinuous Conduction Mode (DCM). In DCM, the inductor current reaches zero during the switching period, which means that the energy stored in the inductor is not enough to supply the load during the off time. This can cause high current and voltage ripples, which can damage the components and reduce the efficiency of the converter. Therefore, it is important to ensure that the inductance value is high enough to operate in Continuous Conduction Mode (CCM).

The equation that describes the inductor is defined as:

$$V_L = L \cdot \frac{di_L}{dt} = L \cdot \frac{\Delta I_l}{\Delta t} \Leftrightarrow L = \frac{V_L \Delta t}{\Delta I_L} \tag{4.8}$$

In the first stage (transistor in conduction mode), the voltage in the inductor is equal to the input voltage and it occurs for $D \cdot T$ s. Therefore, the inductor ripple current can be defined as follows:

$$\Delta I_L = \frac{V_{in} \cdot D \cdot T}{L} \tag{4.9}$$

The current ripple can be described using the ripple factor and the average inductor current as follows.

$$r = \frac{\Delta I_L}{I_{L,avg}} \tag{4.10}$$

$$I_{L,avg} = \frac{I_o}{1 - D} \tag{4.11}$$

Using Equation 4.11, 4.10, 4.8 and 4.2, the following equation is achieved:

$$L = \frac{V_L \Delta t}{\Delta I_L} = \frac{V_{in} DT}{\frac{I_o r}{1-D}} = \frac{V_{in}(1-D)DT}{I_o \cdot r} = \frac{V_{out}(1-D)^2 DT}{I_o \cdot r} \tag{4.12}$$

The value of D that maximizes Equation 4.12 is 1/3, which results in the critical inductance formula:

$$MAX[(1-D)^2 \cdot D] = (1-1/3)^2 \cdot 1/3 = \frac{4}{27} \tag{4.13}$$

$$L_{critical} = \frac{4}{27} \cdot \frac{V_o}{I_o \cdot r \cdot f} \tag{4.14}$$

Using the maximum battery voltage (50.4 V) and the maximum current ($I_{bat} + Margin = 10\ A$), and a ripple factor of 2, which is equivalent to the CCM limit, the following result is achieved:

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 2 \cdot 100 \times 10^3} = 3.73\ \mu\text{H} \tag{4.15}$$

This is a reference value, if used, it would highly stress the components, generating heat, magnetic interference and high current ripples. Therefore, the calculation was repeated, this time with a ripple factor of 0.4, which is more reasonable.

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 0.4 \cdot 100 \times 10^3} = 18.66\ \mu\text{H} \tag{4.16}$$

As it is explained in the simulation section (Section 5), a higher value will be beneficial to reduce current ripple and noise. For that reason, the final inductor value was defined as $68\ \mu\text{H}$, which results in a ripple factor of 0.11.

To select the appropriate inductor, the rated current and the saturation current must be taken into account. The rated current is the maximum current that the inductor can handle without overheating. In this case, the rated current must be higher than the maximum short-circuit current of the biggest solar panel configuration (40 cells in series), which is 6.374 A. Some margin must be added to this value, so the rated current must be higher than 10 A.

The saturation current is the maximum current that the inductor can handle without saturating. When the inductor saturates, the inductance drops significantly, which can cause high current ripples and damage the components. Considering the Open circuit voltage of a 40 cells solar panel (worst case), the inductor ripple current is 4.26 A at 100% duty cycle. Therefore, the saturation current must be:

$$I_{sat} > I_{in} + \frac{\Delta I_L}{2} = 6.374 + \frac{4.26}{2} = 8.50\ A \tag{4.17}$$

It is physically impossible to have a solar panel on both short-circuit current and open-circuit voltage at the same time. And for this reason, the calculation is overdimensioned and it is only used as a reference instead of a strict limit. The inductor chosen was the 74437529203680 from Würth Elektronik, which has an inductance of $68\ \mu\text{H}$, a rated current of 16 A and a saturation current of 7.2 A ($|\Delta L/L| < 10\%$) and 10.6 A ($|\Delta L/L| < 30\%$). The saturation current is too close to the objective, however, since the inductance is higher than what is needed, the inductance would still be acceptable even at -30% inductance.

Additionally, it has a low Equivalent Series Resistance (ESR) of only $10.3\ \mathrm{m\Omega}$, a high self-resonant frequency of 5.7 MHz, and a high maximum voltage of 250 V.

4.1.2.B Capacitor

The output capacitor is responsible for smoothing the output voltage and reducing the voltage ripple. The capacitor's electrical characteristic can be described by the following equation:

$$i_c = C\frac{dv_o}{d_t} \tag{4.18}$$

Based on this equation, the capacitor value can be calculated based on the desired voltage ripple and the load current as follows:

$$i_c = C_o\frac{\Delta V_o}{\Delta t} = C_o\frac{\Delta V_o}{D \cdot T} \Leftrightarrow C_o = \frac{I_o D}{\Delta V_o \cdot f} = \frac{I_{in}(1-D)D}{\Delta V_o \cdot f} \tag{4.19}$$

Using the maximum input current of 10 A, a duty cycle of 0.5 (the value that maximizes the formula), a voltage ripple of 0.1 V and a switching frequency of 100 kHz, the following value is achieved:

$$C_o = \frac{10 \cdot (1-0.5) \cdot 0.5}{0.1 \cdot 100 \times 10^3} = 250\ \mu\mathrm{F} \tag{4.20}$$

The same calculation applies to the input capacitor, but the voltage ripple is less critical than for the output capacitor because the input voltage is not directly connected to the load. Therefore, a voltage ripple of 0.5 V was used:

$$C_{in} = i_{in} \cdot \frac{\Delta t}{\Delta V_i} = \frac{i_{in} \cdot D}{\Delta V_{in} \cdot f} = \frac{10 \cdot 0.5}{0.5 \cdot 100 \times 10^3} = 100\ \mu\mathrm{F} \tag{4.21}$$

To choose the appropriate capacitors, the voltage rating, current ripple and ESR must be taken into account. The voltage rating must be higher than:

$$V_{rating} > MAX(V_{in}) + \Delta V_{in} = 29 + 0.5 = 29.5\ V \tag{4.22}$$

Where 29 V is the maximum voltage of a solar panel (40 cells in series) and 0.5 V is the voltage ripple defined in Table 4.2.

The current ripple must be higher than the RMS input current, which can be calculated based on the RMS formula for a triangle wave as follows:

$$I_{ripple} > \frac{\Delta I/2}{\sqrt{3}} = \frac{4.26/2}{\sqrt{3}} = 1.23\ A \tag{4.23}$$

As for the ESR, the value must be as low as possible to reduce power losses.

After analyzing the market, the input capacitor selected was the 870575875006, which is a $56\ \mu\mathrm{F}$ (2 in parallel) with a voltage rating of 63 V and a ripple current of 2.4 A each (total of 4.8 A). The main reason for this choice was its low ESR ($22\ \mathrm{m}\Omega$) compared to other capacitors with similar values. Additionally, using two capacitors in parallel further reduces the ESR, increases the current ripple they can withstand, and improves system reliability.

The same process was repeated for the output capacitor, but in this case the voltage rating must exceed the maximum battery voltage, which is 50.4 V. The current ripple must be higher than the RMS output current, which can be calculated as follows:

$$V_{rating} > MAX(V_{out}) + \Delta V_{out} = 50.4 + 0.1 = 50.5\ V \tag{4.24}$$

$$I_{ripple} > \sqrt{\frac{I_{in}^2}{4} + \frac{\Delta I^2}{12} \cdot (1-D)^2} = \sqrt{\frac{6.374^2}{4} + \frac{4.26^2}{12} \cdot (1-0)^2} = 3.42\ A \tag{4.25}$$

The ripple current in the output capacitor is much higher than in the input capacitor, and it is not a common value for electrolytic capacitor, therefore, instead of using a single capacitor, multiple capacitors in parallel will be used to achieve the desired current ripple rating.

After simulations and market analysis, the output capacitor value was increased due to the output current ripple dependency on the output voltage ripple. The final capacitor chosen was the 860040875005 with $100\ \mu\mathrm{F}$ (5 in parallel, so $5 \times 100\ \mu\mathrm{F}$) with a voltage rating of 100 V and a ripple current of 0.8 A each (total of 4 A). Some other options were considered, and compared for optimization in section 5.1.2.

Additionally, the input and output need to be filtered to reduce the high-frequency noise generated by switching. This is achieved by adding a small capacitor in parallel with the main capacitors. The values use are $1\ \mu\mathrm{F}$ and $0.1\ \mu\mathrm{F}$. For this use case, ceramic capacitors are better since their stacked multilayer construction provides extremely low ESR and Equivalent Series Inductance (ESL), which overall lets them maintain low impedance up into the MHz range.

4.1.2.C Transistor and gate driver

The transistor is the switching device in a synchronous boost converter. The choice of transistor strongly influences the converter's behavior and performance.

When selecting a transistor for a switching converter, not only the maximum rating matters, but the switching speed and internal resistance are also quite important.

There are many types of transistors (IGBT, MOSFET, GaN, etc.), however, at this frequency (100 kHz) and power level the choice is between MOSFETs and GaN transistors (Figure 4.3). The GaN (Gallium Nitride) transistor technology appeared in 1993 in lab experiments, however, it reached the market only more recently, first for radio frequency products in 2004 and only in 2009 for power electronics. GaN

transistors have low internal resistance and very fast switching speeds, which can significantly reduce switching losses and therefore increase converter efficiency. Additionally, they have a high energy density and lower Vgs(on). However, they are more expensive and require a precise gate driver to operate correctly because their $V_{\mathrm{GS,max}}$ is usually low.

In applications where the efficiency is a key factor in the design like solar inverters or MPPT chargers, it is better to use the component that has lower Internal (On) Resistance (RdsOn) and supports higher switching frequency. In this case, GaN is the better fit. MOSFETs would take second place since they can handle higher switching frequencies than IGBTs [10].

There are a lot of options in the market, however, there was one in particular that stood out due to its compact size, while having an integrated gate driver and two transistors in the same package. The IC is the EPC23102, which was designed for power applications, with an incredibly low RdsOn (only $5.2\ \mathrm{m\Omega}$), 35 A of maximum current, and a maximum operating voltage of 100 V.

Additionally, it incorporates shoot-through protection that protects the transistor from a control error, however, a dead time needs to be introduced in firmware. It uses a DFN package, which has a good thermal performance.

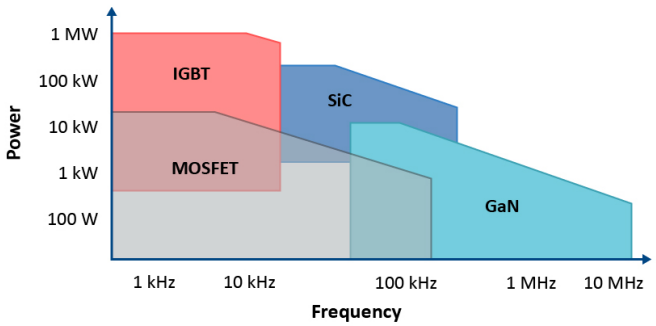

Figure 4.3: Technologies selection based on power and switching frequency [10].

4.2 Processing unit

The processing unit runs the MPPT algorithm, generates the PWM signals and handles sensing, protection and communication. This section describes the microcontroller selection and its configuration.

4.2.1 MCU selection

A Microcontroller Unit (MCU) typically consists of three fundamental components: a central processing unit (CPU), memory, and peripheral modules. The CPU executes program instructions and performs arithmetic and logical operations. Memory is generally divided into non-volatile memory (Flash) for program storage and volatile memory (SRAM) for data and runtime variables. Peripheral modules provide interfaces to the external environment and may include analog-to-digital converters (ADCs), timers, pulse-width modulation (PWM) units, communication interfaces (such as UART, SPI, I²C and CAN), and general-purpose input/output (GPIO) ports. Integrating these components enables standalone operation without extensive external circuitry [33].

When selecting a MCU for a specific application, the first factor to consider is to use an 8, 16, or 32 bit MCU. For this application, a 16-bit MCU would probably be sufficient, however to ensure future-proof and compatibility with other TSB systems, the choice is for a 32-bit MCU. Besides, nowadays, 32-bit MCUs are considered the industry standard since most embedded system Research and Development (R&D) effort is focused on 32-bit cores, and thus both architectures can achieve similar power consumption [34]. There are two main architectures for 32-bit MCUs: AVR and ARM. Being the most widely used and most advanced architecture in the industry, ARM was chosen for this project [35].

STM32 is a family of MCUs that offers a wide range of solutions with different characteristics, making it suitable for various applications, including this project. TSB uses this family of microcontrollers, so it will be the one used in this project to maintain the standard.

The MCU requirements for this project are:

- Fast clock frequency: 80 MHz or higher;
- Support 1x CAN peripheral, 1x I2C, 1x UART;
- Able to read 4 or more analog signals;
- 12bit ADC or more.

The final decision was the STM32L476. Although it is not the most suitable STM32 microcontroller for this application, it was selected due to its low cost and because it was already widely used within TSB, reducing development time and associated risks.

The STM32L476 is based on an ARM Cortex-M4 core with floating-point support and operates at frequencies up to 80 MHz. The device provides 1 MB of Flash memory and 128 kB of SRAM, which is more than sufficient for implementing the MPPT algorithm, data acquisition, communication protocols, and debugging features. Regarding peripherals, it includes three 12-bit ADCs with up to 16 external channels each, multiple general-purpose and advanced timers capable of high-resolution PWM generation, several communication interfaces (CAN, SPI, I^2C, UART, USB), and DMA controllers that allow peripheral operation with minimal CPU intervention.

These features make the STM32L476 particularly attractive for power electronics applications that require simultaneous voltage and current measurements, fast PWM generation, and real-time control execution. Furthermore, the hardware floating-point unit simplifies the implementation of control algorithms and numerical calculations.

If there were no limitations, a microcontroller with a higher clock frequency would have been selected, increasing both PWM duty-cycle resolution and available computational power. Additionally, a device with native 16-bit ADCs would improve measurement accuracy and reduce quantization effects. Nevertheless, the STM32L476 provides sufficient performance and peripheral resources to meet this application's requirements.

4.2.2 Configuration

Since the designed board is custom-made, there is no pre-configuration of the MCU in use. In this particular case, the STM32 project needs to be configured in a STM32CubeMX, which is a tool that simplifies the configuration process. This tool is used to define the clock frequency, peripherals working frequency, pin mapping (GPIOs, ADCs, UART, CAN, etc.) and respective configuration, Interrupts, Timers, PWMs, etc. This tool generates a project based on C programming language with all drivers and configurations needed. These files serve as a base for firmware development and can be modified during the project.

As it is not feasible to describe all the configurations used, the following section focuses on those most relevant to the project.

The MCU runs at 80 MHz with an external 16 MHz ceramic resonator. This frequency is enough for the computational time and a minimal PWM resolution. It can be flashed using JTAG and it uses an Independent Watchdog (IWDG) time interrupt. The IWDG is a safety feature that resets the MCU in case of a crash or infinite loops. It works by resetting the timer periodically in a specific part of the code, if the timer is not reset during a specified time period, the interrupt is triggered and the MCU resets automatically.

As for timers, one timer was configured with 100 kHz for PWM and two pins were configured as complementary PWMs with a specific dead time. To prevent cross conduction (both transistors in conduction at the same time), the dead time was used as a safety feature too. The dead time is calculated using Formula 4.26, where the "value" must be in the range 0 to 255.

$$DeadTime = Value/F_{CLK} \tag{4.26}$$

For this project, several values were tested using an oscilloscope. The smallest value that ensures that both signals are never high at the same time is 2, however, 3 was used to add margin. The measured dead time was 32 ns, and the rise and fall times were 19 ns and 22 ns, respectively (Figure 4.4).

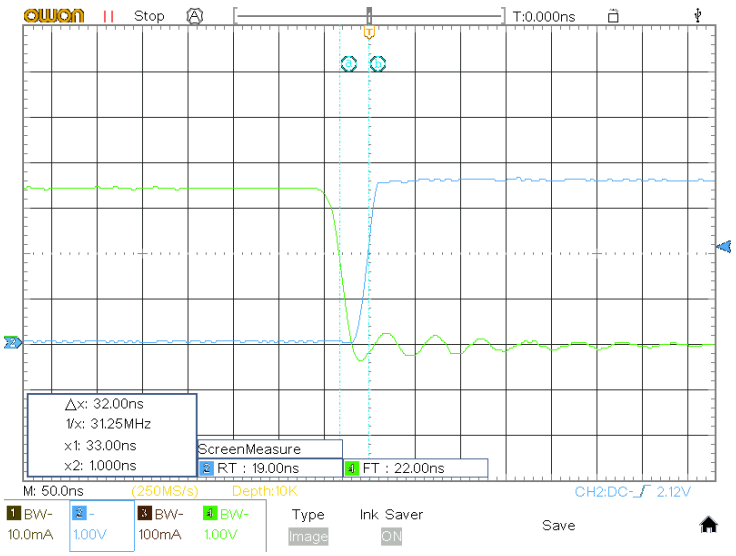

Figure 4.4: Dead time measurement with oscilloscope (value=3).

In simulations, the dead time values of 2 and 3 were compared, and the difference in efficiency is only 0.01%, and therefore the margin is justified.

With 100 kHz PWM and an 80 MHz internal clock, the duty cycle resolution is 0.125 %.

Another timer was configured to trigger every millisecond, and it is used to serve as task interrupts however, each task is called at different periods (always around milliseconds), which is implemented with a counters and comparators.

The ADC can be configured in a variety of different ways, i chose to use the scan mode which reads multiple inputs in sequence. To read the measured values i am using the poling method however in theory if the DMA is used instead, the results could be better since it takes less time to read the ADC values and therefore a better average could be made. In practice i could not get reliable results and therefore the pooling method was adopted. The average is made with all data acquired between each MPPT algorithm call.

4.3 Sensors

The MPPT algorithm and the monitoring system rely on measurements of the electrical and environmental conditions of the system. This section describes the sensors used to measure the voltage and current at the converter's input and output, as well as the temperature and irradiance.

4.3.1 Voltage and current sensing

Voltage sensing is generally straightforward and can be implemented using resistive voltage dividers followed by an ADC. Additionally, it is suggested to use an op-amp as a buffer to reduce the impact of the ADC input capacitor, while charging. This buffer reduces noise and improves the accuracy of the measurement while also allowing the use of higher sampling rates.

Current sensing, however, is more critical and requires careful consideration of the available measurement techniques.

The two main techniques are:

- Hall effect current sensor;
- Shunt resistor sensing with operational amplifiers.

The Hall effect current sensor is a non-invasive method that uses the Hall effect to measure current. It provides galvanic isolation and can do bidirectional measurements. However, it is typically more expensive, has lower bandwidth, and may introduce additional noise due to its magnetic sensing nature.

As for shunt resistor sensing, it is a simple and cost-effective method that uses a low value resistor to convert the current into a voltage drop. This voltage is then amplified using operational amplifiers to match the ADC input range. This method provides high accuracy, low cost, and fast-response time, but it requires careful design to minimize power loss, since it dissipates power proportional to the current.

In the prototype, an IC based on a shunt resistor sensing method was used, which has a built-in amplifier. The chosen sensor was the INA253A2IPWR, which has a PWM rejection block and a small form factor that will be extremely useful for this application. However, this IC is more expensive than a simple shunt resistor and op-amp solution, but it is more accurate and has better noise rejection. In the future, it can be swapped for a lower cost solution if the performance is acceptable.

Both sensors use the internal ADC from the STM32L476, which has a 12 bit ADC (with 3.0 V as the reference). The current sensor outputs an analog voltage that is equal to the current multiplied by 0.2 V/A. This means that the resolution of the measured current is 3.0/(4096 * 0.2) = 3.66 mA. As for the voltage sensor, it outputs an analog voltage that is equal to the voltage multiplied by 10/210. The resolution is 21 * 3.0/(4096) = 15.38 mV, which is more than enough for this application.

Both circuits are represented in Appendix A.

4.3.2 Temperature and irradiance sensors

For security reasons, it is important to monitor the temperature of the PCB. Also, monitoring the temperature and irradiance of the PV panel is important for performance analysis and defect detection. These measurements can also be used to improve the MPPT algorithm, since the MPP is dependent on temperature and irradiance.

For temperature sensing, there is a wide range of options, from simple thermistors to more complex digital temperature sensors. For this application, it was used a simple thermistor because it is cheap and easy to implement. The thermistor is connected to a voltage divider and the voltage is read by the ADC. The thermistor used to measure ambient temperature and transistor temperature was the ERT-J1VS104FA, which has a resistance of $100\ \mathrm{k\Omega}$ at $25\ ^{\circ}\mathrm{C}$ and a B value of 4330 K. The thermistor used to measure the PV panel temperature is the 104JT-025, with a resistance of $100\ \mathrm{k\Omega}$ at $25\ ^{\circ}\mathrm{C}$ and a B value of 4390 K. The temperature can be calculated using the Steinhart-Hart equation [36].

$$\frac{1}{T} = A + B \cdot ln(R) + C \cdot ln(R)^3 \tag{4.27}$$

This equation uses 3 parameters (A, B and C) that are usually not provided in the datasheet but can be calculated using 3 measurements of temperature and resistance. However, the most common approach is to use a simplified version of the equation, which uses only 2 parameters (B and R0). The temperature can be calculated using the following equation:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \cdot ln(\frac{R}{R_0}) \tag{4.28}$$

The thermistor that measures the transistor temperature is also used in analog protection. This protection uses a comparator to disable the transistors after a temperature threshold that can be modified by changing one resistor in the comparator circuit (Appendix A).

As for the irradiance sensor, the board was designed to use a TSL2591, which is a digital light sensor with a wide dynamic range and high sensitivity. It can measure light intensity from 0.000118 lux to 88,000 lux, making it suitable for a wide range of lighting conditions. The sensor communicates with the MCU via the I2C protocol, allowing for easy integration and data acquisition. However, this sensor is not ideal for outdoor use, since it can easily saturate and its spectrum is not ideal for solar irradiation measurement. The best application for this sensor is in laboratory, however, it was not needed, and therefore it was not used

In the future, if the irradiance value needs to be used in the MPPT algorithm, a more precise irradiance sensor is needed. The most precise sensors use a PV cells, which based on its temperature, short circuit current and open circuit voltage can provide a good estimation of the irradiance. The design of this circuit using the same cell model used in the PV panel would accurately measure irradiance and have exactly the same spectral distribution as the PV panel, which can be used for certain algorithms like LUT. This was not implemented in this project due to time constraints, but it is a good option for future work.

4.4 Software

This section explains how the firmware inside the MCU is structured and how it works. Also, the basic idea behind the communication protocols used and how they are applied to this project as well as the Graphical User Interface (GUI) developed for configuration and data analysis.

4.4.1 Firmware overview

The firmware uses a flag-based scheduler with four tasks, shown in Figure 4.5: ADC (1 ms), MPPT (20 ms), CAN (100 ms) and serial (1 s). The periods are user configurable, originally the MPPT period was set to 10 ms (as it is in simulations) and increased to 20 ms to improve the measurement accuracy by increasing the number of samples taken. A 1 s watchdog timer resets the MCU if a task crashes or enters infinite loops.

The fastest of these is the ADC task, which converts the analogue readings into current, voltage and temperature. If any value exceeds the configured limits, a fault is raised and the charging state machine is blocked for 5 s. After this delay the ADC task runs again and checks whether the faults have cleared.

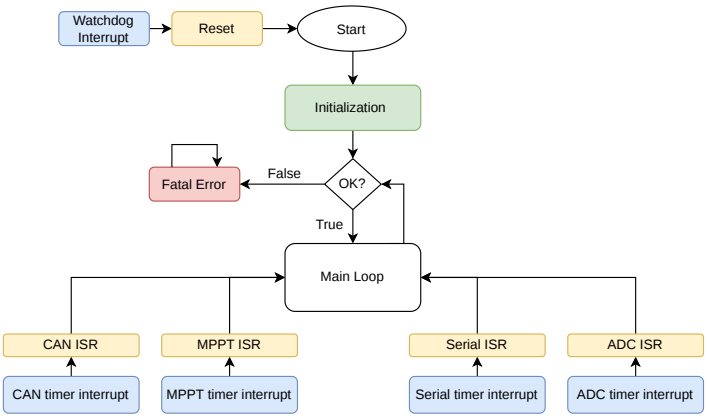

Figure 4.5: Firmware flowchart.

The MPPT task is responsible for computing an average of the last measurements and refreshing the state machine, which changes the operation mode depending on the measurements and outputs a PWM duty cycle value. In Figure 4.6, is represented the flow chart of the MPPT task state machine.

The states are divided into charging and not charging. There are 5 ”not charging” states:

- **Delay**: waits 200 cycles (4 s) to skip the transients and then decides the next state. This state is important to let the output capacitors discharge when load is disconnected, therefore enabling the detection of the ”No load” state;
- **Disable**: disables the DC-DC;

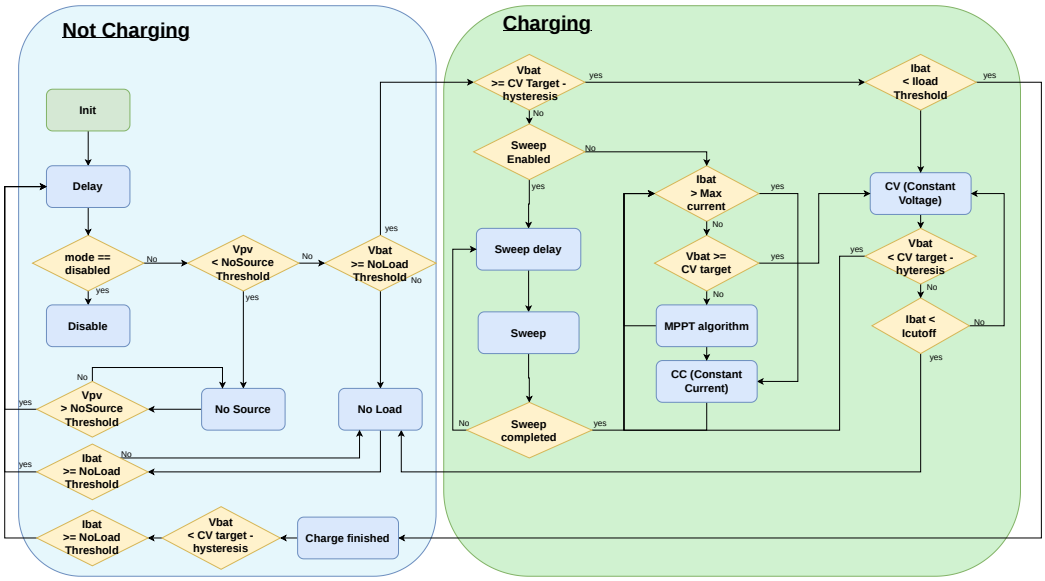

Figure 4.6: Charging state machine.

- **No Source**: the solar panel is disconnected and therefore the DC-DC is disabled;
- **No load**: the battery is disconnected, and therefore the output voltage is set to the maximum battery voltage to prevent the need for a precharge circuit;
- **Charge finished**: similar to ”No load”, runs a proportional controller to follow the maximum battery voltage to prevent the need for a precharge.

As for the charging states, there are also five:

- **Sweep delay**: wait between sweep steps to ignore transients;
- **Sweep**: decrease the duty cycle until the end of the sweep. The sweep is used to find a true MPP and ignore local maximums generated by partial shading;
- **MPPT algorithm**: runs the MPPT algorithm chosen;
- **CC**: if the current exceeds the battery limit, runs a proportional controller to follow the maximum current allowed;
- **CV**: runs a proportional controller to control the PV side by following the battery voltage.

4.4.2 Serial and CAN communication

One of the most important features of this circuit board is the communication with the rest of the boat. This will enable better decision making relating to the energy budget. Also, it will be easier to detect broken or malfunctioning solar panels or human errors such as bad or forgotten connections.

The main communication protocol is CAN. This is the TSB standard protocol and is widely used in the automotive industry due to its high robustness. The protocol was developed by Bosch in 1986 to let microcontrollers and devices talk to each other without a central host computer.

From a hardware point of view, CAN uses a bus instead of wire to wire communication. It also uses twisted wires (CAN High and CAN Low) to reduce electrical noise. From a software point of view, it uses an error checking system (CRC) ensuring reliability.

For TSB electrical system, it was developed a CAN protocol on top of the standard CAN communication. Each message is structured as shown in Figure 4.7 following the CAN 2.0A standard.

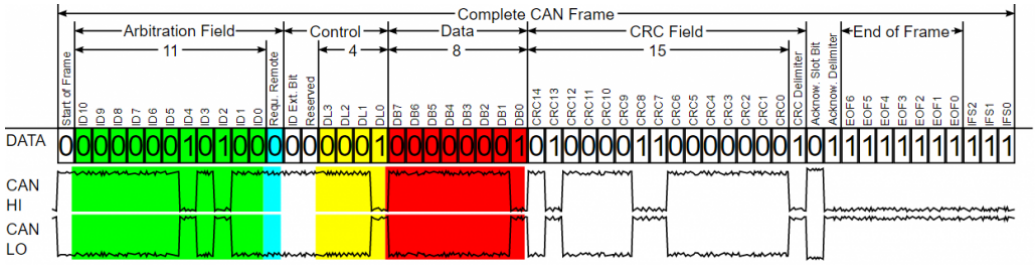

Figure 4.7: CAN protocol structure.

The 11 bits marked in green (Arbitration Field) can be used by each protocol differently. Usually, it is used to send priority bits, device ID and message ID. In the current vessel (SG01), the priority bits were removed and used to increase the number of devices on the bus. With the current structure, represented in Table 4.3, the protocol supports up to 32 devices, each with 63 messages.

Table 4.3: TSB CAN messages arbitration field.

Message ID	Device ID
6 bits	5 bits

In the Control field, the first bit indicates whether to use Extended IDs, and the next bit is reserved. The yellow field is called Data Length Code (DLC) and is where the length of the data is defined. The maximum length is 8 bytes. Then there is the Data field (represented in red).

The rest of the fields are treated by the CAN library which is an adaptation of the open source library called `STM32_CAN` by pazi88.

For the MPPT it was defined an ID structure, where the device ID is maintained intact, and Message ID is divided in two. The 3 most significant bits represent the MPPT number from 0 to 7, and the remaining bits identify the messages. This structure provides enough message IDs for each MPPT as well as not occupying a lot of Device IDs. Table 4.4 shows an example of the corresponding IDs when the Device ID is 0x0D (default ID).

The Device ID can be configured/changed via GUI however, the logic is maintained.

The CAN protocol is excellent for communication between boards, however, it is not the most practical

Table 4.4: CAN IDs for each MPPT number and message.

MPPT Number	MSG:MPPT_bat	MSG:MPPT_pv	MSG:MPPT_aux1	MSG:MPPT_aux2
0	0x00D	0x02D	0x04D	0x06D
1	0x10D	0x12D	0x14D	0x16D
2	0x20D	0x22D	0x24D	0x26D
3	0x30D	0x32D	0x34D	0x36D
4	0x40D	0x42D	0x44D	0x46D
5	0x50D	0x52D	0x54D	0x56D
6	0x60D	0x62D	0x64D	0x66D
7	0x70D	0x72D	0x74D	0x76D

protocol for configuration because the CAN IDs can change between vessels and imply the use of an external hardware to communicate between the board and a computer.

In contrast, serial communication is done by just connecting a cable to the board, and since the protocol is only used for configuration and data analysis, it can be the same forever, and if it needs to be changed, it does not depend on any vessel.

There is no default protocol to send data over serial, however, during the BMS thesis a serial protocol was created by the team and will serve has based for this project. Unfortunately, the firmware library is not compatible with STMs, so it was created from scratch.

The protocol uses the structure defined in Table 4.5. It begins with the Start Byte, which is defined as 0xAC, followed by the device ID, which defines the source of the message, and the message ID, defining what message the device is sending, the data length, the data, the checksum byte, and the end byte (0xAD) marking the end of the message.

Table 4.5: TSB Serial messages format.

Start	Device ID	MSG ID	LEN	Data	Checksum	END
1 Byte	1 Byte	1 Byte	1 Byte	8 Bytes	1 Byte	1 Byte

Similar to the CAN protocol, for each message, a Message ID was defined respecting the same logic. This ID, alongside with the Device ID and length, is initialized at the beginning of the code, the rest of the variables are handled by the library, except the data field which is changed before sending any message.

The Data of each message is structured in the same way as in CAN protocol however it was added 4 messages for configuration. These messages enable the change of current limits, voltage limits, MPPT parameters, CAN settings and CC-CV targets.

4.4.3 EEPROM emulation

To save the configuration received from the serial protocol, a library was used that emulates an EEPROM, which is a type of non-volatile memory that retains its contents after power down. EEPROMs have

been used for many years in a wide range of devices to store settings, calibration data, and backups. In this project, no external EEPROM was added to the board, and the MCU does not include one internally. However, the library can use internal flash memory, which is normally reserved for the flashed code, to store data, thereby emulating EEPROM behavior.

The main disadvantages of this approach are limited storage capacity and reduced write endurance compared with a dedicated EEPROM. The STM32 internal flash can typically withstand on the order of 10,000 erase cycles per page, whereas external EEPROM devices often support roughly one order of magnitude more. Even so, this is not a practical limitation in this application. Configuration data is stored as a small set of 16-bit variables, and each save operation appends new records to flash rather than overwriting the previous value in place. Each page can hold 252 variables. For this project, a full board configuration requires 13 write operations, therefore, one flash page can hold many configuration updates before it becomes full and must be erased. Furthermore, the emulation is configured to use ten flash pages, with wear leveling implemented through page transfer and cleanup. When a page is full, the latest values are copied to another page and the old page is eventually erased. This distributes erase cycles across multiple pages and extends the effective lifetime of the memory.

In conclusion, neither storage capacity nor endurance is expected to be a problem for this project. Under the current configuration, a complete configuration upload would need to be performed hundreds of thousands of times before the theoretical endurance limit of a flash page is reached, which is far beyond the expected use case of occasional configuration updates over serial.

Since the emulated EEPROM shares the same physical flash as the firmware, care must be taken to ensure that program code is never placed in the reserved EEPROM region. This was achieved by modifying the linker script, which defines the memory map used during the build process. The length of the program flash region was reduced so that firmware is linked only below the first address saved for EEPROM emulation.

Two additional mechanisms were implemented to protect the emulated EEPROM from accidental corruption. The first is a magic word that is written to the first address of each page. This is a fixed 16-bit signature that indicates that the page was written by valid firmware rather than left as uninitialized or partially erased flash. The second is a validation process. Each variable is only considered valid if it is within the specified range.

4.5 Graphical User Interface

In parallel with the design and test of the MPPT board, a GUI was designed to make the project more accessible and easy to use. The application was built from scratch, however it was inspired in the application built during BMS thesis. It was built using QT framework, which is a powerful, cross platform

software toolkit used to build high performance applications and GUIs that run on multiple operating systems like Windows, macOS and Linux.

This GUI communicates with the board with Serial or CAN communication. To use CAN communication, the computer should be connected to the bus with a tool called PEAK-CAN.

After connecting the computer to the board, the application identifies the connected board and shows it to the user on the Dashboard page (Figure B.1).On the main page, the user selects which board he is working on. The application is prepared for several PCBs, however, only the BMS and MPPT are correctly working. The rest are placeholders for future improvements.

When the MPPT page is selected (Figure 4.8), a new tab opens showing the most important variables, such as voltage, current, power, temperature, efficiency, irradiance, and duty cycle. Additionally, two visual representations are shown. On the left, a P-V graph shows theoretical curves for different irradiance values, with a blue dot at the current operating point that updates with the received data.

On the right, the CC-CV curve is represented, where the received battery side data is shown over time.

Figure 4.8: MPPT monitor tab.

On top of the page, the user can switch to the configuration tab (Figure B.3). In this tab, the user can read the configuration currently installed on the board, reset this configuration, or upload a new one.

The last page is the debug page (Figure B.4), where the user can visualize both Serial protocol messages and serial prints at the same time.

5

Simulations

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Simulations were used to validate the converter design and the MPPT algorithm before hardware implementation. This chapter presents LTspice simulations of the photovoltaic model and the boost converter, an analytical power-loss estimate in MATLAB, and the evaluation of the tracking algorithm in Simulink.

5.1 DC-DC converter

The converter was simulated in LTspice to confirm the component selection under representative charging conditions. This section presents the photovoltaic cell model and the boost converter simulations used to compare inductor and capacitor options and to estimate ripple and efficiency.

5.1.1 Solar panel model

The simulations started in LTspice to test the converter under different conditions. To achieve that, a PV electrical model of the cells was designed using a five parameter model (1M5P) configuration. There are more electrical models that can simulate a solar panel, some with higher accuracy but greater complexity. This model was ideal for the simulation because it is both reliable and simple.

The 1M5P model represents the solar cell as a single-diode equivalent circuit with a current source, a diode, a series resistance R_s and a shunt resistance R_{sh}, as shown in Figure 5.1. The current source models the photogenerated current I_{ph}, the diode the nonlinear I-V characteristic of the p-n junction, R_s the ohmic losses in the material and contacts, and R_{sh} the leakage current.

The output current is given by Equation 5.1, where I_0 is the diode saturation current, n the ideality factor and V_t the thermal voltage [37]. Together with I_{ph}, R_s and R_{sh}, these five parameters give the model its name.

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \tag{5.1}$$

These parameters are not directly provided by manufacturers, but can be extracted from the standard datasheet values, namely the open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), voltage and current at the MPP (V_{mp}, I_{mp}), and the respective temperature coefficients, through a set of equations that ensure the model matches these reference points under STC. There are several ways to do it, however, it is outside the range of this work. To simplify, it was used an online tool for the calculations, which we will not explain in detail here.

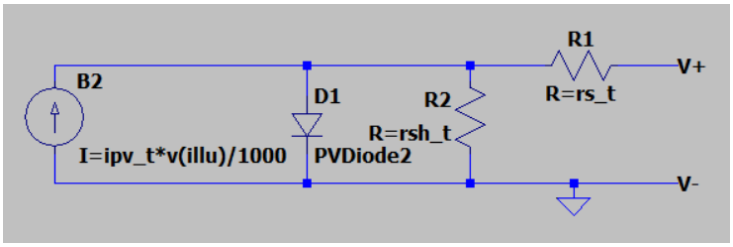


Figure 5.1: Solar panel model diagram (1M5P configuration).

5.1.2 Boost converter

To validate the theoretical calculations, simulations in LTspice were carried out using the boost converter in steady state mode and a fixed duty cycle.

Considering a solar panel operating at its MPP with 35 cells, five points of the CC-CV charging curve were chosen to test the performance of the converter. Figure 5.2 shows these five points.

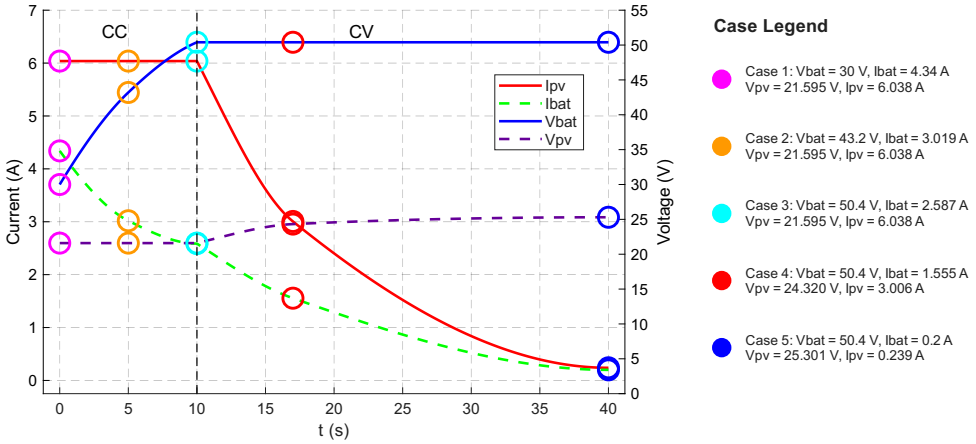


Figure 5.2: Draft of CC-CV charge characteristics and analysis point selection.

It is important to note that, in this CC-CV charging profile, the constant current is determined by the solar panel rather than the battery, as is typically assumed. This is only possible because the maximum charging current of the battery is lower than the maximum current that can be supplied by the solar panel. Therefore, during the CC phase, the converter operates at the MPP of the solar panel.

The timescale of the charging process depends on the battery capacity, with larger capacities requiring longer charging times under equivalent conditions.

With the chosen five points, some simulations were carried out using capacitors and inductors that would fit the requirements calculated earlier. The objective of these simulations is to optimize the performance of the converter, leading to better choices.

The model was built in LTspice using manufacturer component models, except for the solar panel, which was implemented as described above. The battery was represented by a voltage source and a

series resistor, and the transistor dead time was set to the prototype value, using the measured rise and fall times. Figure 5.3 shows the simulation circuit and his parameters.

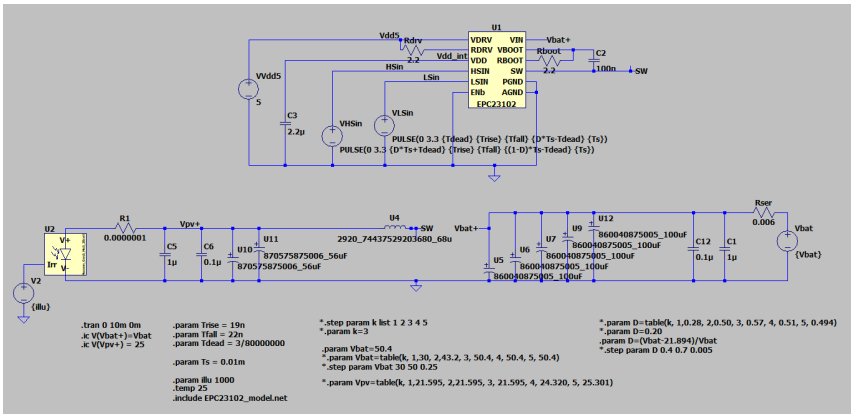


Figure 5.3: Simulation circuit schematic.

The first simulations were carried out to choose the input and output capacitors. The idea is to simulate the circuit using a known inductor ($68\ \mu\mathrm{H}$) and change the capacitors for different configurations. All configurations are overdimensioned to give low ripples (current and voltage). So the main factor is the efficiency of the converter.

Table 5.1 shows the results. The efficiency of all combinations is very high, however, the highest is the one with two $56\ \mu\mathrm{F}$ as input capacitors and three $220\ \mu\mathrm{F}$ capacitors as output capacitors. Unfortunately, the $220\ \mu\mathrm{F}$ capacitors were not available at the time, so the second best option was chosen: five $100\ \mu\mathrm{F}$ capacitors.

It is worth noting that this is a simulation and therefore the efficiency is higher than what will be measured in laboratory results. Additionally, the difference between each combination is very small. To make the product cheaper, the third option would be as good as the one chosen.

Table 5.1: Loss (%) for different capacitor configurations (Cin, Cout).

Case #	$2 \times 56\ \mu\mathrm{F}$, $3 \times 220\ \mu\mathrm{F}$	$100\ \mu\mathrm{F}$, $3 \times 220\ \mu\mathrm{F}$	$100\ \mu\mathrm{F}$, $5 \times 100\ \mu\mathrm{F}$	$2 \times 56\ \mu\mathrm{F}$, $5 \times 100\ \mu\mathrm{F}$
1	0.5729	0.5796	0.5856	0.5788
2	0.6762	0.6985	0.7069	0.6844
3	0.7359	0.7655	0.7743	0.7449
4	0.6654	0.7091	0.7161	0.6735
5	3.4177	3.7148	3.7554	3.4671

As in the previous simulation, the inductor was chosen by keeping all components with the same values (the capacitors used were $C_{\mathrm{in}} = 2 \times 56\ \mu\mathrm{F}$ and $C_{\mathrm{out}} = 5 \times 100\ \mu\mathrm{F}$) and changing the inductor. Similar to before, all inductors simulated are within the minimum required.

Table 5.2 shows the losses and the inductor current ripple for each charging case. As before, the efficiency difference is very low but less linear than in the previous simulation, likely due to the inductor

models chosen. The chosen inductor strongly influences the input ripple, which can generate heat and electrical interference. The $22\ \mu\mathrm{H}$ inductor showed a higher ripple current, reaching values close to the input current. The inductor chosen was $68\ \mu\mathrm{H}$, since it has overall good efficiency and a reasonable inductor current ripple.

Table 5.2: Loss (%) and Δi_L ($I_{L,\max} - I_{L,\min}$) for different inductance values.

Case #	Loss (%)					Δi_L (A)				
	$22\ \mu\mathrm{H}$	$33\ \mu\mathrm{H}$	$47\ \mu\mathrm{H}$	$68\ \mu\mathrm{H}$	$100\ \mu\mathrm{H}$	$22\ \mu\mathrm{H}$	$33\ \mu\mathrm{H}$	$47\ \mu\mathrm{H}$	$68\ \mu\mathrm{H}$	$100\ \mu\mathrm{H}$
1	0.4206	0.4439	0.5453	0.5788	1.2659	2.6939	1.8906	1.3256	0.9781	0.6251
2	0.6645	0.6793	0.6627	0.6844	1.3664	4.7934	3.3416	2.3354	1.7066	1.0707
3	0.7883	0.8076	0.7290	0.7449	1.4193	5.4733	3.8253	2.6813	1.9644	1.2839
4	1.0556	1.0583	0.6867	0.6735	0.9624	5.5596	3.8565	2.7075	2.0139	1.2925
5	6.7045	6.6230	3.6556	3.4671	3.8177	5.5520	3.8510	2.6903	1.9636	1.2344

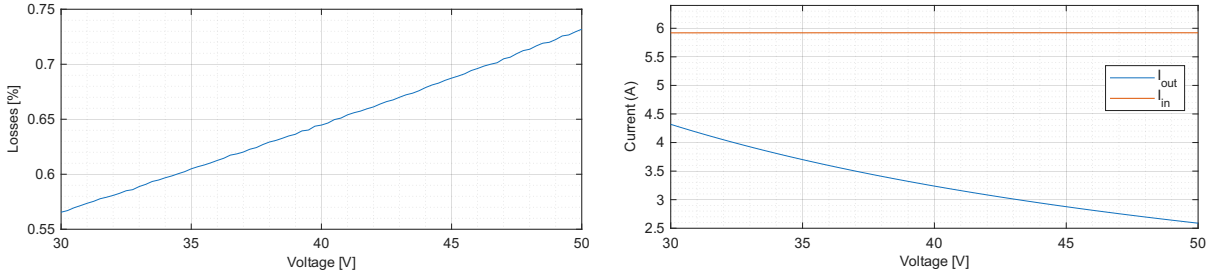
Now that the converter components are all chosen, the following measurements were taken in LTspice (Table 5.3).

Table 5.3: Ripple and power loss summary for each operating point.

Case #	ΔV_{bat} (V)	ΔV_{pv} (V)	ΔI_L (A)	Loss (%)
1	0.0406	0.0145	0.9781	0.578
2	0.0491	0.0212	1.7066	0.684
3	0.0529	0.0244	1.9644	0.745
4	0.0406	0.0220	2.0139	0.674
5	0.0302	0.0215	1.9636	3.464

Both voltage ripples are lower than the goal defined in Table 4.2. As for the current ripple, it is still reasonable. The losses/efficiency are also higher than what was aimed for, reaching efficiency values up to 99.4%.

A voltage sweep was performed, and as expected, the efficiency drops with the increase of output voltage (Figure 5.4). Also, as mentioned earlier, the output current drops because, in CC mode, the solar panel current is constant, so the output current decreases as battery voltage increases due to power conservation.



(a) Losses (%) over battery voltage.

(b) Output and input current over battery voltage.

Figure 5.4: Battery voltage sweep simulation results (solar panel in MPP).

Note: two Schottky diodes added in prototype version 1.1 were not included in these simulations.

They increase the simulated efficiency by only $+0.03\%$, which is negligible given the short dead time (0.375% of the PWM period).

5.2 Power loss estimation

Although LTspice provides an accurate estimate of converter efficiency, it does not directly identify each component's contribution to total power dissipation. Therefore, an analytical power loss estimation was developed in MATLAB to estimate the individual losses of each component of the designed synchronous boost converter. This estimation uses the electrical parameters of the selected components from their datasheets, along with the converter operating conditions.

The total prototype losses are obtained by summing the losses associated with the GaN transistors, inductor, capacitors and auxiliary electronics:

$$P_{loss} = P_{tran} + P_L + P_C + P_{aux}. \tag{5.2}$$

The prototype efficiency is then calculated as:

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100\%. \tag{5.3}$$

The transistor losses can be split in three. The conduction, switching and deadtime losses:

$$P_{tran} = P_{cond_{HS}} + P_{cond_{LS}} + P_{sw_{HS}} + P_{sw_{LS}} + P_{dead} \tag{5.4}$$

The conduction losses of each GaN transistor are given by:

$$P_{cond} = I_{RMS}^2 R_{DS(on)}. \tag{5.5}$$

These losses are primarily caused by the on state resistance of the transistors, resulting in power dissipation whenever current flows through the transistors.

The switching losses are estimated using Equation 5.6, which takes into account the losses during the period when the transistor is neither fully on nor fully off. They also include the losses associated with charging and discharging the output capacitance of each transistor.

$$P_{sw} = \frac{1}{2} V_{out} I_{L_{AVG}} (t_r + t_f) f_{sw} + \frac{1}{2} C_{oss} V_{out}^2 f_{sw} \tag{5.6}$$

The deadtime losses are calculated using Equation 5.7. These losses originate from the conduction of the reverse conduction path during the dead time interval required to prevent shoot through. This

reverse conduction path is either from the third-quadrant conduction of the GaN device or from the parallel Schottky diode added later in version 1.1 of the prototype.

$$P_{dead} = 2 \cdot V_f \cdot I_{L_{AVG}} \cdot t_{dead} \cdot f_{sw}. \tag{5.7}$$

The inductor and DC capacitor losses are estimated using the RMS current passing through them and their ESR. The inductor manufacturer does not provide the required magnetic material parameters to estimate the AC losses. However, an employee suggested using their own tool called REDEXPERT, which can simulate both DC and AC losses. Therefore, the value AC losses were copied from their tool

$$P_L = I^2_{L_{RMS}} ESR + P_{L_{AC}}. \tag{5.8}$$

$$P_C = I^2_{C_{RMS}} ESR, \tag{5.9}$$

The gate driver losses are estimated using Equation 5.10 which takes into account the energy needed to drive the transistors. Although the EPC23102 has an internal gate driver, this is still relevant.

$$P_{gate} = (Qg_{LS} + Qg_{HS}) * Vdrv * fsw. \tag{5.10}$$

Last but not least, the rest of the electronics (MCU, DC-DC, sensors, etc.) were considered too. The sum of the maximum currents of all components was calculated. Then the power losses was estimated based on the power and efficiency of each DC-DC converter, Equation 5.11.

$$P_{aux} = (I_{3v3} * 3.3) * eff_{dcdc} + (I_5 * 5) * eff_{dcdc} + I_{in} * R_{shunt} + I_{out} * R_{shunt} + P_{gate} \tag{5.11}$$

Figure 5.5 shows the resulting power-loss distribution. The calculated total power loss is approximately 1.98 W, corresponding to 98.50% efficiency. This is about 1% lower than the LTspice result, and a further reduction is expected in the laboratory because PCB parasitics were not included.

The inductor DC losses are the largest term, followed by the auxiliary electronics (mainly the auxiliary DC–DC conversion), the GaN transistor losses (24.4%) and the capacitors (23.5%). Overall, nearly half of the losses come from passive components and about one quarter from the semiconductors. Despite this, both the simulation and the analytical model indicate a highly efficient converter.

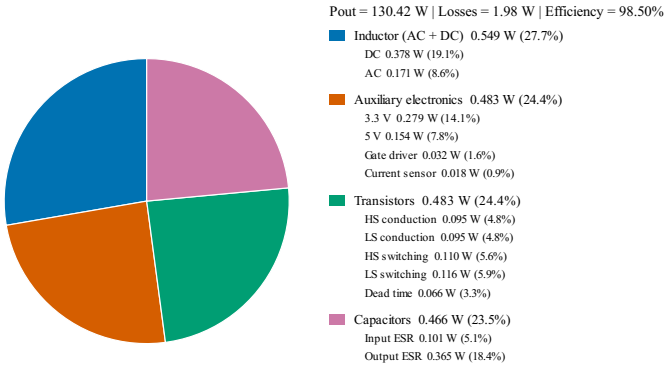


Figure 5.5: Prototype analytical power loss distribution at Case 2.

5.3 MPPT algorithm

The algorithm parameters (switching and sampling frequencies, step size and power dead band) must be tuned for speed without compromising stability. A MATLAB Simulink model of the boost converter, solar panel and battery was therefore used to evaluate the tracker performance (Figure 5.6). To simplify the model, a diode replaces the synchronous high-side transistor, which does not affect the algorithm comparison.

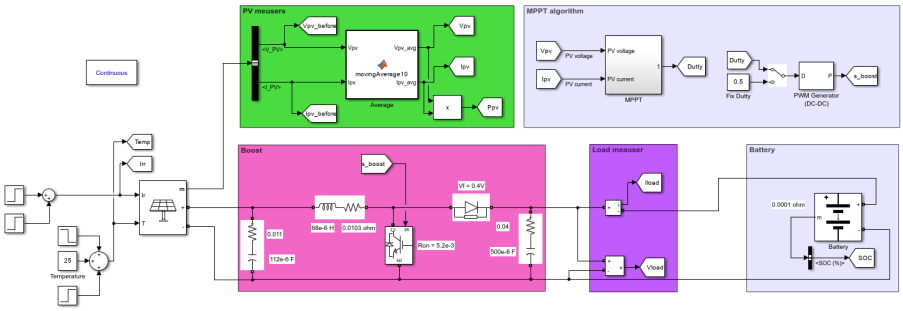


Figure 5.6: MATLAB Simulink MPPT simulation model.

The solar panel (35 cells) and battery were configured to match a 35 cells solar panel and SG01 battery. As in the real system the PV side measurements include noise, which is filtered by averaging the last 10 samples at 1 kHz before being sent to the algorithm, which runs at 100 Hz and implements both the sweep and P&O.

To understand the impact of each environment variation on the MPP point, a sweep was performed for different irradiance values and temperatures (Figures 5.7 and 5.8). For irradiance variations, the maximum duty cycle difference is 3%. In contrast, the maximum duty cycle difference for temperature variations is 9%. This means the algorithm will always be slower to track temperature variations than to irradiance variations. However, it is more common to have a fast changing irradiance than temperature. The solar panels will heat slowly, and when covered in water, they cool by around 10 °C to 20 °C (values

measured during tests).

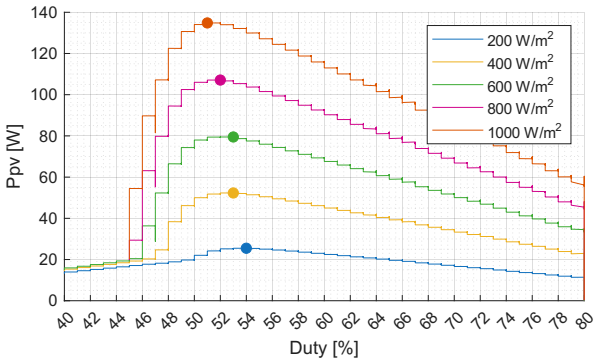


Figure 5.7: Duty cycle sweep with different values of irradiance.

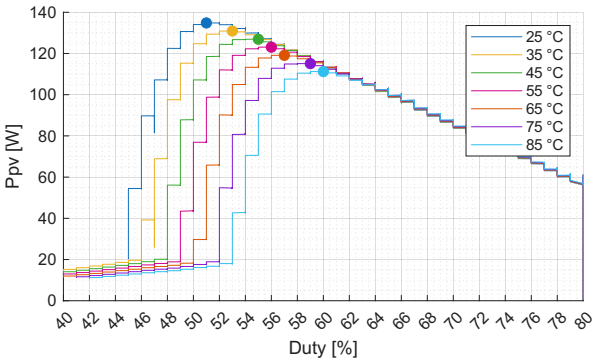
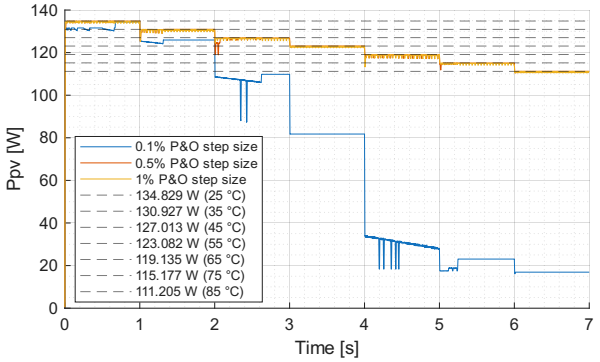


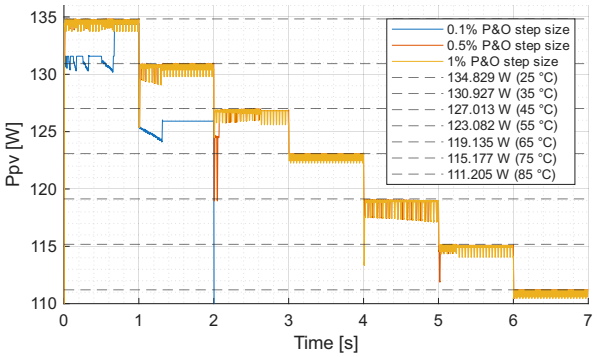
Figure 5.8: Duty cycle sweep with different values of temperature.

Assuming the worst-case variation of 9%, a combination of the step size and operating frequency must be selected to achieve the desired tracking time of 0.5 s. For example, with an operating frequency of 100 Hz, the algorithm has 50 steps to reach the MPP within 0.5 s. Therefore, if the algorithm uses a step size of $9\%/50 = 0.18\%$, it could ideally reach the target tracking time in every case. However, such a small step size may make the algorithm more susceptible to measurement noise, potentially causing the perturbations to be indistinguishable from noise. Therefore, the step size and operating frequency must be tuned according to the characteristics of the system in which the algorithm is implemented.

With the frequency fixed at 100 Hz, three step sizes were compared under successive temperature steps (Figure 5.9). The smallest step never reaches the MPP, most likely because of noise. The other two perform similarly, but the larger step oscillates more around the MPP.



(a) Complete response.



(b) Zoomed view.

Figure 5.9: Power response for different P&O step sizes during a temperature variation of $10\ ^\circ\mathrm{C/s}$.

The analysis was then extended to more severe operating-point variations. For a temperature step from $25\ ^\circ\mathrm{C}$ to $85\ ^\circ\mathrm{C}$ (Figure 5.10), the 0.5% step, which had performed well previously, required about 1 s to reach within 5% of the MPP. This meets the 1 s limit but not the typical 0.5 s target. The 1% and

2% steps were faster but oscillated more around the MPP. A step between 0.5% and 1% is therefore the best compromise. Occasional stalls are caused by measurement noise and the P&O deadband and can be reduced by further tuning.

A rapid irradiance variation was also simulated (Figure 5.11). The MPP is reached almost instantaneously for all step sizes, as expected, because the duty-cycle change is about three times smaller than in the temperature case.

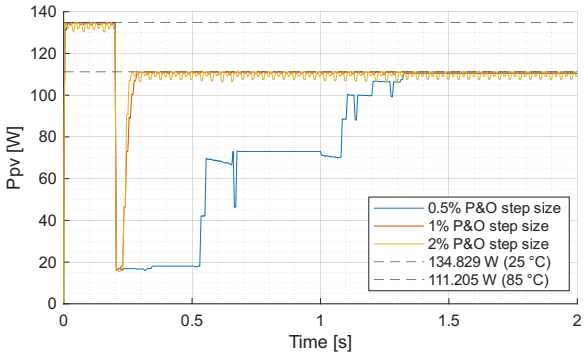


Figure 5.10: Power response for different P&O step sizes during a temperature variation from $25\ ^{\circ}\mathrm{C}$ to $85\ ^{\circ}\mathrm{C}$.

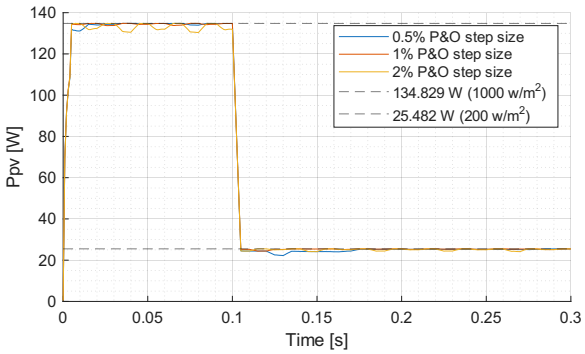


Figure 5.11: Power response for different P&O step sizes during an irradiance variation from $1000\ \mathrm{W/m^2}$ to $200\ \mathrm{W/m^2}$.

6

Experimental validation

Contents

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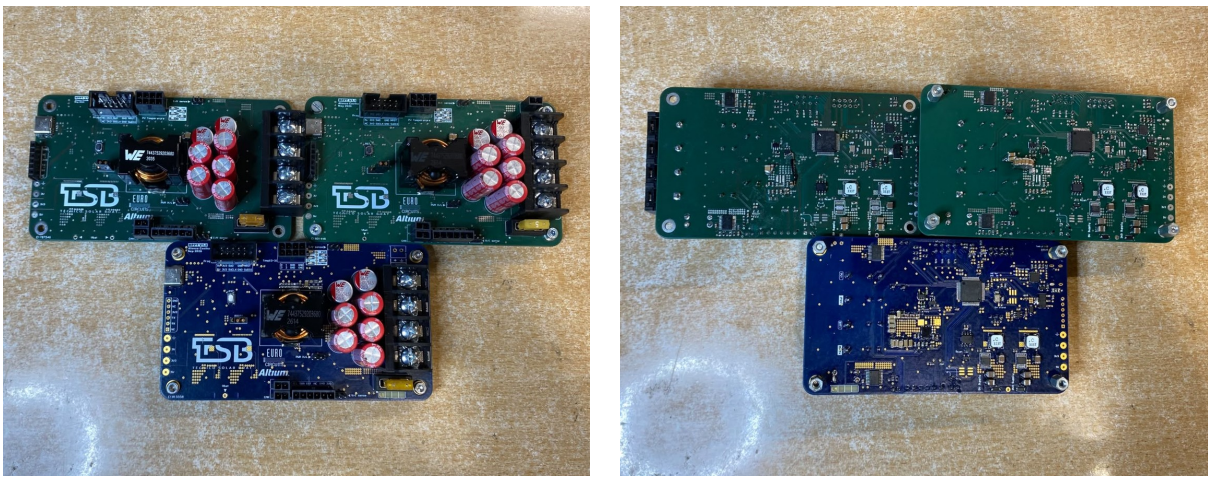
Following the design and simulation stages, a prototype was manufactured and assembled. However, practical implementations may differ from simulation results, as simulations generally rely on idealized component models and do not fully account for parasitic effects associated with the components and the PCB. Therefore, laboratory testing was conducted to experimentally validate the design and verify that the prototype operates as expected.

6.1 Design and assembly

All boards developed for this project, shown in Figure 6.1, were designed in Altium Designer, manufactured by Eurocircuits, and assembled by hand to minimize production costs. The total cost of the electronic components is approximately 90 €, although this value may vary with purchase quantities. The PCB manufacturing cost ranges from approximately 3 € to 35 €, depending on the manufacturer.

Consequently, the total cost of a complete prototype is approximately 125 €, which is lower than the cost of the product currently in use (240 €). Furthermore, the effective cost can be further reduced through component and manufacturing sponsorships.



(a) Front side. **(b)** Back side.

Figure 6.1: All prototype versions (V1, V1.1 and V1.2).

The **first prototype (V1)** iteration had several issues identified during the initial testing phase. For instance, an incorrect 3D model caused the main connector to be positioned incorrectly (Figure 6.1). Additionally, the flash connector did not provide 5 V, which made testing more difficult. The UART signals were swapped, and therefore the communication was only possible using an external USB to UART adapter.

Although none of these issues were critical for the fundamental operation of the circuit, they made the testing phase more time-consuming and complex. Therefore, a second iteration was designed and

manufactured with all issues fixed.

Besides the errors found in the first prototype, in the **second iteration (V1.1)**, two Schottky diodes were placed in parallel with the boost transistors, which, according to the transistor datasheet, should improve the boost converter's efficiency. After some simulations, it was found that this statement does not apply in this case due to the short dead time. Furthermore, during testing of the first iteration, an increase in the inductor temperature was observed. To address the possibility of requiring additional cooling, a connector was added to place a fan if needed. These will also be analyzed in this section.

Near the end, a **final version (V1.2)** was made, which includes the extra filter capacitors to filter noise, which will be discussed later. Additionally, the sensors filters were placed near the sensors. Since switching devices generate electromagnetic noise, the filters should be close to the MCU pins.

Due to time constraints, the majority of the results presented in the following sections were obtained using version V1.1. Whenever results obtained with the latest version are presented, this will be explicitly stated, otherwise, the results refer to version V1.1.

6.2 Voltage and current sensor

Both current and voltage sensors were tested in the laboratory to validate measurement performance and accuracy.

The voltage sensor test was performed by applying a known voltage to the input and measuring MCU readings.

The test covered the full voltage range required for the application, from 0 V to 51 V. The results demonstrate that the voltage sensors provide accurate measurements, with average absolute errors of 60 mV for the battery voltage measurement and 112 mV for the solar panel voltage measurement. The error is different for each sensor because it depends on the actual resistance of the resistors used in the voltage dividers and the PCB layout. For each board, the error will be different, but it is expected to be in the same order of magnitude since the tolerance of the resistors is equal.

These could be improved even further by using resistors with lower tolerance and a lower temperature coefficient, or by increasing the ADC resolution. Another option that does not increase system cost is to subtract the linear error from the results. The magnitude of the error is small, and therefore it does not affect the MPPT operation, however, for the purpose of this project, the linearization was implemented in firmware to improve the accuracy of the measurements made during the testing phase.

In Figure 6.2 the error of the voltage readings is represented. Since the error is linear, the trend line is also shown and used to correct the results and improve measurement accuracy. The figure also shows the results after subtracting the linear error, and the error is significantly reduced. The average error was reduced to 28.325 mV and 29.825 mV for the battery and solar panel, respectively.

The same test was performed for the current sensors, the average error was 99.93 mA and 8.5 mA for the solar panel and battery, respectively, which is acceptable for this application. Figure 6.3 shows the current sensor error and indicates that it is linear on the solar panel side and almost null on the battery side. This can be explained by the fact that at low currents, the error is smaller, and probably the shunt resistor value on the solar panel side is more accurate than the shunt resistor on the battery side. However, it is still small and does not affect the MPPT operation.

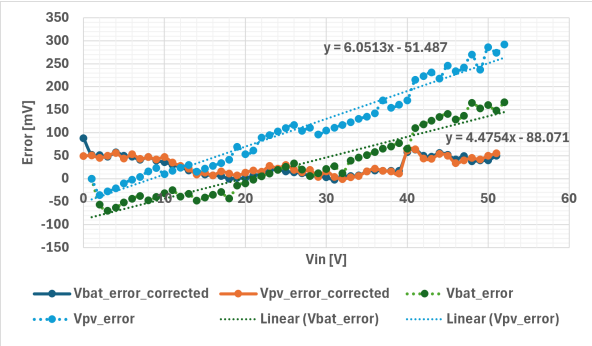


Figure 6.2: Voltage sensor error over voltage.

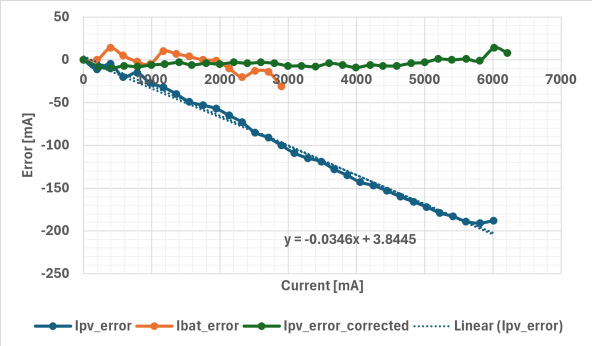
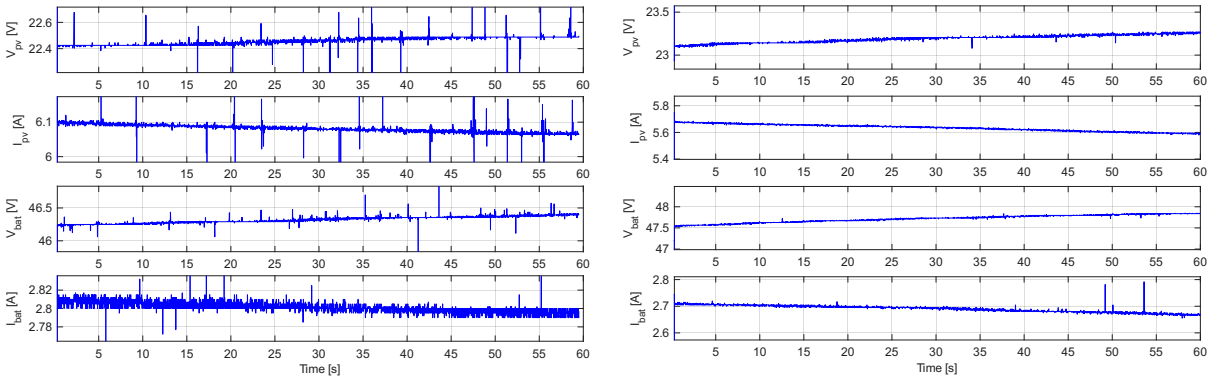


Figure 6.3: Current sensor error over current.

As in the case of the voltage sensor, the subtraction of the linear error would greatly increase the accuracy of the measurements. However, it takes a lot of time to do the measurements and calculate the linearization for each PCB which is not realistic for a student prototype project like TSB. However, to increase the accuracy of the laboratory results, error correction was implemented for the PV side current sensor. The figure also shows the results after subtracting the linear error, and the error is significantly reduced. The average error was reduced to 5.28 mA.

Another factor affecting the accuracy of the measurements is electrical noise. In prototype version 1.1, the filtering components were placed close to the sensors. In practice, this configuration primarily attenuated noise generated locally by the sensors themselves. However, switching converters are also significant sources of electrical noise, which should be attenuated as close as possible to the ADC/MCU. Therefore, the filter placement was modified in version 1.2. The resulting difference in measurement quality is shown in Figure 6.4, where the measurements were collected at a fixed duty cycle.

The difference between the two versions is clearly noticeable. In version 1.1, all measurements exhibited significant noise. In contrast, the modified filter configuration in version 1.2 substantially reduced the observed noise to an acceptable level. The battery current measurement represented the worst-case condition, with two distinct spikes occurring during the one minute measurement interval.



(a) Version 1.1.

(b) Version 1.2.

Figure 6.4: Comparison of sensor readings over a one-minute interval at a fixed duty cycle.

6.3 Boost

The Boost test is essential to identify the efficiency and the correct operation of the synchronous boost converter without the influence of the control system. Additionally, it is essential to confirm the correct dimensioning of the power converter, enabling a comparison with simulations and theoretical calculations.

The test was performed using a voltage source on the input side and two power resistors on the load side. Both duty cycle and load resistance were varied to obtain the results in all different points of the charging curve.

All the measurements were performed with a high precision multimeter. The voltage readings were performed as close as possible to the PCB, whereas the current readings they have to be performed in line.

It is important to note that this board uses the battery side to power itself, however, in this test the battery side is not connected to a battery but since the PV side is connected to a voltage source, this energy flows to the battery side through the boost converter diodes/body diodes and therefore powers the control system.

6.3.1 Switching noise

During the initial experimental tests, significant electrical noise was observed in all measured signals. The resulting current fluctuations repeatedly exceeded the configured Over Current (OC) protection thresholds, causing the converter to enter an undesirable protection cycle. Firmware modifications were therefore implemented, including increasing the ADC sampling time and the number of samples used for each measurement, with the final value obtained by averaging the samples. The fault-handling logic was also modified to introduce a $5\ \mathrm{s}$ delay before verifying whether the fault condition had been cleared.

Although these changes improved measurement stability, they did not eliminate the excessive noise.

The input current was observed to oscillate significantly during OC events, with values ranging approximately from $4\ \mathrm{A}$ to $12\ \mathrm{A}$. Initially, the laboratory power supply was considered a possible cause, as it was operating close to its current limit. However, comparison with the reference evaluation board revealed a significant difference in the output decoupling network. The initial prototype used only a single $1\ \mu\mathrm{F}$ ceramic capacitor, whereas the reference board employed ten $1\ \mu\mathrm{F}$ capacitors in parallel.

The converter was therefore characterized before and after increasing the output decoupling capacitance. Figure 6.5 shows the initial measurements. Nine additional $1\ \mu\mathrm{F}$ ceramic capacitors were then connected in parallel with the existing capacitor, reproducing the total capacitance of the reference board. This modification resulted in a substantial reduction in the observed switching noise, as shown in Figure 6.6.

Additional high-frequency decoupling was subsequently implemented by adding six $100\ \mathrm{nF}$ ceramic capacitors, replacing the additional $220\ \mathrm{nF}$ capacitors used in the reference design due to component availability. The resulting waveforms are shown in Figure 6.7. A further reduction in current noise was observed, although the measured input voltage noise increased slightly. This difference is likely related to measurement inaccuracy.

The influence of the Schottky diodes placed in parallel with the boost transistors was also evaluated. The resulting waveforms are shown in Figure 6.8. The addition of the diodes did not significantly affect the measured noise, although a slight increase in input voltage noise was observed. The decision was to not include them since neither the noise nor the efficiency improved significantly, and they increased the cost of the prototype.

The approximate noise amplitudes measured for the different hardware configurations are summarized in Table 6.1. The addition of the nine $1\ \mu\mathrm{F}$ capacitors produced the most significant improvement, particularly in the input current and voltage measurements. The subsequent addition of the $100\ \mathrm{nF}$ capacitors provided a further reduction in current noise.

Overall, the experimental results indicate that insufficient output decoupling was the primary contributor to the excessive switching noise observed in the initial prototype. Increasing the local output capacitance substantially reduced the noise and improved the stability of the converter. The noise amplitudes presented in Table 6.1 should, however, be considered approximate, as a more controlled measurement procedure with an appropriate oscilloscope trigger would be required to accurately quantify the peak noise amplitudes.

Since the precision of the previous measurement is insufficient to enable a meaningful comparison with other devices, an additional test was performed to characterize the conducted noise spectrum of the prototype.

The test procedure was based on the CISPR 25:2021 standard, which specifies methods for measur-

Table 6.1: Approximate noise amplitudes measured for the three hardware configurations.

Quantity	Original	+9 $\times$ 1 μF	+9 $\times$ 1 μF +6 $\times$ 100 nF	All + schottky diode
I_{in}	1.8 A	640 mA	292 mA	140 mA
I_{out}	0.8 A	300 mA	220 mA	272 mA
V_{in}	2.0 V	350 mV	380 mV	490 mV
V_{out}	8.0 V	2.3 V	2.2 V	1.440 V

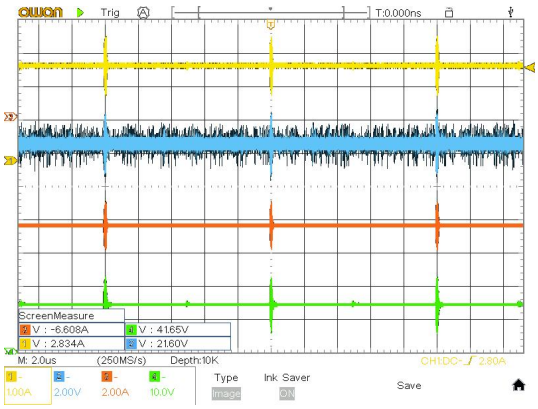


Figure 6.5: Measured waveforms before adding the additional ceramic capacitors.

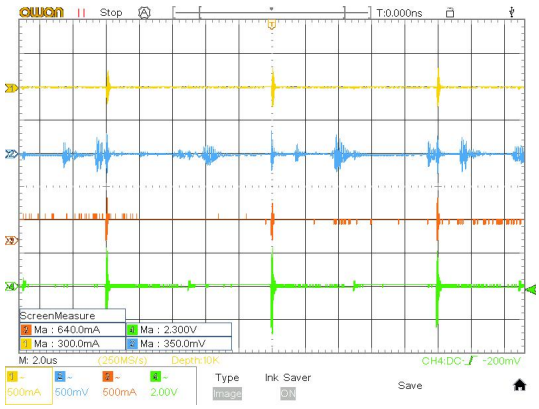


Figure 6.6: Measured waveforms after adding nine additional $1\ \mu$F ceramic capacitors.

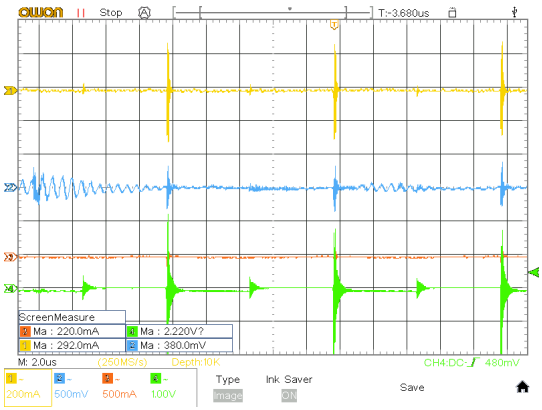


Figure 6.7: Measured waveforms after adding six additional 100 nF ceramic capacitors.

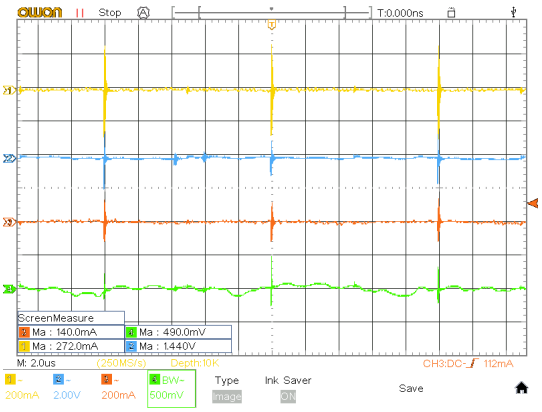


Figure 6.8: Measured waveforms after adding two Schottky diodes (channels 4 and 2 are swapped compared to the other figures).

ing radio disturbances generated by electrical and electronic components intended for use in vehicles. This standard was selected because one of the main concerns associated with electromagnetic noise in the prototype is the potential interference with communication systems used in the vessel, such as CAN and VHF communication systems. CISPR 25 defines measurement procedures and limits for conducted disturbances on power supply lines. The limits are divided into five EMC classes, with Class 5 representing the most stringent limits. The limits considered in this analysis are presented in Table 6.2.

The prescribed test procedure was not fully reproduced due to equipment limitations. In particular,

Table 6.2: CISPR 25 EMI noise limitations.

EMC Class	Magnitude (dBμV)									
	0.15–0.3 MHz		0.53–2 MHz		5.9–6.2 MHz		30–54 MHz		68–108 MHz	
	Peak	Average	Peak	Average	Peak	Average	Peak	Average	Peak	Average
1	113	100	95	82	77	64	77	64	61	48
2	103	90	87	74	71	58	71	58	55	42
3	93	80	79	66	65	52	65	52	49	36
4	83	70	71	58	59	46	59	46	43	30
5	73	60	63	50	53	40	53	40	37	24

a Line Impedance Stabilization Network (LISN), which is normally used to provide a defined impedance and isolate the measurement from disturbances originating from the external power supply, was not available. Furthermore, the measurements were not performed inside a shielded enclosure, such as a Faraday cage, and were therefore potentially affected by external electromagnetic disturbances. Consequently, the results should not be interpreted as a formal CISPR 25 compliance test. Nevertheless, the measurements provide a useful indicative characterization of the conducted noise generated by the prototype. Since the absence of a LISN and shielding can introduce additional external disturbances, measured values below the specified limits provide an indication that the prototype's conducted emissions are unlikely to exceed those limits under the tested conditions.

The test setup consisted of a DC power supply set to 22 V and a resistive load of 10 Ω. The prototype operated with a duty cycle of 52.5%. The resulting noise spectrum is presented in Figure 6.9. The measurement was subsequently repeated with no external load while maintaining a fixed output voltage of 50.4 V. The corresponding spectrum is presented in Figure 6.10.

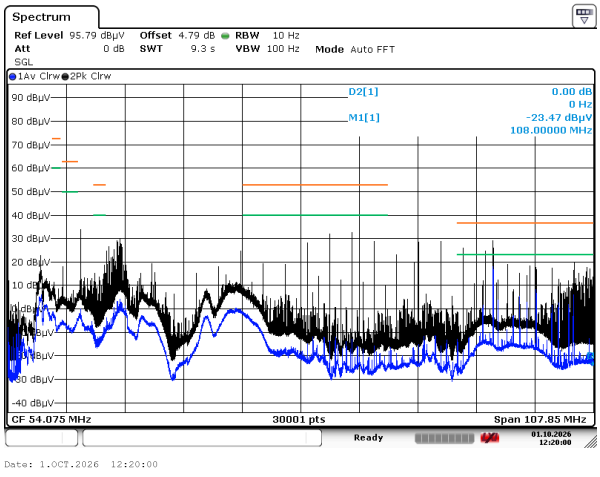


Figure 6.9: Noise spectrum with a 10 Ω load.

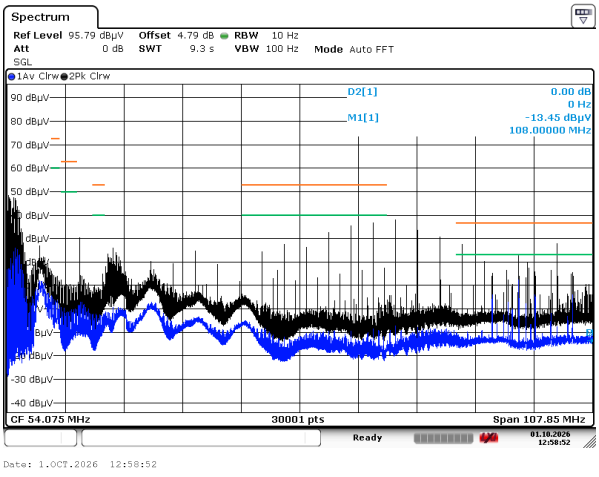


Figure 6.10: Noise spectrum with no external load.

Both measurements show that the measured noise remains below the Class 5 limits over the frequency ranges considered. Therefore, despite the limitations of the experimental setup, the results

indicate a low level of conducted emissions from the prototype when compared with the most stringent limits considered in CISPR 25.

The average noise level is higher in the no-load condition. This behavior can be associated with the different operating conditions of the converter at light load. In particular, reduced load current can lead to changes in the switching current waveform and operating regime of the converter, potentially increasing the relative contribution of high frequency components to the measured spectrum.

6.3.2 Efficiency

The efficiency of the MPPT charger is one of the most important parameters to measure because it directly affects the power effectively extracted from the solar panel and delivered to the battery. As in simulations, the efficiency was measured for each point of the charging curve. All the measurements were taken with the multimeter and the results are presented in Table 6.3.

Table 6.3: Efficiency measurements for the different test cases.

	Multimeter			- OC power	- OC-Transistors power
Case	Power in	Power out	Efficiency	Efficiency	Efficiency
Case 1	127.444	123.413	96.84%	97.23%	97.64%
Case 2	129.108	124.685	96.57%	96.96%	97.36%
Case 3	127.523	123.232	96.64%	97.02%	97.43%
Case 4	73.1277	70.818	96.84%	97.52%	98.24%
OC	0.5103	0	–	–	–
OC+Transistors	1.04279	0	–	–	–

The measured efficiency exceeds 96% across all test cases, which is higher than the goal (90%). This efficiency accounts for all converter losses, including the measurement and control systems. Compared with the power loss estimate in section 5.2, the efficiency is about 2% lower than expected. This difference can be explained by the fact that the components used can have slightly different characteristics. Also, the estimation did not account for losses caused by the parasitic elements of the PCB.

The PCB was also tested in open circuit with and without driving the transistors. The results show the losses of the circuit without the boost converter. Around 0.5 W of losses are caused only by the measurement and control system. When the transistors are driven without any load, the converter losses are negligible because the current is too low. So the measured losses are approximately equal to the driving circuit + the measurement and control system. Based on these values, Table 6.3 also presents an estimate of the efficiency without the measurement and control system, and another without the driving circuit. The converter efficiency is around 97% to 98% for all test cases, which is close to the simulation measurements. From another perspective, this shows how significant the losses are in the rest of the circuit.

6.3.3 Temperature rise

The temperature of each element was measured before each test and after 10 min. All test show similar result, a temperature rise of up to $10\ ^{\circ}\mathrm{C}$. The inductor is the main heat element, reaching $43\ ^{\circ}\mathrm{C}$, the capacitors also reach around $40\ ^{\circ}\mathrm{C}$. As expected, the transistors do not overheat. The maximum temperature measured was $40\ ^{\circ}\mathrm{C}$.

The results are consistent with the expected behavior. In Section 5.2, it was concluded that the inductor exhibits the highest losses, followed by the transistors and the capacitors. Since the temperature rise is directly related to the power dissipated by each component, a similar trend was expected in the thermal measurements. The results confirm this expectation, with the inductor exhibiting the highest temperature.

Thermal images of the circuit were also obtained, as shown in Figures 6.11 and 6.12, allowing identification of the areas with the highest temperatures. However, image quality is limited, and a vertical offset is visible between the thermal and visible images. Furthermore, the vertical lines present in the thermal images are measurement artifacts caused by the thermal imaging device and do not represent actual temperature variations.

Despite these limitations, the results indicate that the inductor reaches the highest temperature, with a noticeable spread of heat to the surrounding area. The capacitors also dissipate some heat, though to a lower extent than the inductor. On the rear side of the PCB, only the transistors show a significant temperature increase.

Figure 6.11: Front side thermal image.

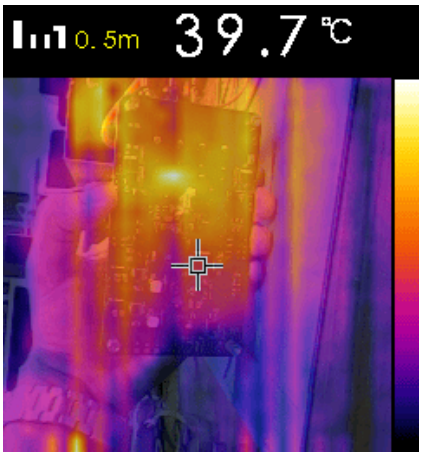


Figure 6.12: Back side thermal image.

To confirm that the temperature does not rise to a concerning level, a test of 3 h was performed with the converter operating at Case 3. Figure 6.13 shows the temperature evolution over the 3 h for the inductor, capacitors, transistors and the ambient temperature. These results were acquired using the thermistors placed on the board, which measure the ambient temperature and the transistors'

temperature. The other measurements were acquired using additional external thermistors that can be read by the board using the connector placed for thermistors.

The measured values were confirmed using an infrared thermometer which show insignificant difference in temperature, confirming the quality of the measurements.

From Figure 6.13, it can be concluded that after 40 min at maximum power, all the temperatures stop rising. Some oscillations were measured during the experiment because the ambient temperature also oscillated. As seen earlier, the inductor is the element that heats the most, followed by the transistor and capacitors. After 3 h, none of the elements reached a concerning temperature.

This test concludes that no cooling is needed. At least under the same test conditions. To confirm this statement, the board was placed inside a cardboard box since TSB usually places boards inside closed boxes. If the components inside the box need cooling, a fan or water cooling needs to be added.

Figure 6.14 represents the described test. With no airflow, the ambient temperature reading is more stable, and so are the others. As in the previous test, after 60 min the temperatures remained approximately constant and the maximum temperature was $51.6\ ^{\circ}\mathrm{C}$. Since none of the temperatures show concerning values, it is safe to conclude that no additional cooling is needed

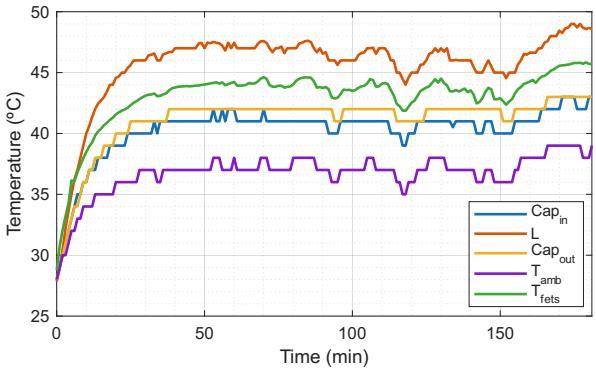


Figure 6.13: Temperature evolution during 3 h at charging case 3.

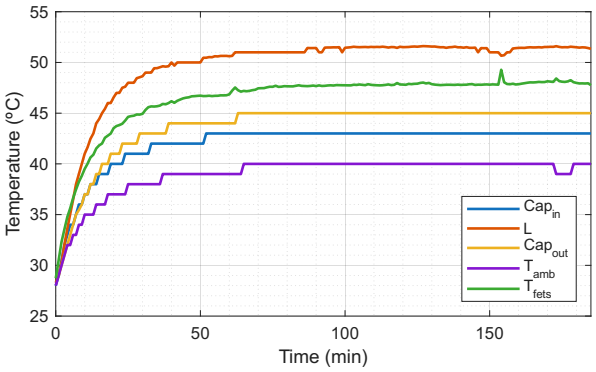


Figure 6.14: Temperature evolution during 3 h at charging case 3 in a closed box.

6.4 MPPT

In this section, the charging state machine is tested, as well as the MPPT algorithm, to ensure the functionality of the prototype.

In the following tests, a laboratory setup was used, where a power supply (the SM 1500-CP-30) was used with an internal function that can simulate the behavior of a solar panel. This function was configured as a 35 cells solar panel at configurable temperature and irradiance controlled by sending TCP messages.

The load was the battery from SR03, which is described in section 2.3.

This setup ensures a controlled environment where the solar panel operates at specified temperature and irradiance, increasing the number of controlled variables. Because the user sets the environmental conditions, this setup can be easily replicated for future comparisons, increasing the scientific value of the tests.

6.4.1 State machine

First, the state machine was tested. Figure 6.15 illustrates the evolution of the different states over time. The state machine responded as expected when both the source and load were disconnected. When both the source and load were connected, the P&O/MPPT state was activated. After a few seconds, the irradiance level of the power supply, which was used to emulate a solar panel, was increased, resulting in an increase in the output current. For this test, the CC current limit was set to $2\ \mathrm{A}$. Consequently, once the output current reached this limit, the state machine transitioned to the CC state.

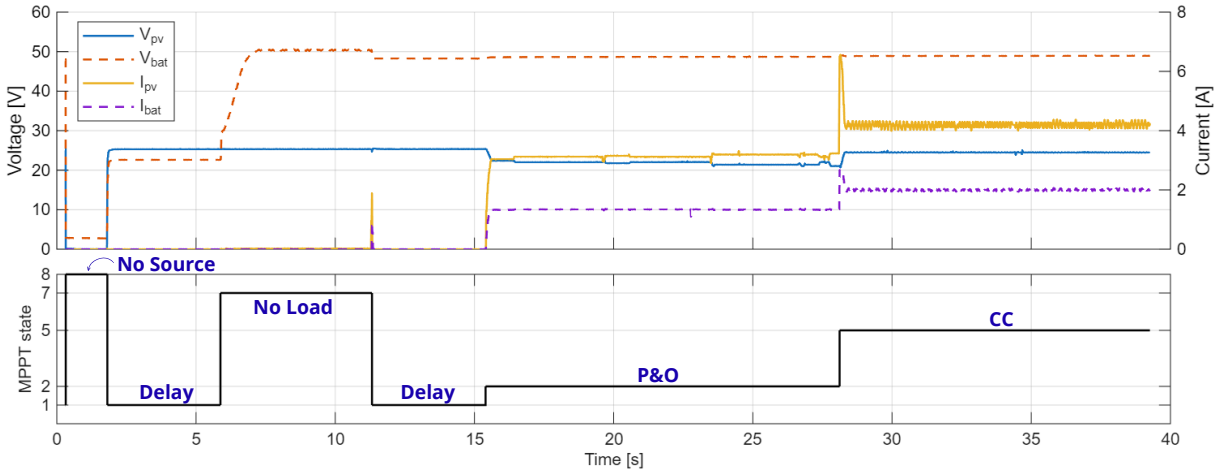


Figure 6.15: State machine operation over time.

The sweep mode was disabled for this test because its operation with a power supply resulted in occasional overcurrent events. However, when using a real solar panel, the sweep mode operated as expected, as shown in Figure 6.16. At approximately $27\ \mathrm{s}$, the sweep was initiated, causing the PV voltage to gradually increase while the current decreased. After a few seconds, the sweep was completed, and the operating point transitioned to the MPP, while the P&O was simultaneously initiated. During the sweep, two states were observed: the sweep delay state, which occupied most of the sweep duration, and the sweep state, during which the duty cycle was progressively decreased.

The CV and charge finished states are demonstrated later during a complete charging cycle.

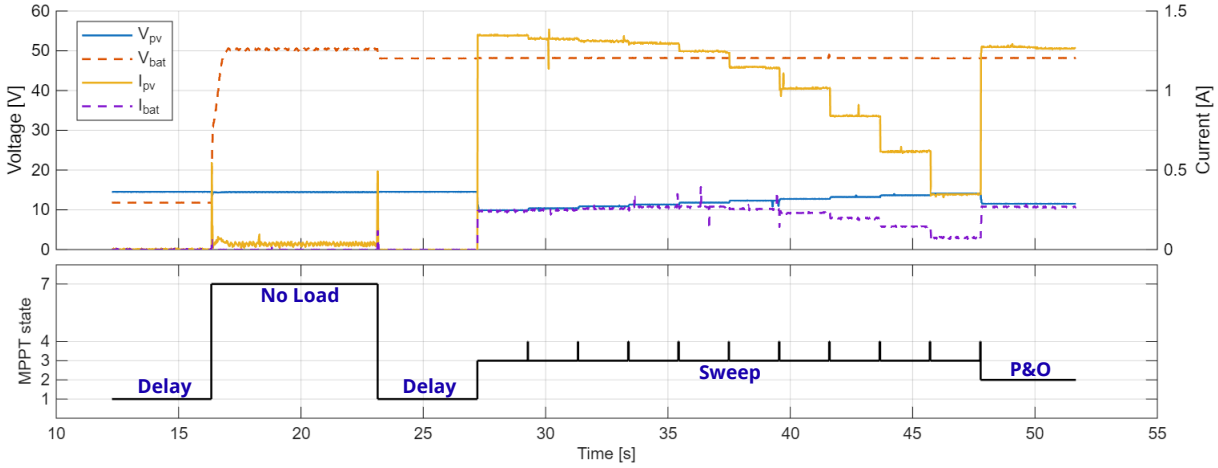


Figure 6.16: State machine operation over time (sweep mode).

6.4.2 Efficiency

To validate the converter's efficiency under different current and voltage inputs, an irradiance sweep was performed with $10\ \mathrm{W/m^2}$ steps from $100\ \mathrm{W/m^2}$ to $1000\ \mathrm{W/m^2}$. Each irradiance level is maintained for 2 s to ensure that the algorithm is stable. All the values measured during this time are used to compute an average. Additionally, outliers were removed for better results.

The result is an efficiency curve (Figure 6.17) that increases with both the input current and the irradiance, which are approximately proportional. Since the test was performed with different solar panel parameters, three curves were obtained. The converter is clearly more efficient at higher input voltages, that is, for panels with more cells in series.

Both trends are expected. At low currents, the efficiency is lower because the constant losses required to power the electronics (MCU, sensors, etc.) have a larger relative impact. As the current increases, those losses become less significant and the efficiency stabilizes. After around 3 A ($400\ \mathrm{W/m^2}$), the converter switching losses become dominant. Since they increase with the input current, a slight downward slope of the curve is expected.

The higher efficiency at higher input voltage is explained by the fact that a boost converter is more efficient at lower duty cycles. From equation 4.2, for a given output voltage the duty cycle decreases as the input voltage increases. Therefore, increasing the number of cells in series reduces the duty cycle and improves the efficiency.

On the other hand, if the input voltage is constant and the output voltage increases (Figure 6.18), efficiency decreases for the same reason. Equation 4.2 shows that for a fixed input voltage, the duty cycle increases with the increase of the output voltage and therefore efficiency decreases.

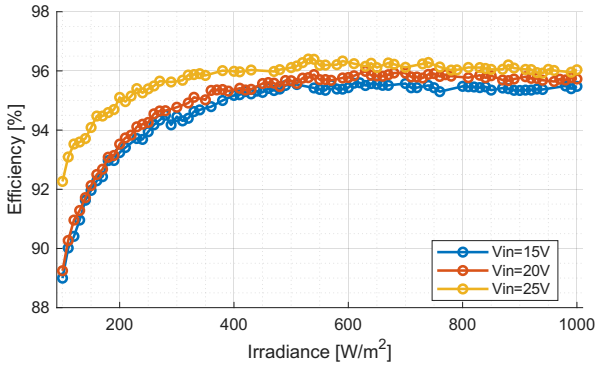


Figure 6.17: Converter efficiency over different input voltages and irradiance ($V_{\text{bat}} = 43.4$ V).

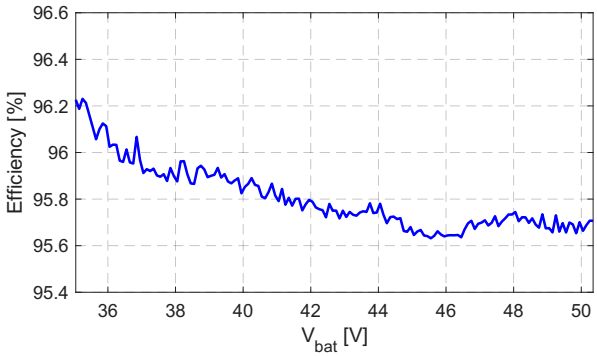


Figure 6.18: Converter efficiency over output voltage ($V_{\text{pv}} = 21.3$ V and $I_{\text{pv}} = 6.5$ A).

6.4.3 Tracking speed

Several configurations were tested to optimize the P&O algorithm. Following the tuning process, the final configuration was defined as shown in Table 6.4.

Table 6.4: MPPT parameters.

Sampling frequency	MPPT frequency	Step size	Power Dead Band
1 kHz	50 Hz	0.7 %	700 mW

To evaluate the performance of the MPPT algorithm under varying environmental conditions, several Python scripts were developed to control the temperature and irradiance of the power source. Through a LAN connection, the scripts transmit TCP messages to the power supply, allowing its environmental parameters to be modified during the tests.

Two test scenarios were defined:

1. **Irradiance staircase:** The test starts at an irradiance of $1000\ \text{W/m}^2$, which is subsequently reduced by $200\ \text{W/m}^2$ at approximately 2 s intervals until reaching $200\ \text{W/m}^2$. The irradiance is then increased to $1000\ \text{W/m}^2$ in one step.

2. **Temperature staircase:** The test starts at $25\ ^\circ\text{C}$, after which the temperature is increased in $10\ ^\circ\text{C}$ steps at 2 s intervals until reaching $85\ ^\circ\text{C}$. The temperature is then decreased to $25\ ^\circ\text{C}$ in one step.

Both tests were divided into two phases. In the first phase, the environmental parameters were varied gradually to evaluate the tracking performance under normal operating conditions. In the second phase, more abrupt changes were introduced to assess the tracking performance under more demanding conditions. This approach allows the response of the algorithm to be evaluated for both gradual and rapid variations in the operating conditions.

The results of the irradiance and temperature staircase tests are presented in Figures 6.19 and 6.20,

respectively. Overall, the algorithm demonstrated satisfactory tracking performance, maintaining the operating point close to the MPP while responding to changes in the environmental conditions.

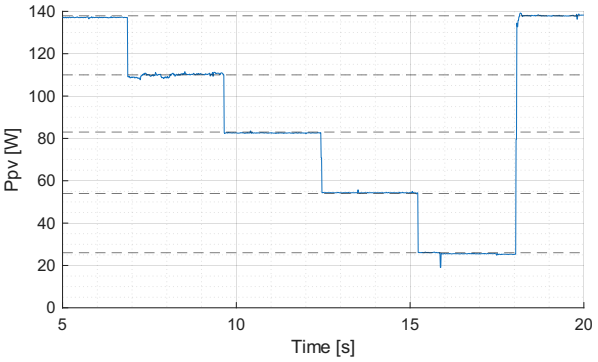


Figure 6.19: Power response for the "Irradiance stairway" test.

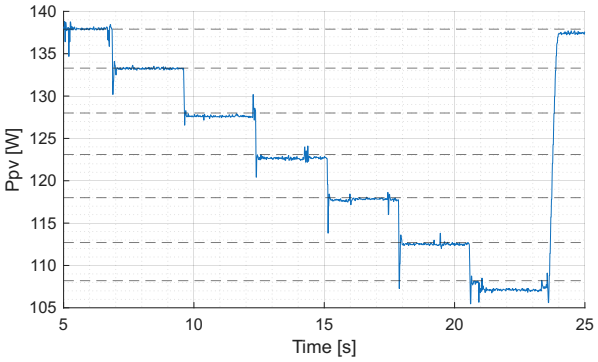


Figure 6.20: Power response for the "Temperature stairway" test.

For both tests, the measured tracking time, defined as the time required to reach within $\pm 5\%$ of the MPP, was typically below 100 ms, based on the measurements obtained during the first phase. When the irradiance was increased from $200\ \mathrm{W/m^2}$ to $1000\ \mathrm{W/m^2}$, the tracking time was slightly higher but remained below 100 ms. The highest measured tracking time occurred when the solar panels were cooled from $85\ ^\circ\mathrm{C}$ to $25\ ^\circ\mathrm{C}$, for which a tracking time of approximately 200 ms was measured. Individual tracking times for each test condition are not presented, as the response was nearly instantaneous in most cases, making accurate measurement difficult with the available instrumentation.

In addition to the tracking speed, the MPPT algorithm demonstrated a high tracking accuracy, with an average value of approximately 99%. Nevertheless, some deviations from the MPP were observed under specific operating conditions, particularly at $85\ ^\circ\mathrm{C}$. These deviations are likely related to measurement noise, although this could not be conclusively verified.

6.5 Charging test

To prove the functionality of the developed prototype, two charging tests were recorded. Both tests use the smaller battery described in section 2.3.

By using the SM 1500-CP-30 power supply in photovoltaic function connected to the MPPT charger, the battery was fully charged. Figure 6.21 shows the result where the CC-CV charging curve was achieved. In CC mode, the algorithm tracked the MPP successfully. At CV, the output voltage is constant, and the output current dropped slowly.

The last test is a field test (on shore) with a solar panel. Several attempts were made to perfect the performance of the prototype. A lot of errors were encountered and fixed. For example, the first time the test was performed, it was used a solar panel of only 24 cells, which was never tested before, and did not

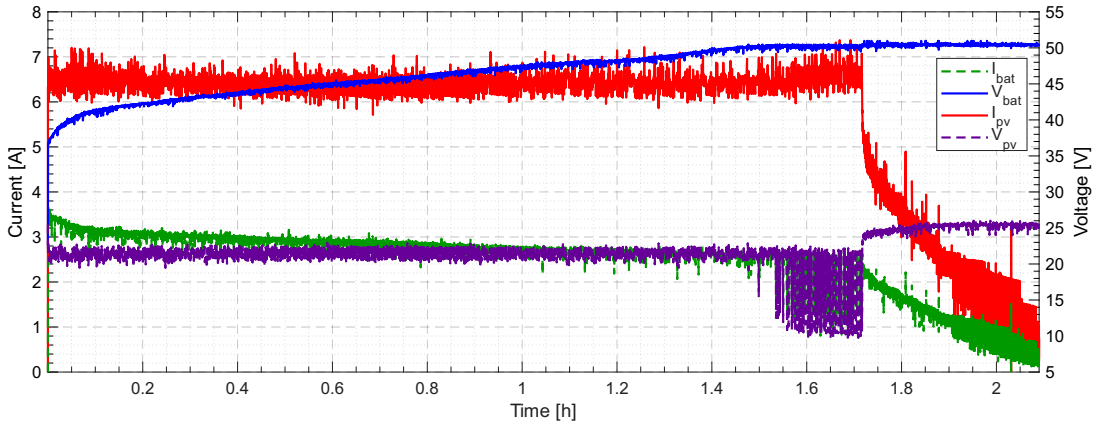


Figure 6.21: Full battery charge in laboratory.

work correctly because the duty cycle upper limit was too low. By increasing the duty cycle upper limit, the issue was solved. Then in prototype version 1.2, the noise was reduced and the power deadband was too high, which made the MPPT algorithm not able to track slow environment changes. After some tests, the power deadband was completely removed. Some other adjustments were made to the state machine and the final test was performed.

The final test was performed with two solar panels in series, with a total of 42 cells. The test was performed in a cloudy day, with a lot of irradiance variations. The results show the correct operation of the CC-CV charging curve, and the correct operation of the MPPT algorithm, Figure 6.22.

Since the test was performed in a cloudy day, the irradiance was very unstable, which induces PV current variations. The temperature of the solar panel was stable and therefore the PV voltage only adjust slightly to reach the MPP.

The test was interrupted one time due to an accidental disconnection of the solar panel.

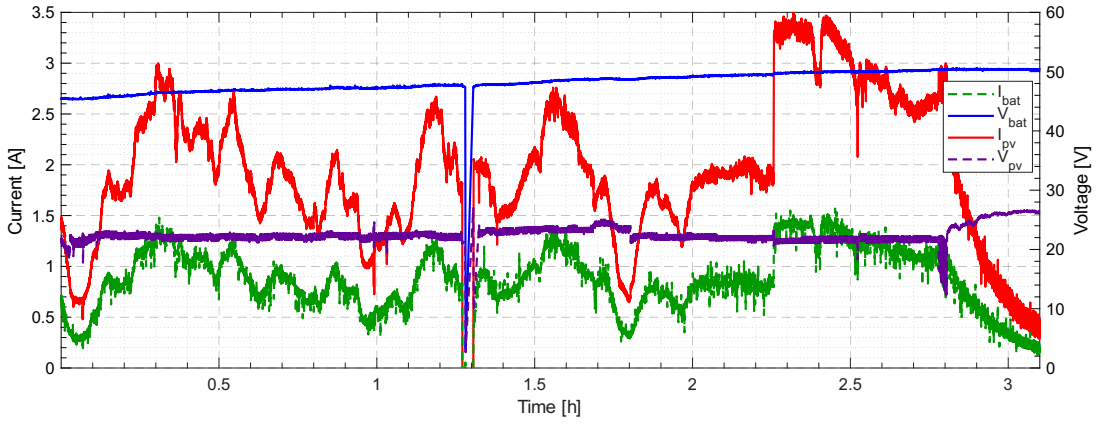


Figure 6.22: Battery charging test. Voltage and current curve.

7

Conclusion

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7.1 Future work 79

The objective of this thesis was to develop a low-cost and high efficiency MPPT charger for SG01, while also increasing the data acquired during operation.

The potential impact of such a prototype was assessed by estimating the time to drain a battery with and without solar panels, with two different batteries. For both batteries, the calculated discharge time can increase up to 200%, although it is highly dependent on the environmental conditions. To establish a target tracking time, environmental variations were analyzed, indicating that the tracking algorithm should be capable of reaching the MPP within $1\ \mathrm{s}$ to remain effective under real operating conditions.

A state-of-the-art review identified suitable MPPT algorithms and DC-DC converters useful for this project. A subsequent market analysis revealed a lack of companies addressing the subject. The few companies that did present some flaws, such as high cost, incompatibility with the system, or limited data logging and communication capabilities.

To meet the project requirements, a synchronous boost converter was carefully dimensioned and validated in LTspice simulations, while the MPPT algorithm was developed in MATLAB Simulink, which helped to understand how to tune the algorithm and what its limitations are. The P&O algorithm was selected because it offers satisfactory tracking speed and accuracy, along with a simple implementation and tuning process.

After selecting the converter components, an STM32 microcontroller and voltage and current sensors, a PCB was designed, manufactured, and hand soldered. The estimated cost of the final prototype is 125 €, which is lower than the equivalent commercial solutions currently used (245 €) and meets the cost requirement set for this project.

Initial tests revealed concerning levels of noise caused by the switching converter. The problem was addressed with several SMD capacitors, which drastically improved the noise. A conducted emissions measurement based on CISPR 25:2021 was then performed to characterize the residual noise against the limits relevant to the vessel communication systems, such as CAN and VHF. In both the loaded and unloaded conditions, the measured spectrum remained below the Class 5 limits, the most stringent class defined by the standard. This is an indicative comparison, not a formal compliance test. Additionally, it was observed that the inductor heats up during operation. A 3 h continuous test, performed both in open air and inside an enclosed box, confirmed that all component temperatures stabilize after approximately 1 h, reaching a maximum of $51.6\ ^{\circ}\mathrm{C}$, within safe operating limits.

Performance testing produced strong results overall. The efficiency was measured under several conditions, such as DC-DC mode, current sweep, or battery sweep. It was measured an efficiency of up to 96.84%. When isolating the losses of the boost converter itself from those of the measurement and control circuit, the converter estimated converter efficiency is 98%. The tracking speed is consistently under 100 ms, reaching up to 200 ms in the most demanding test case, while averaging 99% tracking accuracy. In all cases the response remains below the design targets of 0.5 s under typical conditions

and 1 s in the worst case. Finally, the prototype successfully charged a battery following the expected CC-CV curve in both laboratory and field tests (on shore), demonstrating that the system performs correctly as an integrated whole and not only at the component level.

These results confirm that the developed prototype meets the efficiency, cost, and tracking speed requirements defined for this project. However, as a prototype, it still has some limitations: measurement accuracy, cost optimization, lack of tests with different battery voltages and solar panels, field tests with the prototype installed in a vessel, etc. These limitations, along with further improvements, are discussed in the following section.

7.1 Future work

The prototype developed in this thesis is not a finished product ready for market and fully tested under all conditions. However, its functionality was proven and can be implemented in SG01. As a prototype, there remains considerable room for improvement.

Due to the limited time available the prototype was not tested in the field, installed in a vessel. This is an important step to validate the system under real operating conditions, which may include more extreme environmental variations.

On the control side, additional MPPT algorithms should be implemented, tuned and compared under the same operating conditions. A dedicated irradiance measurement, used with the existing temperature sensing, would also enable a more accurate estimate of conversion losses and could serve as input to more advanced tracking methods.

Regarding the processing unit, a MCU with a native 16-bit ADC would improve measurement resolution, while a higher clock frequency would increase the PWM duty cycle resolution, which is particularly relevant to increase accuracy.

Finally, both cost and output voltage ripple could be reduced by coordinating the MPPTs that operate in parallel. If the converters are interleaved, that is, operated with a controlled phase shift, the individual output ripples tend to cancel rather than add. A common output capacitor bank could then be shared among chargers, reducing the required capacitance per unit and the overall cost of the installation.

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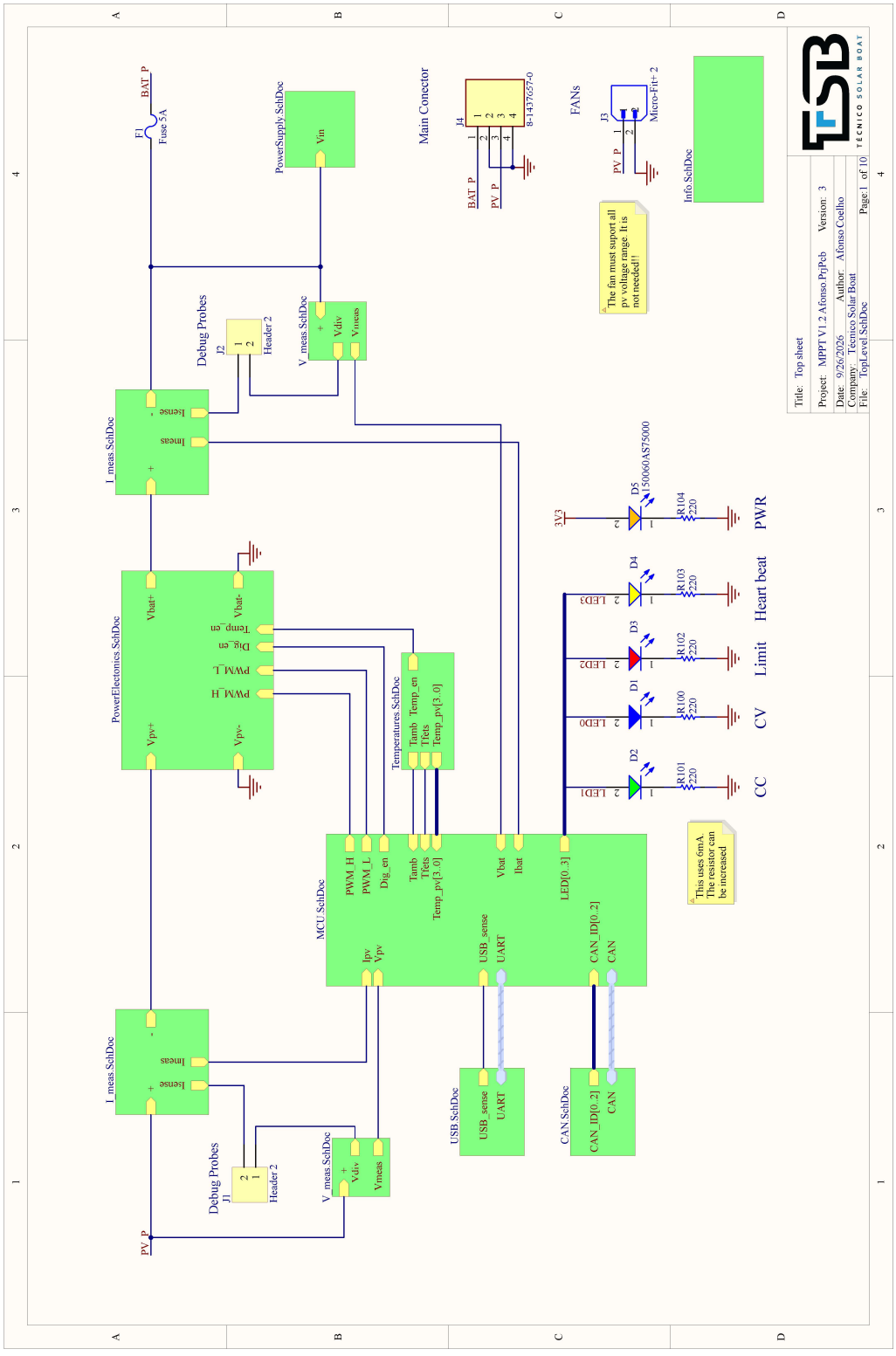
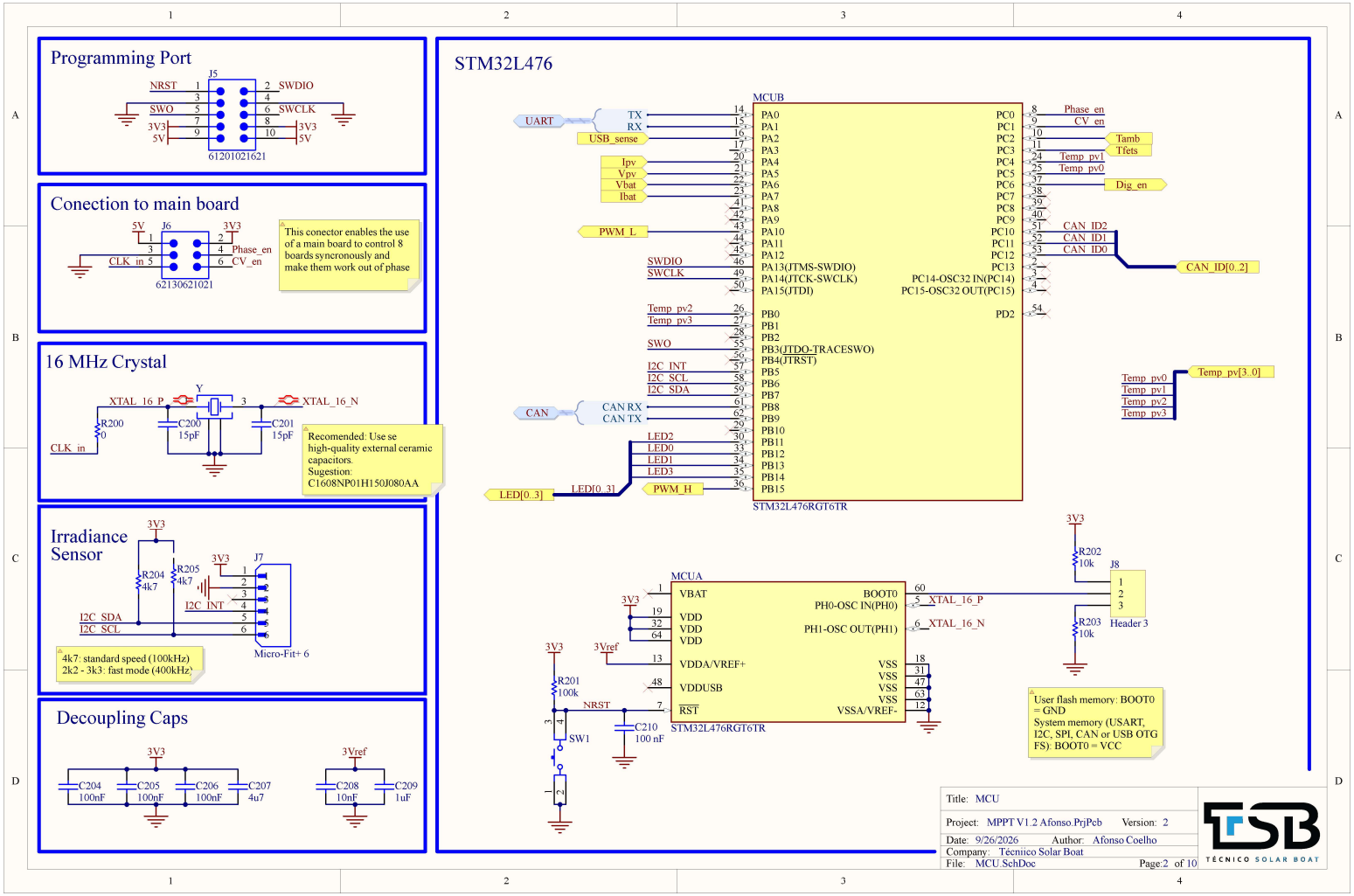
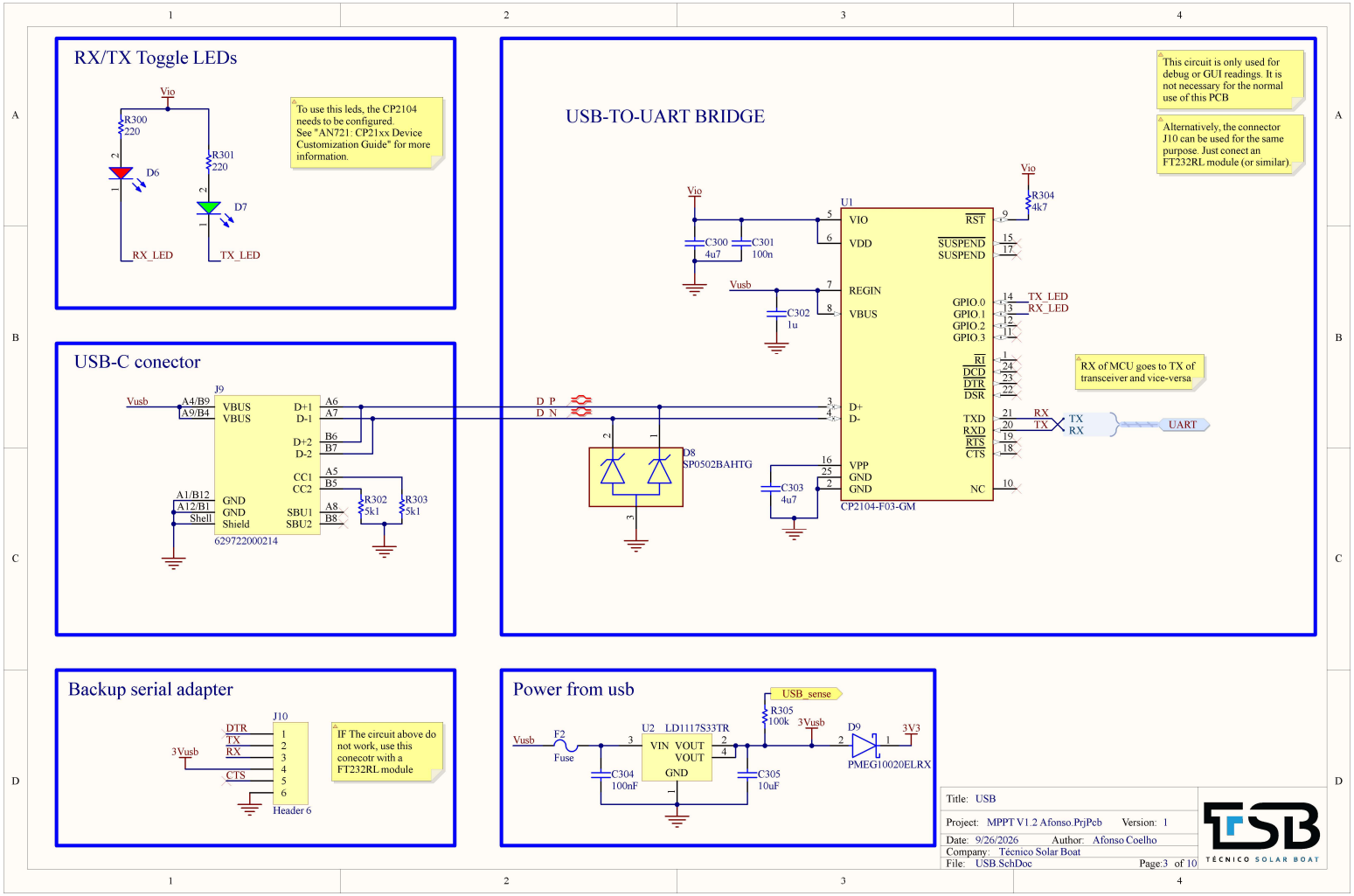
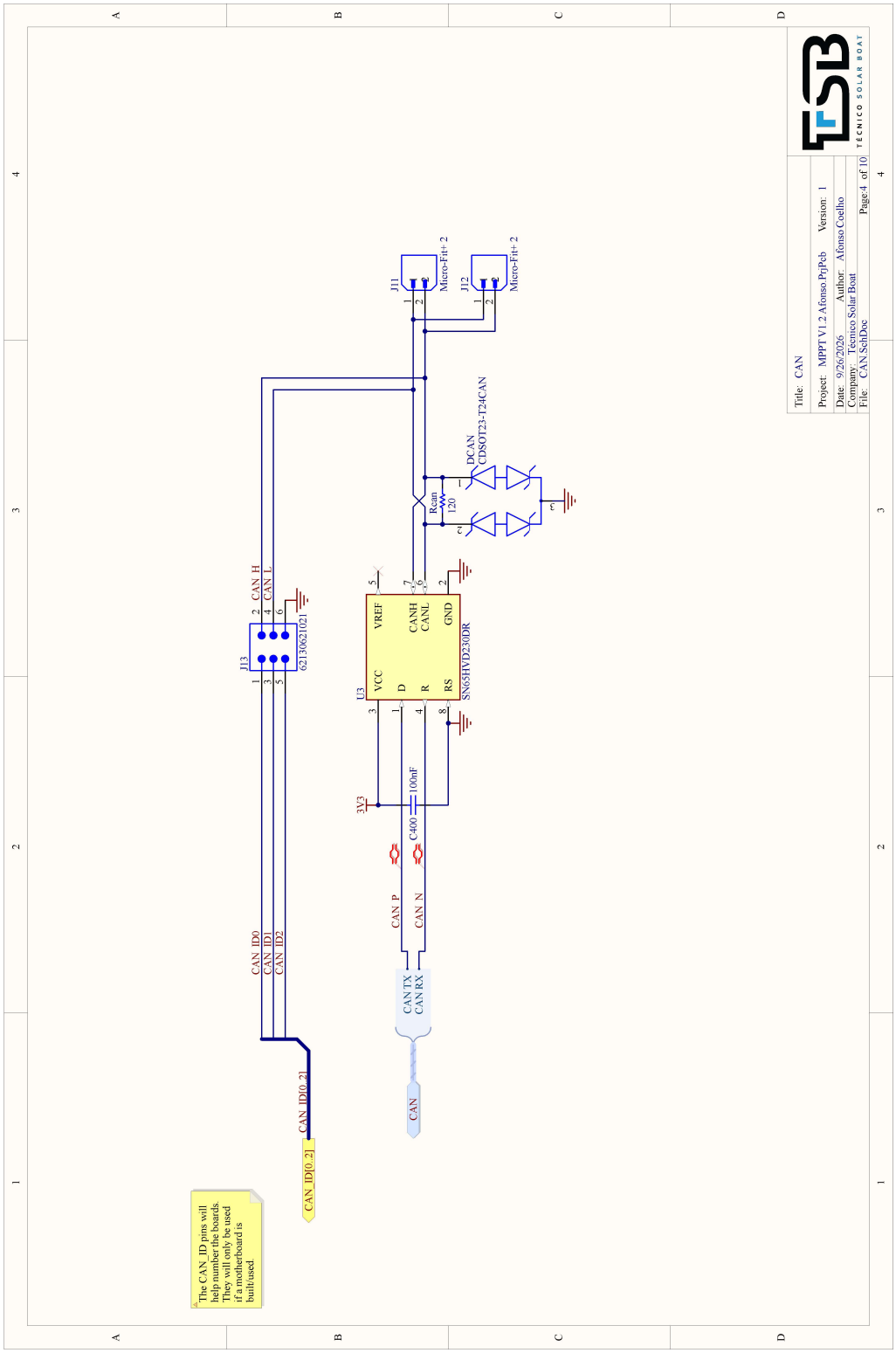
A

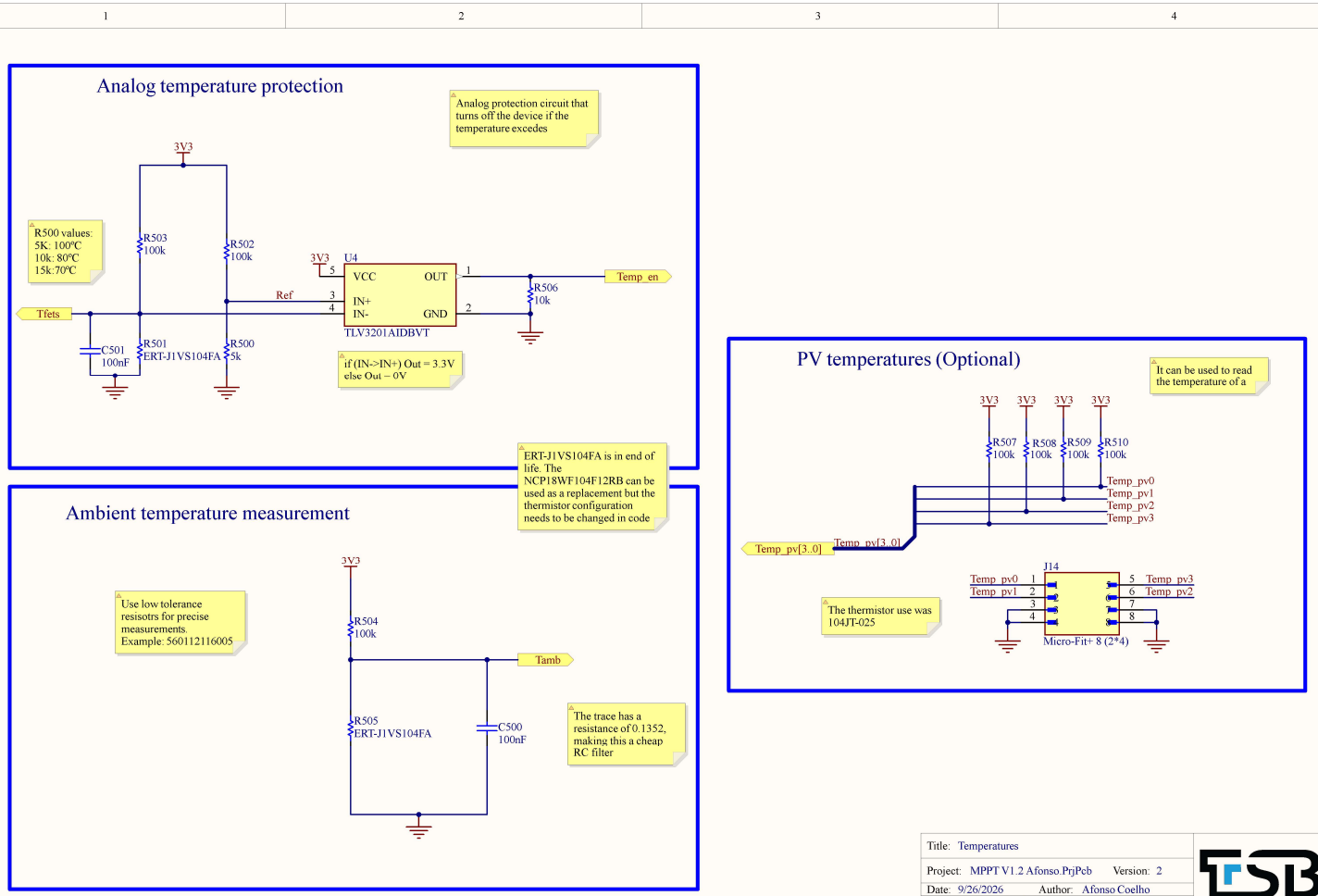
MPPT V1.2 schematics

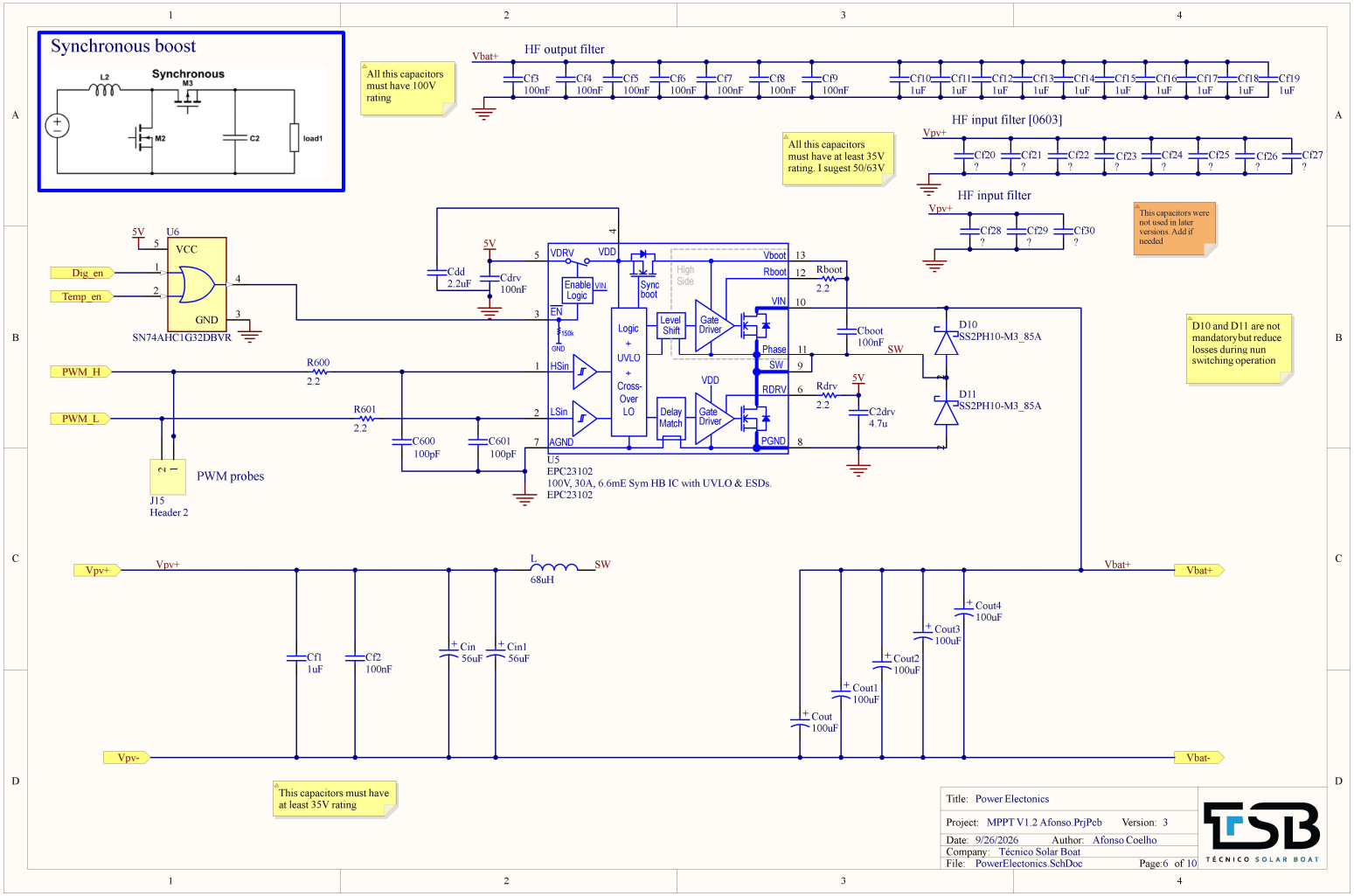

Title: Top sheet
Project: MPPT V1.2 Afonso.PrjPcb Version: 3
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: TopLevel.SchDoc Page 1 of 10
TÉCNICO SOLAR BOAT
Main Conector
FANs
Debug Probes
Debug Probes
PowerSupply.SchDoc
I_meas.SchDoc
I_meas.SchDoc
V_meas.SchDoc
V_meas.SchDoc
PowerElectronics.SchDoc
Temperatures.SchDoc
MCU.SchDoc
USB.SchDoc
CAN.SchDoc
Info.SchDoc
The fan must suport all pv voltage range. It is not needed!!!
This uses 6mA. The resistor can be increased
CC
CV
Limit
Heart beat
PWR

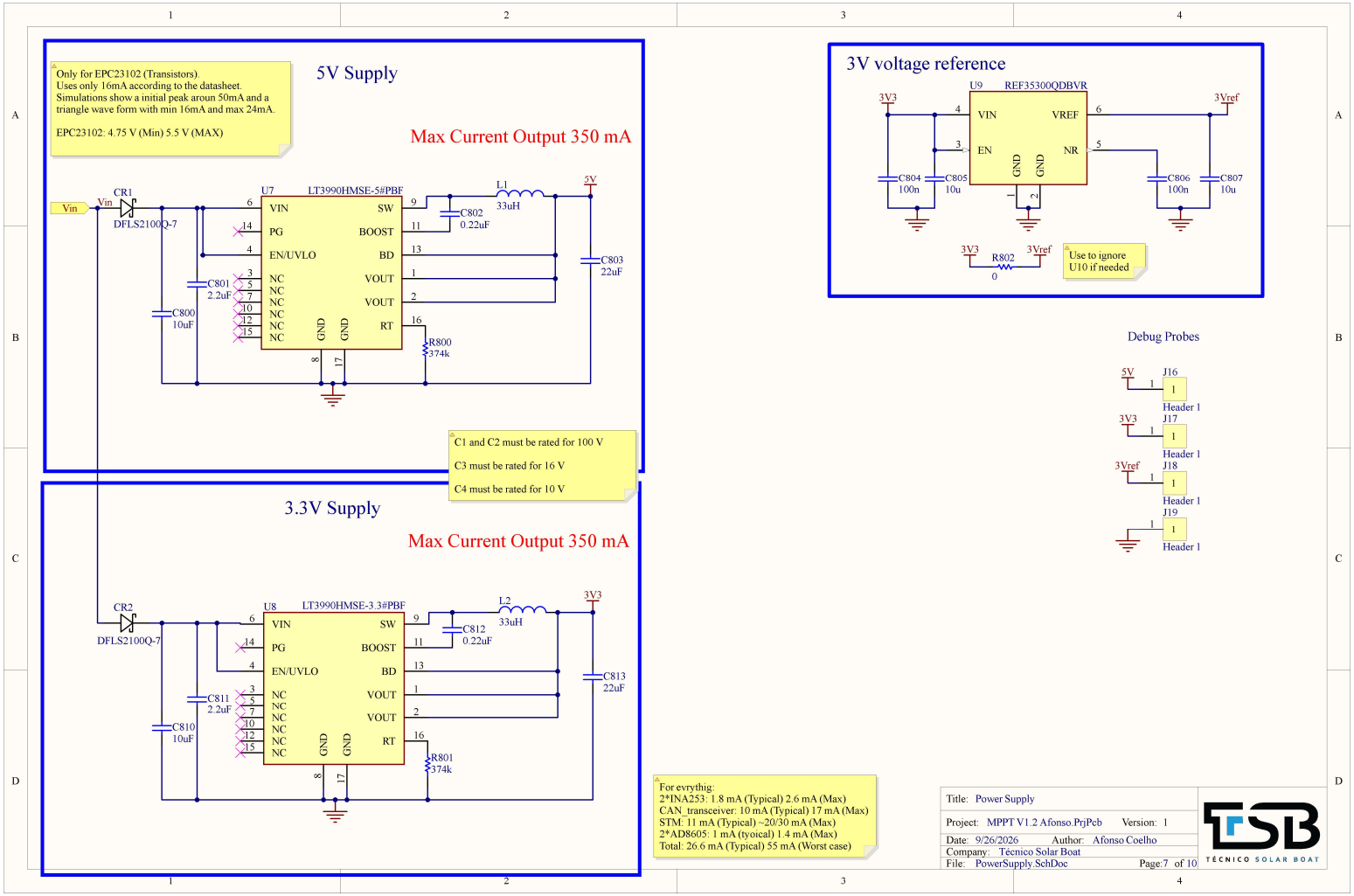

Programming Port
J5
NRST
SWO
3V3
5V
SWDIO
SWCLK
61201021621
Conection to main board
J6
CLK_in 5
Phase_en
CV_en
62130621021
This conector enables the use of a main board to control 8 boards syncronously and make them work out of phase
16 MHz Crystal
XTAL_16_P
XTAL_16_N
R200
0
C200
15pF
C201
15pF
CLK_in
Recomended: Use se high-quality external ceramic capacitors.
Sugestion:
C1608NP01H150J080AA
Irradiance Sensor
R204
4k7
R205
4k7
I2C_INT
I2C_SDA
I2C_SCL
J7
Micro-Fit+ 6
4k7: standard speed (100kHz)
2k2 - 3k3: fast mode (400kHz)
Decoupling Caps
C204
100nF
C205
100nF
C206
100nF
C207
4u7
3Vref
C208
10nF
C209
1uF
STM32L476
MCUB
UART
TX
RX
USB_sense
Ipv
Vpv
Vbat
Ibat
PWM_L
SWDIO
SWCLK
Temp_pv2
Temp_pv3
SWO
I2C_INT
I2C_SCL
I2C_SDA
CAN
CAN RX
CAN TX
LED2
LED0
LED1
LED3
LED[0..3]
LED[0..3]
PWM_H
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2
PB3(JTDO-TRACESWO)
PB4(JTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32 IN(PC14)
PC15-OSC32 OUT(PC15)
PD2
Phase_en
CV_en
Tamb
Tfets
Temp_pv1
Temp_pv0
Dig_en
CAN_ID2
CAN_ID1
CAN_ID0
CAN_ID[0..2]
Temp_pv0
Temp_pv1
Temp_pv2
Temp_pv3
Temp_pv[3..0]
STM32L476RGT6TR
MCUA
VBAT
VDD
VDD
VDD
VDDA/VREF+
VDDUSB
RST
BOOT0
PH0-OSC IN(PH0)
PH1-OSC OUT(PH1)
VSS
VSS
VSS
VSS
VSSA/VREF-
XTAL_16_P
XTAL_16_N
STM32L476RGT6TR
R201
100k
NRST
C210
100 nF
SW1
R202
10k
R203
10k
J8
Header 3
User flash memory: BOOT0 = GND
System memory (USART, I2C, SPI, CAN or USB OTG FS): BOOT0 = VCC
Title: MCU
Project: MPPT V1.2 Afonso.PrjPcb Version: 2
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: MCU.SchDoc Page:2 of 10
TÉCNICO SOLAR BOAT

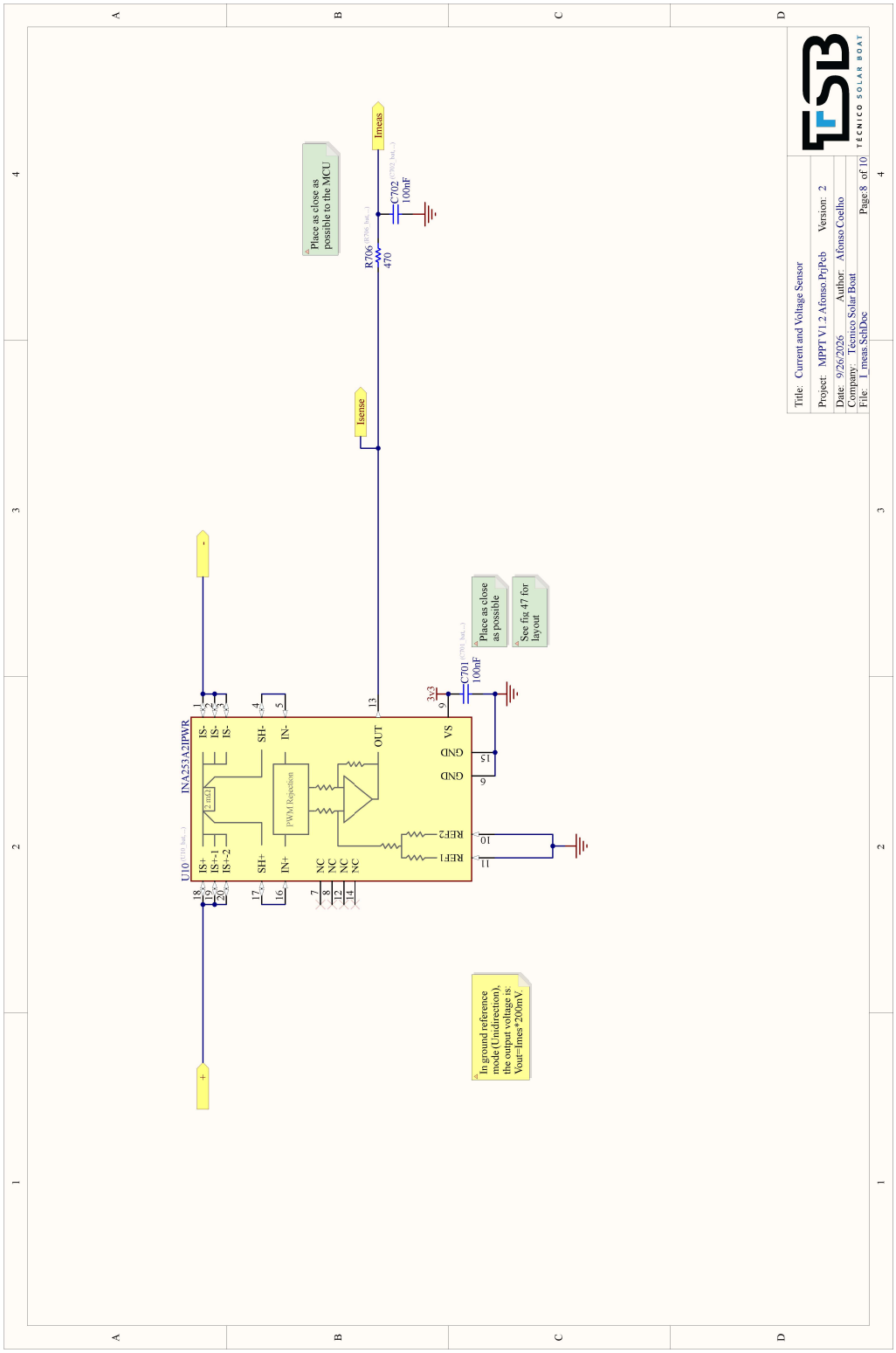

RX/TX Toggle LEDs
Vio
R300
220
R301
220
D6
D7
RX_LED
TX_LED
To use this leds, the CP2104 needs to be configured.
See "AN721: CP21xx Device Customization Guide" for more information.
USB-TO-UART BRIDGE
This circuit is only used for debug or GUI readings. It is not necessary for the normal use of this PCB
AIternatively, the connector J10 can be used for the same purpose. Just conect an FT232RL module (or similar)
U1
VIO
VDD
REGIN
VBUS
RST
SUSPEND
SUSPEND
GPIO.0
GPIO.1
GPIO.2
GPIO.3
RI
DCD
DTR
DSR
D+
D-
TXD
RXD
RTS
CTS
VPP
GND
GND
NC
CP2104-F03-GM
C300
4u7
C301
100n
C302
1u
C303
4u7
R304
4k7
TX_LED
RX_LED
RX of MCU goes to TX of transceiver and vice-versa
RX
TX
TX
RX
UART
D_P
D_N
D8
SP0502BAHTG
USB-C conector
J9
Vusb
A4/B9
A9/B4
VBUS
VBUS
D+1
D-1
D+2
D-2
CC1
CC2
GND
GND
Shell
SBU1
SBU2
A6
A7
B6
B7
A5
B5
A8
B8
A1/B12
A12/B1
R302
5k1
R303
5k1
629722000214
Backup serial adapter
J10
3Vusb
DTR
TX
RX
CTS
Header 6
IF The circuit above do not work, use this conecotr with a FT232RL module
Power from usb
USB_sense
R305
100k
3Vusb
Vusb
F2
Fuse
U2 LD1117S33TR
VIN
VOUT
VOUT
GND
C304
100nF
C305
10uF
D9
3V3
PMEG10020ELRX
Title: USB
Project: MPPT V1.2 Afonso.PrjPcb Version: 1
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: USB.SchDoc Page:3 of 10
TÉCNICO SOLAR BOAT

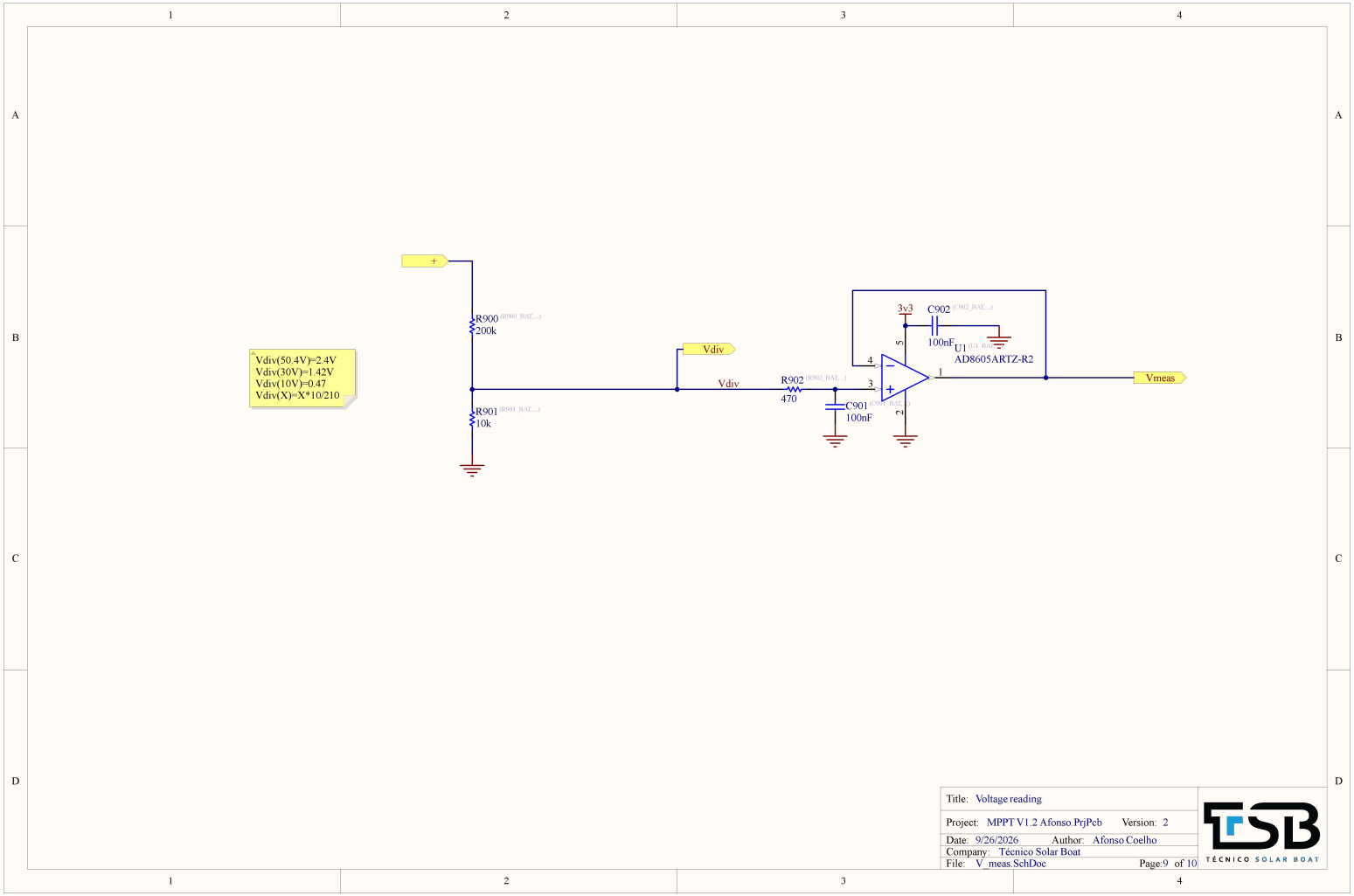

Title: CAN
Project: MPPT V1.2 Afonso.PrjPcb Version: 1
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: CAN.SchDoc Page 4 of 10
TÉCNICO SOLAR BOAT
The CAN_ID pins will help number the boards. They will only be used if a motherboard is built/used.
CAN_ID[0..2]
CAN_ID0
CAN_ID1
CAN_ID2
CAN_H
CAN_L
J13
6213662102L
J11
Micro-Fit+ 2
J12
Micro-Fit+ 2
Rcan
120
DCAN
CDSOT23-T24CAN
U3
SN65HVD230DR
VCC
D
R
RS
VREF
CANH
CANL
GND
3V3
C400
100nF
CAN P
CAN N
CAN TX
CAN RX
CAN


Analog temperature protection
Analog protection circuit that turns off the device if the temperature excedes
3V3
R500 values:
5K: 100ºC
10k: 80ºC
15k:70ºC
R503
100k
R502
100k
3V3
U4
5
VCC
OUT
1
Temp_en
Ref
3
IN+
4
IN-
GND
2
R506
10k
Tfets
C501
100nF
R501
ERT-J1VS104FA
R500
5k
TLV3201AIDBVT
if (IN->IN+) Out = 3.3V
else Out = 0V
ERT-J1VS104FA is in end of life. The NCP18WF104F12RB can be used as a replacement but the thermistor configuration needs to be changed in code
Ambient temperature measurement
Use low tolerance resisotrs for precise measurements.
Example: 560112116005
3V3
R504
100k
Tamb
R505
ERT-J1VS104FA
C500
100nF
The trace has a resistance of 0.1352, making this a cheap RC filter
PV temperatures (Optional)
It can be used to read the temperature of a
3V3
3V3
3V3
3V3
R507
100k
R508
100k
R509
100k
R510
100k
Temp_pv0
Temp_pv1
Temp_pv2
Temp_pv3
Temp_pv[3..0]
Temp_pv[3..0]
J14
Temp_pv0
1
Temp_pv1
2
3
4
5
Temp_pv3
6
Temp_pv2
7
8
Micro-Fit+ 8 (2*4)
The thermistor use was 104JT-025
Title: Temperatures
Project: MPPT V1.2 Afonso.PrjPcb
Version: 2
Date: 9/26/2026
Author: Afonso Coelho
Company: Técnico Solar Boat
File: Temperatures.SchDoc
Page:5 of 10
TÉCNICO SOLAR BOAT

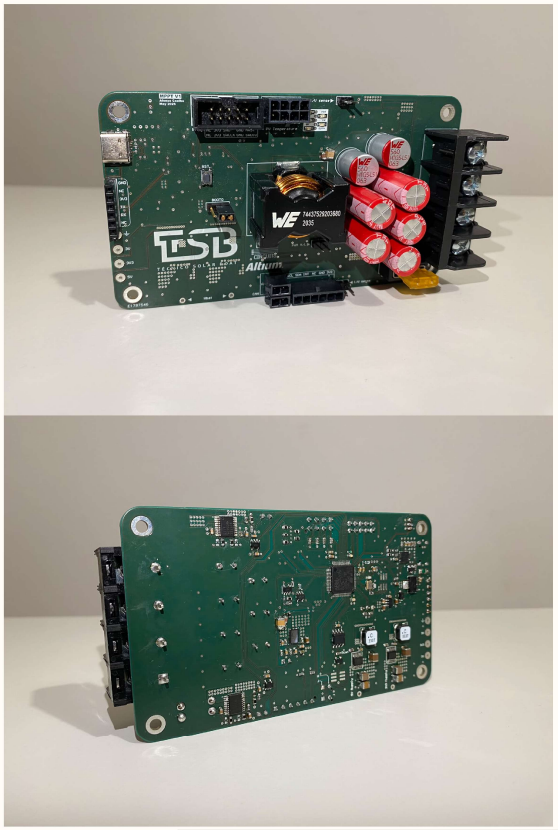

Synchronous boost
Synchronous
L2
M3
M2
C2
load1
All this capacitors must have 100V rating
HF output filter
Vbat+
Cf3 100nF
Cf4 100nF
Cf5 100nF
Cf6 100nF
Cf7 100nF
Cf8 100nF
Cf9 100nF
Cf10 1uF
Cf11 1uF
Cf12 1uF
Cf13 1uF
Cf14 1uF
Cf15 1uF
Cf16 1uF
Cf17 1uF
Cf18 1uF
Cf19 1uF
All this capacitors must have at least 35V rating. I sugest 50/63V
HF input filter [0603]
Vpv+
Cf20 ?
Cf21 ?
Cf22 ?
Cf23 ?
Cf24 ?
Cf25 ?
Cf26 ?
Cf27 ?
HF input filter
Cf28 ?
Cf29 ?
Cf30 ?
This capacitors were not used in later versions. Add if needed
5V
U6
VCC
GND
SN74AHC1G32DBVR
Dig_en
Temp_en
Cdd 2.2uF
Cdrv 100nF
VDRV
VDD
Enable Logic
VIN
EN
150k
GND
Sync boot
High Side
Vboot
Rboot
Rboot 2.2
Logic + UVLO + Cross-Over LO
Level Shift
Gate Driver
Phase
SW
RDRV
Rdrv 2.2
Delay Match
PGND
HSin
LSin
AGND
Cboot 100nF
D10
SS2PH10-M3_85A
D11
SS2PH10-M3_85A
D10 and D11 are not mandatorybut reduce losses during nun switching operation
PWM_H
PWM_L
R600 2.2
R601 2.2
C600 100pF
C601 100pF
C2drv 4.7u
PWM probes
J15
Header 2
U5
EPC23102
100V, 30A, 6.6mE Sym HB IC with UVLO & ESDs.
EPC23102
L
68uH
SW
Vpv+
Vpv-
Vbat+
Vbat-
Cf1 1uF
Cf2 100nF
Cin 56uF
Cin1 56uF
Cout 100uF
Cout1 100uF
Cout2 100uF
Cout3 100uF
Cout4 100uF
This capacitors must have at least 35V rating
Title: Power Electonics
Project: MPPT V1.2 Afonso.PrjPcb Version: 3
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: PowerElectonics.SchDoc Page:6 of 10
TÉCNICO SOLAR BOAT


Only for EPC23102 (Transistors).
Uses only 16mA according to the datasheet.
Simulations show a initial peak aroun 50mA and a triangle wave form with min 16mA and max 24mA.
EPC23102: 4.75 V (Min) 5.5 V (MAX)
5V Supply
Max Current Output 350 mA
CR1
DFLS2100Q-7
U7
LT3990HMSE-5#PBF
VIN
SW
PG
BOOST
EN/UVLO
BD
NC
VOUT
RT
GND
L1
33uH
C802
0.22uF
C803
22uF
C800
10uF
C801
2.2uF
R800
374k
C1 and C2 must be rated for 100 V
C3 must be rated for 16 V
C4 must be rated for 10 V
3.3V Supply
Max Current Output 350 mA
CR2
DFLS2100Q-7
U8
LT3990HMSE-3.3#PBF
L2
33uH
C812
0.22uF
C813
22uF
C810
10uF
C811
2.2uF
R801
374k
3V voltage reference
U9
REF35300QDBVR
VIN
VREF
EN
NR
C804
100n
C805
10u
C806
100n
C807
10u
R802
0
Use to ignore U10 if needed
Debug Probes
J16
J17
J18
J19
Header 1
For evrythig:
2*INA253: 1.8 mA (Typical) 2.6 mA (Max)
CAN_transceiver: 10 mA (Typical) 17 mA (Max)
STM: 11 mA (Typical) ~20/30 mA (Max)
2*AD8605: 1 mA (tyoical) 1.4 mA (Max)
Total: 26.6 mA (Typical) 55 mA (Worst case)
Title: Power Supply
Project: MPPT V1.2 Afonso.PrjPcb Version: 1
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: PowerSupply.SchDoc Page:7 of 10
TÉCNICO SOLAR BOAT

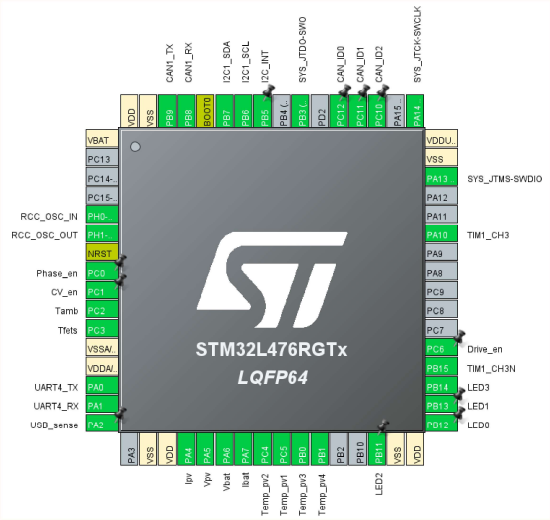

Title: Current and Voltage Sensor
Project: MPPT V1.2 Afonso.PrjPcb Version: 2
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: I_meas.SchDoc Page 8 of 10
TÉCNICO SOLAR BOAT
Imeas
Isense
R706
470
C702
100nF
Place as close as possible to the MCU
U10
INA253A2IPWR
IS-
IS-
IS-
SH-
IN-
OUT
VS
GND
GND
REF2
REF1
IS+
IS+-1
IS+-2
SH+
IN+
NC
NC
NC
NC
2 and 1
PWM Rejection
3v3
C701
100nF
Place as close as possible
See fig 47 for layout
In ground reference mode (Unidirection), the output voltage is Vout=Imes*200mV.


Vdiv(50.4V)=2.4V
Vdiv(30V)=1.42V
Vdiv(10V)=0.47
Vdiv(X)=X*10/210
+
R900
200k
R901
10k
Vdiv
Vdiv
R902
470
C901
100nF
3v3
C902
100nF
U1
AD8605ARTZ-R2
Vmeas
Title: Voltage reading
Project: MPPT V1.2 Afonso.PrjPcb Version: 2
Date: 9/26/2026 Author: Afonso Coelho
Company: Técnico Solar Boat
File: V_meas.SchDoc Page:9 of 10
TÉCNICO SOLAR BOAT

PCB (V1)



STM32CubeMX



For detailed design thinking and results go to: GitHub

COMPONET LABELS
U: ICs
J: Conectors
D: Diode/LED
Rxyy: Resistor (x=page number, yy: componet number)
Cxyy: Capacitor (x=page number, yy: componet number)

Difrences betwen V1 and V1.1:
- Solved UARTH signals swaped;
- Usb working;
- 3D model correction of main conector;
- 5V in programing port;
- 5V in or gate;
- Label sizes;
- Added diodes to power electronics circuit (incresed eff);
- Added diodes in DCDCs for reverse current protection;
- Reduced pcb cost;
- Added FANs conector;

Difrences betwen V1.1 and V1.2:
- Added capacitors to filter the high frequency noise generated by the switching converter;
- Changed the placement of the filters to reduce measurements noise;
- Fixed the missing 5V label in the flash conector;
- Changed the led labels;
- 0 Ohm resistors in current sensor circuit removed;
- Added a capacitor for Tfet measurement

Title:	Additional Information
Project:	MPPT V1.2 Afonso.PrjPcb Version: 3
Date:	9/26/2026 Author: Afonso Coelho
Company:	Técnico Solar Boat
File:	Info.SchDoc Page: 10 of 10

TÉCNICO SOLAR BOAT

B

Graphical User Interface

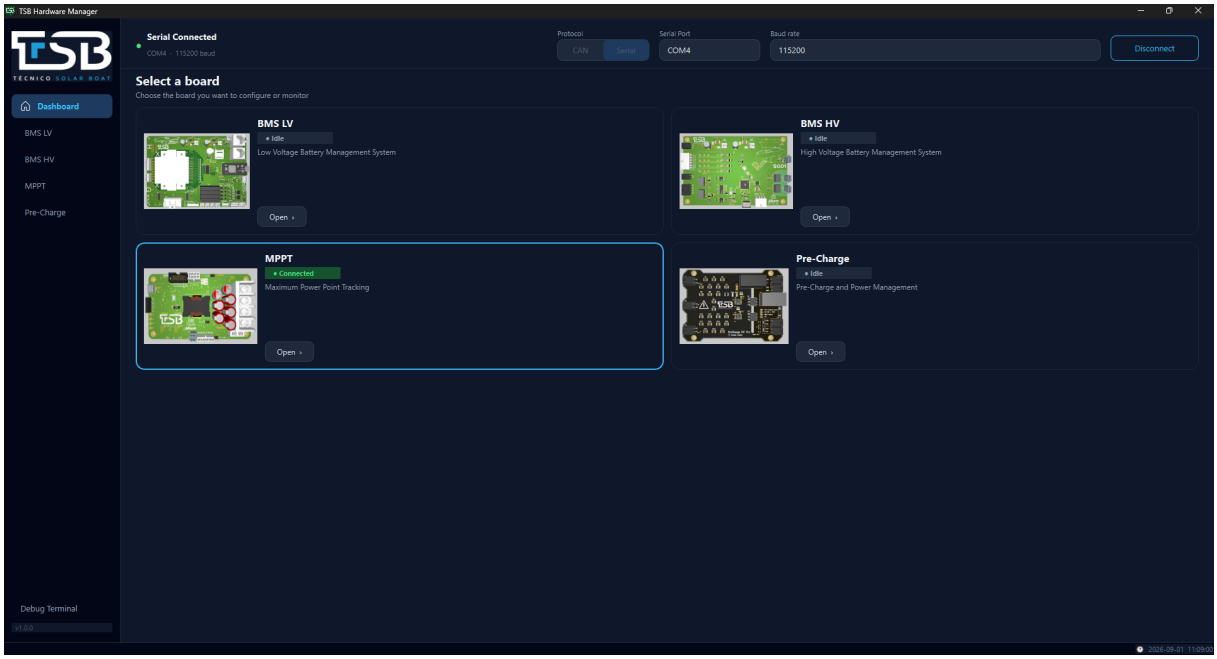


Figure B.1: Dashboard page.

Figure B.2: MPPT tab.

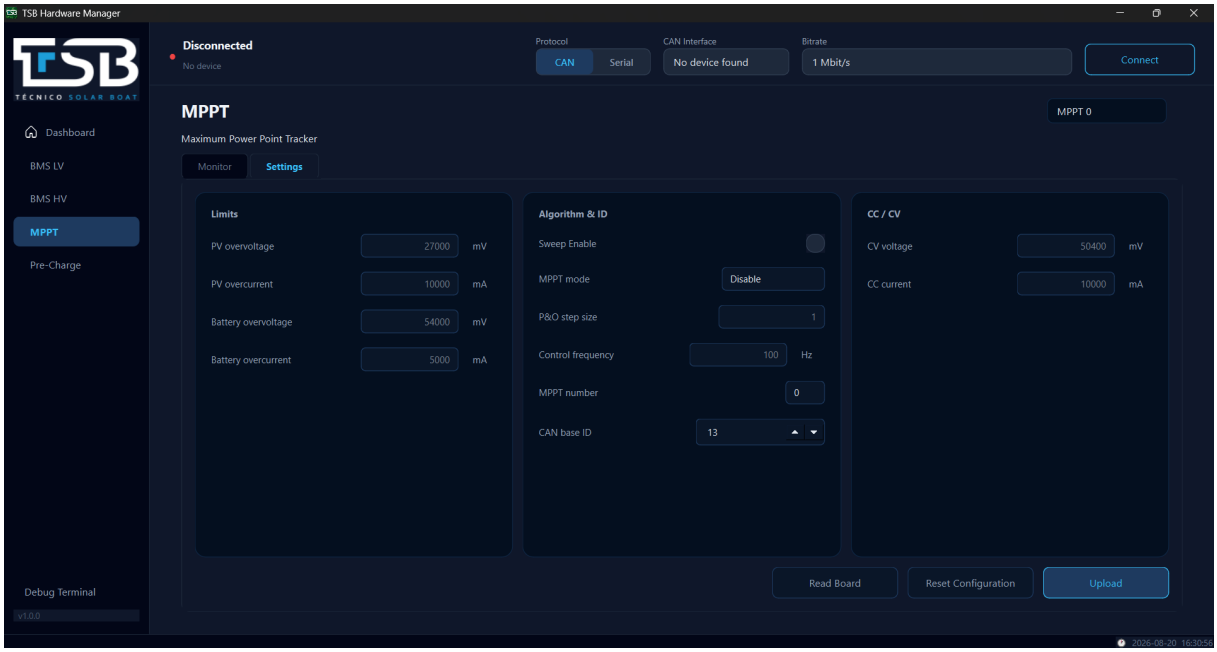


Figure B.3: MPPT configuration tab.

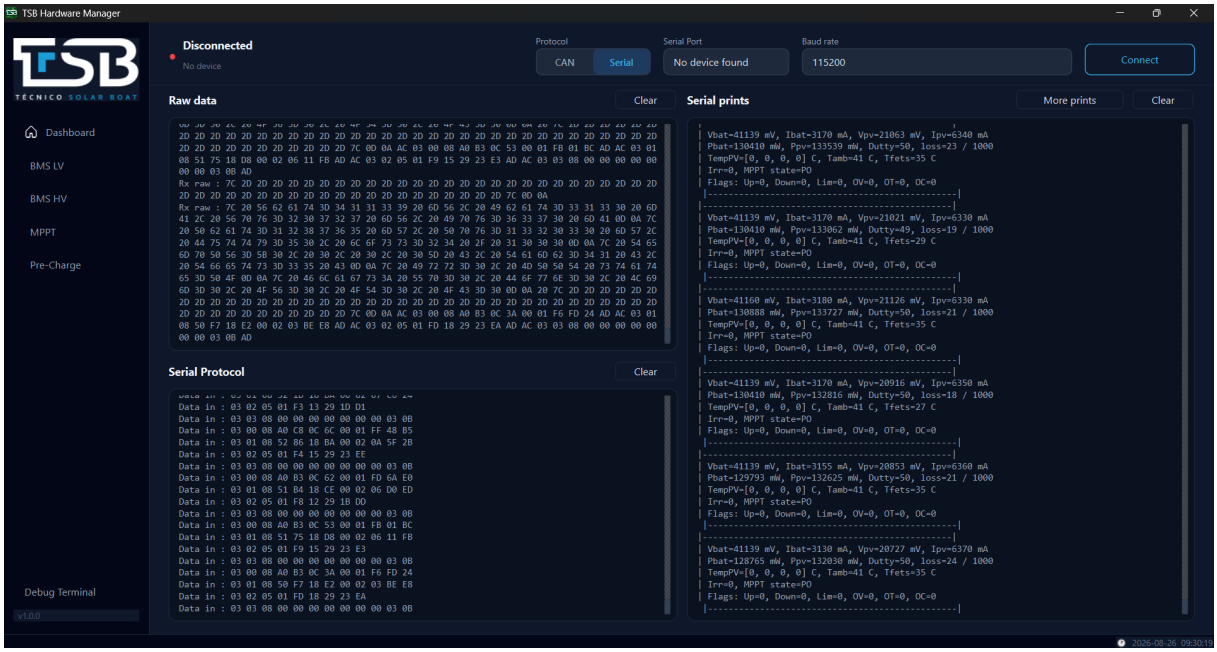


Figure B.4: Debug page.